Building Segregation: The Long-Run Neighborhood Effects of American Public Housing*

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Abstract

This paper studies the long-term neighborhood effects of the construction of mid-century American public housing. I construct a new national dataset tracking the locations, completion dates, and characteristics of over 1 million public housing units built between 1935 and 1973, which I link to neighborhood-level data from 1930 to 2010. I document that public housing projects were systematically targeted towards poorer, more populated neighborhoods with higher Black population shares, consistent with the program's slum clearance goals and racialized site selection politics. Using a stacked matched difference-in-differences design, I find that public housing construction caused large, persistent increases in Black population shares and substantial declines in median incomes and rents. Geographic spillovers to nearby neighborhoods were modest: Black population shares increased slightly, driven primarily by white population decline rather than Black inflows. I find evidence consistent with neighborhood tipping: neighborhoods with initially moderate Black shares experienced substantial white population outflows. Finally, linking to intergenerational mobility data, I show that children from low-income families who grew up in tracts containing public housing experienced significantly lower rates of upward mobility than those in comparable control areas. These findings demonstrate that mid-century public housing, despite intentions of neighborhood revitalization, reinforced existing patterns of racial and economic segregation with lasting consequences for economic opportunity.

JEL Codes: H42, J15, J62, R23, R38
Keywords: public housing, neighborhood effects, racial segregation, urban policy, intergenerational mobility

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1 Introduction

> The Congress hereby declares that the general welfare and security of the Nation and the health and living standards of its people require housing production and community development sufficient to remedy the housing shortage, eliminate substandard housing through the clearance of slums and blighted areas, and achieve as soon as feasible the goal of a decent home and suitable living environment for every American family.
>
> — *Housing Act of 1949*

Between the 1930s and 1970s, the United States constructed approximately 1.4 million federally-funded public housing units in one of the most ambitious urban policy initiatives of the 20th century (Schwartz 2021). Designed to clear slums, address urban housing shortages, and provide affordable housing to low-income families, the American public housing program has, to many, become synonymous with policy failure. Prominent accounts in urban history and sociology have argued that the program created and entrenched racial and economic segregation (e.g., R. Rothstein 2017; Hirsch 1998; Massey and Denton 1993), and accelerated mid-century urban decline (Jackson 1985). Large, megablock projects such as Cabrini-Green, Pruitt-Igoe, and the Robert Taylor Homes became infamous, criticized for concentrating poverty, promoting crime, and destroying the urban fabric of neighborhoods (Jacobs 1961; Newman 1972). Federal policy since the 1970s has reflected this negative view through a retreat from public housing: The government has shifted funding toward market-based alternatives like Housing Choice Vouchers and the Low Income Housing Tax Credit, while simultaneously dismantling the existing stock through the HOPE VI program, which has funded the demolition or removal of roughly 30% of the original federal public housing stock since the early 1990s (Schwartz 2021).

Yet the evidence for public housing's failure is mixed. Some scholars contend that the program has been unfairly maligned by its most notorious examples and argue its poor reputation partially reflects broader social and economic challenges faced by American cities in the post-war period (Bauman 1994; Bloom, Umbach, and Vale 2015; Goetz 2013). Indeed, despite widespread criticism of the program, approximately two-thirds of public housing residents reported being "satisfied" or "very satisfied" with their housing as late as 1999 (Schwartz 2021). Some causal evidence from economists suggests that living in public housing either has neutral or positive effects on individual outcomes (Chaudhry and Eng 2024; Currie and Yelowitz 2000; Jacob 2004; Pollakowski et al. 2022), which could indicate that public housing may not have been as detrimental as some narratives suggest. At the same time, disentangling the local effects of public housing from pre-existing neighborhood conditions remains challenging. Given that projects were often sited in already-distressed areas as part of slum clearance efforts, did public housing cause neighborhood decline, or simply reflect the targeting of struggling areas? These debates have practical stakes as cities grapple with housing affordability and revisit the question of what role public housing should play.

Resolving these debates requires evidence on where public housing was built and how it shaped neighborhoods over the long run, but no comprehensive dataset linking the projects to their construction dates and precise locations. This paper addresses this research gap by constructing a new dataset on American public housing. Combining previously digitized data, public data sources, and newly digitized materials, I obtain the locations, construction dates, and characteristics, such as project size and racial composition, of over 8,000 public housing projects containing over 1 million units nationwide. I then integrate this dataset with a consistent-area panel of Census tract-level data from 1930-2010 that I build using both existing Census tract data and geolocated full count Census data from 1930 and 1940, all concorded to consistent tract boundaries. I also merge these data with three other sources: Home Owners' Loan Corporation (HOLC) "redlining" maps from the late 1930s, maps with the locations of Urban Renewal projects, and the locations of interstate highways. With these data, I examine the siting and long-run neighborhood consequences of the U.S. public housing program.

First, what neighborhood characteristics influenced public housing site selection? This is important for understanding the political economy of public housing and for informing my identification strategy for estimating the neighborhood effects of public housing construction. In particular, I am interested in whether public housing was explicitly targeted towards Black neighborhoods, as some historical accounts suggest (e.g., Hirsch 1998; R. Rothstein 2017), or whether it was more broadly targeted towards poor and working-class neighborhoods. To answer this, I estimate linear probability models to explore how pre-existing neighborhood characteristics, including demographic, socioeconomic, housing, and institutional factors, predict the placement of public housing projects. I find that public housing projects were more likely to be built in neighborhoods that were initially poorer, more populated, had higher shares of Black residents, had lower homeownership rates, and had a higher share of children, consistent with the program's slum clearance goals and the racialized politics of housing policy in the mid-20th century. I also find that neighborhoods that received public housing were more likely to be affected by urban renewal, another large mid-century urban policy that targeted predominantly Black neighborhoods (LaVoice 2024). Finally, I find that public housing projects built in poorer neighborhoods and neighborhoods with higher Black population shares tended to have higher shares of Black residents themselves, indicating that public housing reinforced rather than disrupted existing residential segregation patterns. I also show that these patterns hold in a specific instance of site selection in Philadelphia by digitizing a map of proposed-but-not-selected public housing sites from Bauman (1987): In particular, neighborhoods proposed but rejected for public housing were initially much whiter than those that ultimately received projects. These results are consistent with historical narratives and case study evidence about the politics of public housing site selection (e.g., Meyerson and Banfield 1955; Bauman 1994; Hirsch 1998).

Second, what were the short- and long-run effects of public housing construction on both the recipient and surrounding neighborhoods? To answer this question, I employ a stacked matched difference-in-differences strategy that explicitly compares each public housing neighborhood to a matched never-treated control neighborhood based on pre-treatment characteristics. For each neighborhood that received a public housing project, I use nearest-neighbor propensity score matching to identify a comparable control neighborhood in the same county based on the pre-treatment characteristics that I demonstrated were key determinants of site selection. I then treat each matched pair as a separate "sub-experiment" and stack these sub-experiments into a single analytic dataset. The specification includes matched-pair-by-year fixed effects, ensuring that each treated neighborhood is compared only to its matched control in each year, while controlling for time-varying shocks that affect both neighborhoods in the same pair. Identification stems from the fact that federal funding for public housing was limited, and local housing authorities could not build projects in every eligible neighborhood, creating variation in project placement across otherwise similar areas. Importantly, this stacked framework also avoids the econometric pitfalls that arise in staggered adoption settings when treatment effects are heterogeneous across cohorts and over time (Wing, Freedman, and Hollingsworth 2024). I extend this methodology to analyze the geographic spillovers of the projects by conducting a similar exercise for neighborhoods adjacent to public housing neighborhoods, thereby estimating the broader geographic effects of the projects.

In the public housing ("treated") neighborhoods, I find large increases in total population driven particularly by increases in the Black population, resulting in substantial increases in Black population shares. I also find large decreases in median rents, along with declines in median incomes and various other measures of economic well-being, including lower labor force participation rates and higher unemployment rates. These results, in conjunction with the site-selection results, suggest that while public housing was targeted toward poorer, minority neighborhoods, the construction of the projects further accelerated demographic changes and socioeconomic decline in these neighborhoods over the long run.

I find modest geographic spillovers to adjacent neighborhoods: median incomes declined and Black population shares increased by 2.0–2.6 percentage points, driven primarily by white population decline rather than Black population inflows.

I also explore the heterogeneity of these effects along several dimensions. First, I explore heterogeneity based on the neighborhood's initial racial composition. I find that treated and nearby neighborhoods that initially had Black population shares within a potential "tipping range" (between 1% and 12%, based on work by Card, Mas, and J. Rothstein 2008) saw outflows of white residents. In contrast, neighborhoods that were initially more Black did not. This suggests that public housing construction triggered racial tipping dynamics in some neighborhoods. I also find that the long-run effects were larger for projects built before 1960 and were driven mainly by neighborhoods not treated by urban renewal.

Finally, I explore the implications of these neighborhood effects for economic opportunity. To do so, I link my dataset to data on the upward mobility of children born to low-income families born from 1978-1983 from Opportunity Atlas (Chetty et al. 2018). I find that low-income children growing up in neighborhoods that received public housing projects experienced significantly lower upward mobility than children in matched control neighborhoods, even after controlling for neighborhood characteristics. I also find small adverse effects on upward mobility in nearby neighborhoods, but these effects are fully accounted for by controlling for neighborhood income and demographics in 1980, with no additional effects from proximity to public housing itself. This pattern suggests that these modest spillovers occurred through earlier neighborhood changes, rather than through persistent or independent effects of public housing on later mobility outcomes in surrounding neighborhoods.

This paper relates to several literatures in urban and public economics and economic history. First, it contributes to work on the local effects of affordable housing programs (Baum-Snow and Marion 2009; Diamond and McQuade 2019) and revitalization policies (Collins and Shester 2013; LaVoice 2024; Rossi-Hansberg, Sarte, and Owens 2010) by estimating the neighborhood effects of one of the largest urban policies in American history. More directly, recent literature has studied the effects of HOPE VI public housing demolitions on neighborhood (Tach and Emory 2017; Blanco and Neri 2025; Aliprantis and Hartley 2015; Sandler 2017; Almagro, Chyn, and Stuart 2023) and individual (Haltiwanger et al. 2024; Chyn 2018) outcomes. This literature has focused on the demolitions of the most distressed public housing projects, and much of it has been focused on a single city (Chicago). There are several reasons to think these estimates of the effects of demolitions may not generalize to those of public housing construction. First, the projects demolished through the HOPE VI program are a selected set of the most distressed projects. They are thus not necessarily representative of the public housing program as a whole. Second, the effects of project construction likely depended on existing neighborhood conditions, which likely varied in ways that they did not for demolitions. New projects may have served as local amenities in some neighborhoods but disamenities in others. Finally, the effects in Chicago might not necessarily generalize to other cities.

My findings also complement earlier work from other social sciences on public housing and the concentration of poverty in central cities, which generally used data compiled from one or a small number of cities (e.g. Carter, Schill, and Wachter 1998; Massey and Kanaiaupuni 1993). Relative to this older literature, I use a dataset covering a much broader set of cities, projects, and time periods, and apply modern econometric techniques to estimate the causal effects of public housing construction on neighborhoods.

My paper adds to the small literature estimating the effects of public housing construction. Shester (2013) uses digitized data from HUD that contain all federally funded public housing projects built until 1973, with locations at only the locality level (discussed in Section 4) to study the effect of public housing construction on a variety of county- and city-level outcomes. This paper finds that cities in the same state with more public housing construction experienced decreases in median property values, median family income, and population density. Shester, Allen, and Handy (2019) uses the same dataset to study the effects of public housing construction on the rise in single motherhood at the MSA level. I build on this work by merging precise locations to these data, allowing me to study neighborhood effects.

The most closely related contribution is that of Guennewig-Moenert (2025), who studies the effects of public housing construction in New York City, specifically on neighborhood racial composition and rents, along with welfare estimates based on a structural model. I compile data on public housing projects across the country, which allows me to characterize the effects of public housing across the United States and speak to broader historical debates about the program. Scholars have argued that the New York City Housing Authority (NYCHA) was in many ways exceptional in its effectiveness at dealing with many of the factors that have plagued public housing authorities in the U.S. (Bloom 2008).[1] My empirical approach also differs: he uses variation in proximity to public housing to define the control group, while I use similarity in pre-treatment characteristics. Another related contribution is contemporaneous work by Harris (2025), who studies the short-run neighborhood effects in the first two decades of the program and finds that public housing initially reduced poverty concentration in recipient neighborhoods.

Finally, my paper contributes to the economic literature on the emergence and consequences of segregation in the United States. While prior research has documented migration responses

[1]For example, NYCHA was a relatively small participant in HOPE VI and did not demolish any of its high-rise projects through the program (Schwartz 2021).

to neighborhood demographic change (Shertzer and Walsh 2019; Boustan 2010), my paper addresses one way in which federal housing policy directly influenced neighborhood sorting. More broadly, this paper contributes to the literature and debates around what role public policy has played in shaping racial and economic segregation in American cities (R. Rothstein 2017; Trounstine 2018; Boustan 2013; T. D. Logan and Parman 2025).

The paper proceeds as follows. Section 2 describes the history of the public housing program. Section 3 outlines the potential channels through which public housing may affect neighborhoods. Section 4 describes data sources and construction. Section 5 analyzes the determinants of public housing site selection. Section 6 presents the empirical strategy and estimates of neighborhood effects. Section 7 explores heterogeneity in these effects and examines potential mechanisms. Section 8 links public housing to long-run economic opportunity using data from the Opportunity Atlas. Section 9 concludes.

2 Historical Background: The U.S. Public Housing Program

Public housing in the United States began during the Great Depression as part of New Deal efforts to reduce urban housing shortages and stimulate the economy. The Public Works Administration built the first projects, producing about 20,000 units before court challenges over land acquisition limited its scope (Jackson 1985).

The program was greatly expanded and decentralized with the passage of the Housing Act of 1937, which promoted the creation of local Public Housing Authorities (PHAs) to build and oversee public housing with federal funds. Specifically, federal grants covered the difference between operating costs and tenant revenue, while PHAs were responsible for site selection and project management. The 1937 Act's stated purpose was slum clearance and redevelopment rather than expanding housing supply, which reflected contemporary beliefs that "slum" neighborhoods reduced nearby property values and caused poor health and behavior among residents (Meyerson and Banfield 1955). Approximately 170,000 housing units were constructed under this act, with nearly 90% built on former slum sites (Schwartz 2021).

The program was greatly expanded as part of a broad urban policy agenda following World War II. Many Northern cities experienced severe housing shortages, which were especially acute for African Americans, while white flight, suburbanization, and urban "blight" challenged cities. These challenges generated a large political coalition for urban redevelopment policy, culminating in the Housing Act of 1949 (von Hoffman 2000). Title III of the 1949 Act provided funding for an additional 810,000 units of public housing and marked a major shift in the program's scope and goals, expanding public housing as a tool for slum clearance, a source of low-income housing, and a potential destination for those displaced by the Urban Renewal slum clearance program, which was funded by Title I of the Act.

The ambitious goals of the 1949 Act, however, quickly met political, fiscal, and social obstacles. Several developments in the 1950s and 1960s transformed public housing into a deeply controversial policy.

First, as detailed in Section 2.1, racial dynamics in site selection caused intense political conflicts. Second, the composition of public housing tenants changed significantly over time. Early projects primarily housed working families and the "deserving poor," but by the 1960s, they increasingly housed the poorest and most disadvantaged households. The rise of suburban homeownership allowed working-class families to leave public housing, while authorities tightened income eligibility limits. Federal regulations also mandated that PHAs prioritize the neediest applicants, resulting in the concentration of poverty within projects (Schwartz 2021). Third, design and construction issues became more evident. To avoid competing with private housing and to meet mandated construction cost limits, projects were intentionally made austere and inexpensive, making them susceptible to rapid deterioration (Schwartz 2021). Many public housing authorities struggled to fund maintenance as middle-class tenants exited the projects. Furthermore, the 1969 Brooke Amendment capped tenant rents at 25% of income, further limiting PHA revenues (Hunt 2018). These challenges are illustrated by well-known public housing failures, such as St. Louis's Pruitt-Igoe Homes, built in 1954. Initially praised as a model development, Pruitt-Igoe quickly faced issues like crime, vandalism, and abandonment as maintenance funding dried up and middle-class residents moved out (Bristol 1991). As a result of these issues, the complex was ultimately demolished beginning in 1972.

In response to mounting criticism, American housing policy has shifted away from direct public housing provision. A 1971 report by the Nixon Administration wrote that "drab, monolithic housing projects, largely segregated...still stand in our cities as prisons of the poor" (Orlebeke 2000). In 1973, President Nixon declared a moratorium on subsidies for traditional public housing. By the early 1990s, the public housing stock faced serious challenges: physical deterioration due to deferred maintenance, extreme concentration of poverty as working families moved out, and, in some cases, rampant crime and gang activity. The federal response to these challenges was the HOPE VI program, launched in 1992, which provided federal funding to demolish distressed projects or transform them into mixed-income developments. The Faircloth Amendment of 1998 further limited public housing growth by capping the total number of units PHAs could operate at their 1999 levels.

2.1 Race, Segregation, and Public Housing

Race and segregation emerged as central and contentious issues throughout the program's history. From the outset, the Public Works Administration followed a so-called "neighborhood composition rule," formally segregating projects based on the demographics of the neighborhoods where they were built. While ostensibly a neutral policy intended to maintain neighborhood "stability", in practice the rule solidified segregation. Notable accounts, such as R. Rothstein (2017), contend that public housing not only reinforced existing segregation but also actively created it by constructing segregated projects in previously integrated neighborhoods. A well-known example is Atlanta's Techwood Homes, built in 1936 as an all-white development, which displaced a previously integrated low-income neighborhood (R. Rothstein 2017). Many large public housing authorities continued to follow an explicit neighborhood-composition rule well into the 1960s.[2] However, even after these legal changes, projects remained highly segregated in practice (Bickford and Massey 1991).

Historical case studies show that issues related to race and segregation also influenced site selection decisions. Additionally, contemporaneous debates among policymakers reveal that they were fully aware of the program's potential to either worsen or improve residential segregation patterns (Hirsch 2000). In Chicago, the Housing Authority initially planned to build projects throughout the city, but strong opposition from white neighborhoods led to the concentration of projects in predominantly Black areas on the South and West Sides (Meyerson and Banfield 1955; Hirsch 1998). Similar conflicts occurred in other major cities, with white residents and politicians successfully blocking projects in their neighborhoods. I examine a specific example of this dynamic in Philadelphia, comparing proposed-but-not-selected areas to public housing locations in Section 5.2. Building on this historical evidence, Section 5.1 tests whether these racial site selection dynamics held systematically nationwide.

2.2 Motivating Example: Hunters Point in San Francisco

The history of Bayview–Hunters Point in San Francisco provides a clear example of how public housing construction can change a neighborhood's long-run trajectory. Before World War II, the area was a racially mixed neighborhood on the city's periphery. During the war, the federal government built thousands of temporary housing units there to accommodate defense workers, causing an initial influx of African American families. After the war, instead of

[2] Although legal challenges began in the mid-1950s, including the U.S. Supreme Court's 1954 decision to let stand a California ruling striking down San Francisco's segregated projects, federal law did not formally prohibit race-based segregation in public housing until the Fair Housing Act of 1968.

dismantling these temporary units, local and federal officials converted many of them into permanent public housing. Throughout the 1950s and 1960s, the San Francisco Housing Authority replaced and expanded these units into large-scale projects and directed Black families into these developments. This expansion coincided with a quick demographic shift: as the projects grew, the neighborhood ceased to be a mixed-income area; white and Asian households left, and the community became increasingly Black and segregated. Figure 1 illustrates the racial transition of these neighborhoods along with the location of public housing projects, showing the sharp rise in the Black population share and the decline in the white population from 1940 to 1970. The rest of this paper tests whether the patterns seen in Hunters Point were systematic across the country.

3 Potential Effects of Public Housing

In this section, I outline the main channels through which the construction of public housing projects might affect neighborhood outcomes.

First, public housing construction may mechanically alter the racial and socioeconomic composition of neighborhoods. Tenant selection policies, income limits, and subsidized rents determined who occupied these projects, thereby altering the demographic makeup of the areas where they were built.

Second, these compositional changes could trigger endogenous sorting by private households. If individuals have preferences for their neighbors' race or income, the arrival of public housing residents might prompt some existing residents to relocate. Historical work, such as Jackson (1985), argues that part of mid-century "white flight" from central cities was a response to the public housing program. However, empirical evidence on this point remains limited.

Third, the physical structure of public housing projects could create built-environment externalities. Replacing substandard or vacant structures with new housing could improve neighborhood quality and increase neighborhood desirability, as shown for private housing upgrades by Rossi-Hansberg, Sarte, and Owens (2010) and for subsidized housing by Ellen et al. (2007). Conversely, large superblocks or architecturally incongruous high-rise towers might be viewed as local disamenities, and indeed, criticism of the mid-century public housing program often focused on the design of the projects (Jacobs 1961; Newman 1997).

Fourth, public housing construction may generate broader social and market externalities. A frequent concern is that concentrated poverty within large projects generated social disorder and elevated crime, with potential spillovers into surrounding neighborhoods (Sandler 2017).

Conversely, visible public investment could signal neighborhood revitalization and spur private reinvestment, improving local conditions (Rossi-Hansberg, Sarte, and Owens 2010). The direction of these effects may vary across contexts. Modern evidence on the LIHTC program suggests that subsidized housing investments may be a positive amenity in distressed areas but may be perceived negatively in more affluent ones (Diamond and McQuade 2019).

I explore these mechanisms empirically in Section 7.

4 Data

4.1 Public Housing Data

A significant challenge in studying the history of public housing is the lack of a comprehensive dataset that includes project lists, construction dates, and precise project locations. Consequently, previous research on the neighborhood effects of public housing has been limited in scope or restricted to a small number of cities where researchers could obtain this information directly from housing authorities. The data limitations have also hindered broader historical analyses of the public housing program, and have been acknowledged by previous work (Ellen et al. 2007; Hunt 2018). My paper addresses this gap by constructing what I believe to be the most complete dataset of mid-century public housing by combining information from a series of administrative datasets, along with previously digitized and newly digitized sources.

The first source I use is the *Consolidated Development Directory* (CDD), published by the US Department of Housing and Urban Development (HUD) in 1973 and digitized by Shester (2013). This data contains the universe of federally funded public housing projects that existed in 1973, along with various project characteristics (e.g., number of units) and, crucially, the year each project was completed. However, the CDD does not contain location information for the projects. Previous work using these data (Shester 2013; Shester, Allen, and Handy 2019) has therefore been limited to studying aggregate city- and county-level outcomes. These data also contain three project number fields, which I combine with the state code to form federal project codes in the standard HUD format (e.g., IL-2-38-1 for a Chicago project). These project codes allow me to link the CDD data to other project-level datasets such as the Picture of Subsidized Households and HUD Form 951. As far as I know, this element of these data has not been previously exploited.

To obtain location information for the projects, I turn to a series of publicly available HUD administrative datasets. The primary of these is the Picture of Subsidized Households (PSH) datasets, which contain a list of federally funded housing projects, various project characteristics (e.g., number of units, demographics), and, from 1997 onward, their locations. I use the PSH datasets from 2000 and 1997, due to limitations in each. I also link to more recent National Data Geospatial Asset (2023) data. Finally, I link to the HUD File 951 dataset, which lists street addresses, latitudes, and longitudes for the stock of multifamily assisted housing projects that existed between 1986 and 1995 (Kucheva 2013). These datasets largely overlap in coverage, but each contains projects missing from the others. This linkage allows me to assign locations to over 90% of the projects in the CDD. I will refer to this as the geocoded CDD dataset.

I supplemented these data with hand-collected information from historical annual reports of local public housing agencies, obtained from various libraries (or directly from the housing authority in the case of San Francisco), and from FOIA requests to public housing agencies. From this effort, I was able to obtain supplementary data from eight major cities: New York, Chicago, Boston, Los Angeles, Washington, DC, San Francisco, Atlanta, and Baltimore.[3] For these cities, I collected the complete set of projects built up to 1973, including their construction dates and locations. I was able to geolocate these projects using the Google Maps API. For projects that I was unable to geolocate successfully, I hand-identified locations based on street address and name.

There were two motivations for collecting these additional data. First, it allowed for a better understanding of the projects in the CDD for which I was unable to assign locations. Public housing demolitions, such as those from the HOPE VI program that began in 1993, may have caused some CDD projects to be absent from later administrative datasets used for geocoding, preventing me from assigning locations to those projects in the merged dataset. Indeed, the incomplete matching between CDD and PSH data suggests that some projects may have ceased to exist when PSH data collection began. Moreover, these missing projects might not be randomly distributed, as demolitions targeted particularly blighted projects.

Second, the CDD-HUD data include only federally funded public housing projects. Little data exist on city- and state-funded public housing projects, but some cities may have funded some public housing construction outside of the federal program. New York City, in particular, has a notable city- and state-funded public housing program, and using only the data on federal projects in New York City misses a substantial number of housing projects and units.[4] The other cities for which I collected data had either none or very few city or state-funded projects. Ultimately, I use my hand-collected data for housing projects in New York City, Chicago, and

[3]I collected digitized reports from other cities, such as Cleveland, Cincinnati, and St. Louis, but not all contained the precise project locations information.

[4]My digitized dataset of NYC housing projects contains 129,430 housing units, compared to 80,000 in the PSH-CDD data.

San Francisco, supplement the geocoded CDD for Boston and Washington, DC, and rely on the geocoded CDD data for all other cities (Los Angeles, Atlanta, and Baltimore).[5] I found little evidence in the historical record suggesting that, by otherwise relying on data on federal projects, I am missing a notable stock of public housing in other cities.

The last step in constructing the public housing dataset is to obtain information on the populations and racial composition of the projects. These data allow me to distinguish between public housing residents and the rest of the neighborhood's population. For most cities, I obtain this information from the 1977 Picture of Subsidized Households dataset, which was cleaned and shared with me by Yana Kucheva (Kucheva 2013). These data report the number of subsidized households by race in each project, but do not include direct population counts. In Appendix A, I describe how I convert these household counts to population counts. I match these data to the geocoded CDD dataset using federal project codes. For New York City, I used population-by-race data from the early 1970s digitized and shared with me by Max Guennewig-Moenert (Guennewig-Moenert 2025). And for Chicago, I obtained population-by-race data from the 1973 digitized Annual Report of the Chicago Housing Authority. While it would be ideal to measure population by race at the time of construction, these data are not available in most cities. Thus, I assume the population and racial composition of each project in each decade after construction is equal to the 1970s values. I use these population estimates to estimate the private (non-public-housing) population in each decade by subtracting the estimated public-housing population from the total Census-reported population.

The result of this process is a dataset containing the construction dates, coordinates, and characteristics of over 8,000 projects and 1 million units of public housing built from 1935 until 1973.

4.2 Neighborhood Data

To study the neighborhood effects of public housing construction, I construct a panel dataset of neighborhood-level characteristics spanning 1930 to 2010. Building such a dataset presents several challenges. First, the number of cities with tract-level data is limited in 1940 and especially in 1930. Second, Census tract boundaries change over time, requiring the construction of a consistent panel. And third, income data was not collected in the 1930 Census, and median income was not reported in the tract-level Census tables in 1940.

I proceed as follows. First, I collect a variety of outcomes at the Census tract level from

[5] For Los Angeles, Atlanta, and Baltimore, I find no additional projects in my digitized data versus the geocoded CDD data.

the 1930 to 2010 decennial censuses. These data include tract-level population counts by race (white, Black, and other), age structure (share under 18 and share 65 and older), several socioeconomic measures (median income, high school graduation rates, labor force participation, and unemployment rates), and housing market measures (median rents, home values, and homeownership rates). All census data and shapefiles were acquired from IPUMS NHGIS (Manson et al. 2022).

I construct a consistent panel of census tracts by concording all tracts to 1950 Census tract boundaries using an area-reweighting approach. I describe the tract harmonization in more detail in Appendix B.2. I chose 1950 as the base year for two reasons. First, to limit concerns about results being driven by public housing-driven changes in tract boundaries, I chose a year early in my analysis period, rather than at the end, as much of the literature does. Second, publicly available 1940 New York City census tract shapefiles do not correspond to actual census tracts, but to much larger health districts, making 1940 a less suitable base year. Given that New York City is a major city in my sample, I chose 1950 as the base year for the entire sample.

I supplement this tract-level data with data from the 1930 and 1940 full-count Census (Ruggles et al. 2022). To convert the full count data to the tract level, I first aggregate the individual-level data to the enumeration district (ED) level using the 1930 and 1940 enumeration district shapefiles from the Urban Transitions Project (J. R. Logan et al. 2024). I then use area reweighting to aggregate these data to 1950 Census tracts. Incorporating these full-count data serves two purposes. First, they allow me to expand my set of cities and neighborhoods for which tract-level data was unavailable in 1930 and 1940. Second, they allow me to include income information for 1930 and 1940, which is not available in the tract-level tables for those years. I proxy for income in 1940 and 1930 using full-count Census microdata. For 1940, I sum individual wage income (INCWAGE) across all household members to calculate total household wage income, then classify households into income bins within each enumeration district. For 1930, I use machine-learning-adjusted occupation scores from Saavedra and Twinam (2020) aggregated to the household level and similarly binned. In both years, I use income bins that match the 1950 census income categories. I concord these enumeration district-level data to 1950 Census tracts using area-reweighting and calculate tract-level medians from the binned distributions. The area-reweighting and median calculation procedures are described in Appendix B.2.

I convert all monetary values to 2000-dollar values using the US CPI from Officer and Williamson (2025). For population counts, monetary variables, and distance measures, I apply the inverse hyperbolic sine (asinh) transformation in much of the analysis. The asinh transformation approximates the natural logarithm for large values while being defined at zero. This transformation compresses the potentially skewed distributions of these variables and allows regression coefficients to be interpreted approximately as percentage changes without requiring me to drop tracts with zero population in some racial groups (Bellemare and Wichman 2020).[6] When reporting percentage changes from asinh-transformed outcomes in the text, I convert regression coefficients using the hyperbolic sine function, $\sinh(\beta) \times 100$.

In addition to Census data, I incorporate several historical datasets that capture other urban policies that may be related to public housing placement and subsequent neighborhood change. First, I incorporate digitized data and shapefiles of the Home Owners' Loan Corporation (HOLC) "redlining" maps drawn in the late 1930s, made available by *Mapping Inequality* (Nelson and Winling 2023). These maps graded neighborhoods from "A" (best) to "D" (hazardous) to indicate perceived mortgage risk. Although many scholars have argued that these maps institutionalized redlining and curtailed investment in minority neighborhoods (Aaronson, Hartley, and Mazumder 2021), recent evidence complicates this view. Fishback et al. (2024) shows that the maps largely reflected existing patterns of racial and socioeconomic segregation rather than shaping new lending decisions. For my purposes, these maps serve as a historical snapshot of neighborhood conditions and perceived credit risk in the late 1930s and thus serve as a proxy for pre-existing discrimination, rather than a direct driver of exclusion. I overlay these maps onto the 1950 Census tract boundaries. Following Weiwu (2025), I classify a neighborhood as "redlined" if at least 80% of the area is designated as "hazardous" (HOLC grade D).

Second, I also incorporate data on the locations of U.S. urban renewal projects (1955-1966) from *Renewing Inequality* (Nelson and Ayers 2025). Urban renewal, established under Title I of the 1949 Housing Act, provided federal subsidies for cities to acquire and clear "blighted" neighborhoods. Urban renewal was closely intertwined with public housing: clearance projects often created sites for new developments or displaced families that public housing was intended to rehouse (von Hoffman 2000). Recent research finds that Black neighborhoods were two to three times more likely to receive urban renewal projects than white neighborhoods, and that urban renewal areas experienced declines in housing and population density alongside rising rents and incomes (LaVoice 2024). To account for potentially confounding effects of urban renewal and to explore the relationship between the two programs, I overlay these maps onto the 1950 Census tract boundaries and classify a tract as an "urban renewal" tract if more than 5% of its area overlaps with an urban renewal project.

[6] I demonstrate that results are qualitatively similar when using outcome variables in levels in Section 6.4.

Finally, I incorporate data on the locations of interstate highways as of 1996 from Weiwu (2025). Interstate highway construction began in the late 1950s, and has been accused of disproportionately displacing minority communities (Rose and Mohl 2012). These data allow me to control for potential confounding effects of highway construction on neighborhood change, as well as to explore the relationship between highway locations and public housing placement.

4.3 Defining Treatment and Spillover Neighborhoods

To study the neighborhood effects of public housing construction, I define a set of *treated* tracts that received public housing projects and a set of *nearby* tracts that may have experienced spillover effects from nearby projects. To ensure that a project represents a meaningful neighborhood intervention, I include only projects with at least 50 housing units.

Because the public housing projects are geocoded at the coordinate level, I use a geographic buffer approach to define treated tracts. Specifically, a tract is defined as treated if any portion of it intersects a 100-meter buffer around a public housing project. This definition captures cases in which a project straddles multiple tracts; when that occurs, I allocate its units and population evenly across the affected tracts.

Since the Census data are decennial, I assign treatment timing by decade. Following the literature, I use the project's completion date as the treatment date (Asquith, Mast, and Reed 2023). A tract is considered treated in year t if its first qualifying project was completed between $t-9$ and t (e.g., tracts receiving projects between 1951 and 1960 are treated in 1960). This timing convention ensures that the treatment occurs after the pre-treatment observation but before the post-treatment observation in the panel. [7] For tracts receiving multiple public housing projects over time, treatment timing is determined by the first project meeting the size thresholds. Subsequent projects in the same tract are not considered separate treatment events, since the tract has already been "treated" by public housing construction.

To examine spatial spillovers, I define a nearby tract as one that shares a border with a treated tract, is not itself treated, and lies within one kilometer of the nearest public housing project. This hybrid contiguity-and-distance definition identifies neighborhoods that are close enough to plausibly experience externalities from nearby developments while excluding tracts that are technically adjacent to project neighborhoods but geographically distant from a project.

[7] If a project was completed in the Census year but after the Census enumeration date, the effects of the project would meaningfully be captured in the second, rather than first, post-treatment decade, and the $t = 0$ effects would be understated.

I adopt a one-kilometer threshold based on prior literature showing the geographic extent of spatial spillovers from public housing interventions (Blanco and Neri 2025). Nearby tracts inherit the earliest treatment year among their treated neighbors. Figure 2 illustrates the treated and nearby classification for Chicago, showing treated tracts and their adjacent spillover areas.

4.4 Sample Selection

My empirical analysis focuses on projects built between 1941 and 1973, with neighborhood outcomes measured from 1930 to 2010. I restrict the sample in several ways to ensure a balanced panel of neighborhoods with sufficient pre-treatment data to conduct my difference-in-differences analysis.

Table 1 shows the impact of each filtering step on the sample composition. Starting from the original sample of 12,062 census tracts representing 38.1 million people in 1940 (approximately 29% of the U.S. population), I apply the following restrictions. First, I require that tracts exist in all years from 1930 to 2010, which drops about 25% of tracts. Second, I drop tracts for which I cannot calculate population by race, median income, median rent, labor force participation rates, and unemployment rates for all years, which removes an additional 5% of tracts. Third, I exclude tracts with public housing built before 1941 to preserve 1930 and 1940 as clean pre-treatment periods, which removes 109 treated tracts containing 109 projects. Fourth, I drop metropolitan areas with fewer than 30 total tracts to ensure sufficient within-city variation. This excludes three small metropolitan areas. Fifth, I exclude population outliers (tracts with populations below the 5th percentile or above the 98th percentile in any year) and geographic outliers (tracts exceeding 10 km^2, which are typically peripheral suburban tracts). Finally, I exclude tracts that contain a modern public housing project in the HUD administrative data but for which I am unable to link to a construction date in the CDD. This prevents tracts that were at some point treated from being included in the control group.

The resulting balanced sample includes 6,080 census tracts across 47 CBSAs, representing 25.3 million people in 1940 (approximately 19% of the U.S. population). This sample contains 781 public housing projects with 295,458 total units located in 816 treated tracts. Figure 3 shows the geographic distribution of CBSAs included in the analysis. These sample restrictions have implications for external validity: Due to data limitations, the analysis focuses on neighborhoods in major metropolitan areas that had established census tracts by 1930, excluding public housing that was built in rural areas, smaller cities, or neighborhoods that were not populated in the beginning of the period. By excluding projects with fewer than 50 units, my analysis also does not speak to the effects of smaller-scale or scattered-site public housing.

5 Site Selection

In this section, I study the placement of the public housing projects in my sample. These dynamics matter for several reasons. First, the site-selection process for mid-century public housing was a significant criticism of the program. Historical case studies in several cities suggest that projects were often targeted to poorer and minority neighborhoods, both because of the program's slum clearance goals and because of local backlash in white neighborhoods against construction of projects (Meyerson and Banfield 1955; Bauman 1994; Sugrue 2005). Second, these site selection dynamics have had important legal implications: Lawsuits in several cities have alleged that public housing site selection was discriminatory and in violation of Civil Rights Law, most famously Gautreaux v. Chicago Housing Authority (1966), but also in cities like Dallas (Walker v. HUD 1985) and Baltimore (Thompson v. HUD 2005). Still, there has been little systematic evidence on the nationwide patterns of public housing site selection. Finally, understanding these site-selection dynamics is important for estimating the neighborhood effects of public housing and interpreting those effects.

To begin, in Section 5.1, I estimate which pre-existing neighborhood characteristics predict the eventual locations of public housing projects. I also test whether the placement of public housing projects was related to the locations of other mid-century urban policies, particularly urban renewal and the interstate highway system. Then, in Section 5.2, I zoom in on a particular case study of site selection by examining the locations of proposed-but-not-built public housing in Philadelphia in Section 5.2. This example illustrates the political dynamics around site selection decisions.

Finally, in Section 5.3, I examine whether the racial composition of the projects themselves varied systematically with the initial characteristics of the neighborhoods in which they were built.

5.1 Where were the projects built?

Table 2 shows the raw comparison of 1940 baseline characteristics between census tracts that eventually received a public housing project and those that did not. Treated neighborhoods had substantially higher Black population shares, lower median incomes and rents, higher unemployment rates, a higher share of homes needing major repairs, and were located closer to a central business district.

To formally test which neighborhood characteristics predict public housing placement, I estimate linear probability models relating pre-treatment tract characteristics to the likelihood of ever receiving a public housing project. Specifically, I estimate:

$$\text{Treated}_{ic} = \gamma_c + \boldsymbol{X}'_{i,1940}\boldsymbol{\beta} + \boldsymbol{Z}'_i\boldsymbol{\delta} + \epsilon_{ic} \tag{1}$$

where Treated_{ic} is an indicator for whether tract i in county c received any public housing project between 1941 and 1973. The vector $\boldsymbol{X}_{i,1940}$ contains 1940 neighborhood characteristics, and $\boldsymbol{Z}_i$ includes distance to the nearest interstate highway and an indicator for whether a tract was treated by urban renewal. I include county fixed effects γ_c to absorb systematic differences in placement across local public housing authorities.

Columns 1-3 of Table 3 estimate the model without the $\boldsymbol{Z}_i$ controls, while Column 4 adds them to examine whether the relationship between 1940 characteristics and public housing placement is robust to controlling for subsequent federal interventions. I adjust standard errors for spatial correlation following Conley (1999), allowing for correlation of residuals across census tracts within a 2-kilometer radius.

I chose variables based on historical narratives of public housing site selection and the program's stated intentions. First, as described in Section 2, the public housing program was largely intended as a "slum clearance" program, so we might expect that neighborhoods with lower socioeconomic status and worse housing market conditions would be more likely to receive public housing. Second, historical case studies in multiple cities have documented the key role of race in determining where public housing was built (e.g. Hirsch 1998; Bauman 1994), so I include the Black population share as a key predictor of public housing placement. I also include the HOLC redlining designation to capture long-term patterns of racial segregation, discrimination, and disinvestment.

The results from these linear probability models are shown in Columns 1-4 in Table 3. Column 1 shows a parsimonious model with several key neighborhood characteristics, while Column 2 shows a more saturated model including all neighborhood characteristics, including homeownership rates and age composition. Column 3 adds the share of housing units deemed in need of major repairs in 1940, which is missing for about 5% of tracts. Consistent with the historical narrative, I find that public housing projects tended to be targeted towards poorer, minority neighborhoods: Census tracts that were initially more populated, had higher Black population shares, lower median incomes and labor force participation rates, and had higher unemployment rates were more likely to receive public housing projects during this period. Neighborhoods with lower homeownership rates and a higher share of children were also more likely to receive public housing. Public housing was also more likely to be built in neighborhoods that were redlined, had lower rents, and had a higher share of homes needing repairs, although these estimates do not reach statistical significance in the fully saturated models. These findings reflect the slum clearance motivation of the public housing program and are also consistent with historical accounts emphasizing the racial targeting of public housing siting: Even after controlling for local economic, housing market, and demographic conditions, the Black population share remains a strong and significant predictor of public housing placement.

Column 4 additionally tests whether neighborhoods that ultimately received public housing were more likely to be affected by two other transformative mid-century urban policies: urban renewal and the interstate highway system. Evidence and historical narrative suggest that these programs may have affected many of the neighborhoods that were also targeted for public housing. The Urban Renewal program, in particular, was directly related to public housing in many cities, as public housing projects were used to house individuals displaced by urban renewal (Bauman 1994; Hirsch 1998). Neighborhoods that received public housing were also much more likely to be affected by urban renewal, whereas proximity to an interstate highway was not significantly related to public housing placement. This result further confirms the interplay between these two programs and motivates accounting for urban renewal in my empirical strategy in Section 6.[8]

Quantitatively, these effects are sizable. The baseline probability that a census tract in the sample received a public housing project between 1941 and 1973 is 13.4%. Based on the estimates in Column (4), a one standard deviation increase in Black population share increased public housing selection probability by 3.5 percentage points (26% increase), a one standard deviation increase in unemployment rate increased the probability by 3.6 percentage points (26.8% increase), and a one standard deviation decrease in median income increased the probability by 1.7 percentage points (12.7% increase). Finally, neighborhoods that were eventually targeted by urban renewal were 10.4 percentage points (77%) more likely to receive public housing.

5.2 A Philadelphia Case Study

Historical accounts of public housing site selection have highlighted the political battles that surrounded the placement of projects in particular cities, for example, in Chicago (Meyerson and Banfield 1955), Philadelphia (Bauman 1994), and Detroit (Sugrue 2005). These accounts have emphasized the conflict between public housing authorities and white working-class neighborhoods that resisted the construction of public housing projects in their communities.

[8]These results are consistent with similar exercises in Harris (2025) and Massey and Kanaiaupuni (1993).

Here, I present a case study of site selection in Philadelphia, drawing on the historical treatment and maps from Bauman (1987). This case study is illustrative of the political dynamics that shaped site selection decisions in many cities (Hunt 2005).

Following the 1954 Housing Act, the Philadelphia Housing Authority proposed 21 public housing projects across the city. Upon announcement, many of these proposed projects faced significant backlash from local white communities, leading to their eventual cancellation. I identified these proposed Philadelphia public housing locations by scanning and georeferencing a historical map from Bauman (1987), shown in Figure 4. I georeferenced the historical map in QGIS, aligning it with a modern basemap using identifiable landmarks, such as major roads and rivers. Then, I located the proposed public housing sites on the georeferenced map and matched these points to 1950 census tracts. In total, I identify 12 neighborhoods with proposed-but-not-built public housing sites in Philadelphia.

I then compare the baseline characteristics of proposed-but-not-built project locations to those of actual public housing locations in Philadelphia. Table 4 presents the balance table, which shows that the initial characteristics of the proposed locations were very different from those of actual public housing locations. Most notably, these proposed-but-not-built project locations had very low initial Black population shares. They were also further from the central business district, less populated, and had lower unemployment rates. These differences illustrate the dynamics of site selection in one particular city, where whiter neighborhoods often resisted public housing construction. They also show that an identification strategy based on proposed-but-not-built locations would likely be invalid, as these locations likely do not represent a good counterfactual for actual public housing locations.

5.3 Did project demographics vary with neighborhood characteristics?

I now examine how the racial composition of the public housing projects varied with the initial neighborhood characteristics in which they were built. The key question is whether projects reinforced existing patterns of residential segregation or disrupted them. I focus on the share of Black residents in each project, using data from treated census tracts in my balanced sample for which project-level racial composition is available from the 1970s. Ideally, I would observe the racial composition of the projects at the time of construction, but I do not have this information. Still, to the extent that the racial composition of the projects was relatively stable over time, the 1970s data should provide a reasonable proxy for the initial demographics of the projects.

Formally, I regress the project Black share in each tract i in county c on the neighborhood characteristics measured in the decade before construction:

$$\text{Project Black Share}_{ic} = \gamma_c + \boldsymbol{X}'_{i,t-1}\boldsymbol{\beta} + \epsilon_{ic} \tag{2}$$

The unit of analysis is the tract, with one observation per treated tract. For the 75% of treated tracts containing a single public housing project, Project Black Share$_{ic}$ is simply the Black share in that project. For tracts with multiple projects, I define t as the construction decade of the first project to open in the tract, and Project Black Share$_{ic}$ is the population-weighted average Black share across all projects that opened in that first decade. Whereas the regressions in Section 5.1 used 1940 covariates (or later urban policy exposure) for all projects, here I match each tract to neighborhood characteristics from the decade immediately preceding its construction to better capture local conditions at the time of development. The results are shown in Table 5. Column (1) presents a parsimonious specification with only baseline Black share and median income, column (2) includes a fuller set of neighborhood characteristics, including rent, population, unemployment, distance to the CBD, and redlining status, while column (3) adds county fixed effects. I again include the distance from an interstate highway and urban renewal designation.

The core result is robust across specifications: public housing projects largely matched the racial composition of their surrounding neighborhoods, reinforcing rather than disrupting existing residential segregation patterns. Based on the coefficient in column (3), a 10 percentage point increase in baseline neighborhood Black share predicts a 2.3 percentage point increase in project Black share. Furthermore, projects built in poorer neighborhoods, even conditional on race, tended to be more heavily Black, as indicated by the negative coefficient on median income and positive coefficient on unemployment rate.

6 The Effect of Public Housing Construction on Neighborhoods

Having established in the previous section that public housing projects were systematically targeted towards poorer, minority neighborhoods, I now turn to estimating the effects of the projects on neighborhood change in subsequent decades.

6.1 Research Design

I employ a stacked matched difference-in-differences approach to estimate the causal effects of public housing on neighborhood outcomes in both the treated and nearby neighborhoods. The approach matches each treated or nearby tract to an observationally similar never-treated neighborhood, then compares changes within these matched pairs over time while avoiding contaminated comparisons across different treatment cohorts and controlling for time-varying shocks that affect both neighborhoods in each matched pair. This design addresses two key challenges: selection bias from non-random project placement and econometric issues from staggered treatment timing. The implementation proceeds in four steps:

Step 1: Defining the Donor Pool. I restrict potential controls to census tracts that never received public housing and are never nearby tracts, as defined in Section 4. This allows me to separately estimate direct effects and geographic spillovers, while ensuring clean controls and reducing concerns about SUTVA violations.

Step 2: Tract-Specific Propensity Score Matching. For each treatment year cohort, I match both directly treated and nearby tracts to controls from the donor pool. For example, for the 1950 treatment cohort, both the directly treated tracts and their nearby tracts are matched based on their 1930 and 1940 characteristics. This ensures control neighborhoods followed similar pre-treatment trajectories over the relevant decades. Within each cohort, I implement nearest-neighbor matching with replacement using propensity scores estimated on the key predictors of site selection from Section 5, measured in each of the two pre-treatment decades: total population, Black population share, median income, unemployment rate, labor force participation rate, homeownership rate, and share of population under 18. Each treated or nearby tract is matched to its closest control based on propensity score distance. I also require exact matches on county and urban renewal designation.

Exact matching on county ensures that treated and control tracts are subject to the same local economic shocks and policy environment. The within-county restriction leverages the fact that federal funding for public housing was limited, and that public housing authorities therefore did not build in every "eligible" neighborhood within their jurisdiction, creating useful within-county variation in treatment status among neighborhoods with similar observed characteristics. In Section 6.4, I relax this constraint and allow matching to neighborhoods in other metropolitan areas.

I require exact matching on urban renewal to reduce the risk of conflating the effects of the two policies. Because the urban renewal program was closely intertwined with public housing, I treat urban renewal exposure as a co-occurring policy environment, rather than as a purely predetermined neighborhood characteristic. My baseline specification exact-matches treated tracts to controls with the same urban-renewal designation so that estimates compare neighborhoods with similar exposure to this related clearance policy. The resulting estimand is therefore the effect of public housing construction conditional on broad urban-renewal exposure, rather than the total effect of the combined public-housing and urban-renewal policy bundle. Because this conditioning may remove part of the broader clearance-policy bundle, Section 7.3 separately examines whether effects differ by urban-renewal status.

These exact matching restrictions mean that 17 of the 816 treated tracts in the balanced panel could not be matched to a suitable control within their county and urban renewal status, leaving 799 treated tracts in the matched sample.

Figure 5 shows absolute standard mean differences across pre-treatment characteristics for the treated neighborhoods and their matched controls. The matching achieves reasonably good balance on all pre-treatment characteristics, though given the restrictiveness of exact matching on county and urban renewal status, some covariates have standardized differences above 0.1. Section 6.4 shows that results are robust to alternative matching procedures that achieve better balance, whether by dropping poor matches or matching across metropolitan areas.

Step 3: Construction of Stacked Matched-Pair Dataset. Following Wing, Freedman, and Hollingsworth (2024), I create a stacked analytic dataset where each matched pair forms an independent "sub-experiment" observed from 1930 to 2010. Within each sub-experiment, the treated (or nearby) tract and its matched control are assigned a common treatment year, the year the treated neighborhood received public housing, and unique matched-pair identifiers. Control neighborhoods may appear in multiple sub-experiments if they serve as matches for multiple treated neighborhoods.

Step 4: Stacked Event-Study Estimation. I stack all matched-pair sub-experiments and implement a stacked difference-in-differences design that compares changes in outcomes over time between each treated neighborhood and its matched control. In particular, for tract i in matched pair m in time t, I estimate the following event-study specification:

$$y_{imt} = \alpha_{im} + \sum_{\tau \neq -1} \beta_\tau (D^\tau_{imt} \times \text{Treated}_i) + \delta_{mt} + \varepsilon_{imt} \tag{3}$$

where y_{imt} is the outcome of interest for tract i in matched pair m at time t, α_{im} are tract-by-pair fixed effects, $D^\tau_{imt} = \mathbb{1}[(t - T_{im}) = \tau]$ are indicators for event time τ relative to treatment year, Treated_i indicates whether tract i is a treated tract, and δ_{mt} are matched pair-by-year fixed effects. Since all fixed effects are implemented within a sub-experiment defined by each matched pair, identification solely relies on within-matched-pair comparisons. This specification implements the stacked difference-in-differences framework of Wing, Freedman, and Hollingsworth (2024), where I use matched pairs to define the sub-experiments. While Wing, Freedman, and Hollingsworth (2024) demonstrate the benefits of stacking to avoid negative weights in staggered adoption settings, my approach combines their framework with propensity score matching to address selection and staggered timing challenges simultaneously.[9]

The matched-pair-by-year fixed effects (δ_{mt}) provide particularly fine-grained controls for potential confounders. Because matching creates pairs of observationally similar neighborhoods within the same county, these fixed effects absorb not only county-level shocks but also shocks that differentially affect specific types of neighborhoods. For instance, if deindustrialization particularly impacted working-class, Black neighborhoods in Chicago in 1980, and both tracts in a matched pair are working-class, Black Chicago neighborhoods, this shock is absorbed by the pair-by-year fixed effect.

The tract-by-pair fixed effects (α_{im}) absorb all time-invariant differences within each matched pair, including unobserved characteristics that may have influenced both site selection and outcomes. Together, this fixed effects structure ensures identification comes exclusively from differential changes within pairs of similar neighborhoods.

The β_τ coefficients capture the average treatment effect at event time τ across all matched pairs. For directly treated tracts, this identifies the effect of hosting public housing. For nearby tracts, it identifies geographic spillover effects. Because each treated unit is matched to a single control unit, I weight each observation equally, yielding a simple average across sub-experiments without the weighting adjustments outlined in Wing, Freedman, and Hollingsworth (2024). Standard errors are adjusted for spatial correlation following Conley (1999) within a radius of 2 kilometers, allowing for dependence between nearby tracts. In Appendix E, I show that results are robust to using tract-level clustered standard errors instead.

The identifying parallel trends assumption operates at the matched-pair level: within each pair, the treated (or nearby) tract would have continued evolving like its matched control absent public housing. While this assumption is fundamentally untestable, several features of the design support its plausibility. First, the event-study specification allows me to test for differential pre-treatment trends between treated neighborhoods and their matched controls, and I find no systematic divergence prior to public housing construction. Second, the matching procedure directly targets the main sources of observed selection documented in Section 5.1 and emphasized in historical accounts. The site-selection models explain about 15% of the

[9]Recent work by Blanco and Neri (2025) and Guennewig-Moenert (2025) use stacked difference-in-differences designs to study public housing interventions, using only distance to select the control group.

variation in placement decisions, indicating that measured neighborhood characteristics did not mechanically determine which neighborhoods received projects. This does not imply random assignment, but it helps motivate comparisons among observationally similar candidate neighborhoods. Finally, Section 6.4 examines whether the estimates are sensitive to the details of the matching procedure, including restricting the sample to closer matches and relaxing the requirement that matched neighborhoods be in the same county.

6.2 Effects on Treated Neighborhoods

Figure 6 presents the effects of public housing construction on the inverse hyperbolic sine of population by race. Following public housing construction, the total population in the public housing neighborhoods increased substantially, rising by approximately 14.6%[10] in the immediate decade following construction, remaining elevated at approximately 18.1% three decades later. This increase was driven by substantial increases in the total Black population, a 27.8% increase relative to the matched control three decades after construction. On average, I find little to no effect on the white population. These population increases partly reflect the mechanical addition of public housing residents themselves: Figure 7 shows results for the non-public-housing population, defined as total Census-reported population minus estimated public-housing population. The figure shows a large average decline in the estimated private population, both white and Black, following public housing construction.[11]

Figure 8 shows that these population changes led to changes in the racial composition of the treated neighborhoods. Relative to the control neighborhoods, Black population shares increased by 2.6 percentage points in the immediate decade following construction, and continued to increase up to approximately 5.9 percentage points three decades after construction. The latter estimate represents a 23.8% increase relative to the average pre-treatment Black population share.

Figure 9 shows the effects of public housing on median rent, income, and homeownership rates, showing broad declines in all three. Median incomes decline sharply following public housing construction, with estimated effects of -7.7% immediately after construction and -15.1% three decades later. Median rents also decline over time, with statistically insignificant

[10]The event study figures display raw asinh coefficients. Percentage changes reported in the text are calculated as $\sinh(\beta) \times 100$, which approximates $\beta \times 100$ for small coefficients and provides a more accurate conversion for larger effects (Bellemare and Wichman 2020).

[11]Because project-level population by race is generally observed only in the 1970s, this decomposition should be interpreted as approximate. It is intended to separate total neighborhood changes from the mechanical addition of public housing residents, but it may mismeasure the timing or magnitude of private-population changes if project occupancy or racial composition changed substantially over time.

changes immediately after construction but an estimated effect of -14.4% three decades later. Homeownership rates drop sharply by -3.2 percentage points immediately after construction and remain depressed, consistent with the addition of rental public housing units to these neighborhoods.

6.3 Effects on nearby neighborhoods

A primary source of backlash against the mid-century public housing program has been the concern that it precipitated a broader urban decline and white flight, with negative spillovers on surrounding communities (Jackson 1985). These potential spillovers matter for assessing the program's overall urban impact. To test this, I estimate the geographic spillover effects of public housing on nearby neighborhoods.

Using the matched nearby tracts defined in Section 6.1, I estimate geographic spillover effects of public housing on surrounding neighborhoods. Balance statistics for the nearby tract matches (Figure 10) show good balance on all pre-treatment characteristics, with only Black population showing a standardized mean difference of slightly above 0.1.

I then estimate the treatment effects for these nearby neighborhoods using the same stacked difference-in-differences design as in Equation 3.

Figure 11 shows the effects of public housing construction on population by race in the nearby neighborhoods. In contrast to the treated neighborhoods, I find that the spillover neighborhoods experienced small population declines overall. I find declines in white population in the long run, but minimal changes in Black population. Consequently, while Black population shares increased by approximately 2.0–2.6 percentage points in nearby neighborhoods, this change appears to be driven primarily by white population decline rather than Black population growth, in contrast to the treated neighborhoods where substantial Black population inflows occurred.

Figure 12 shows the effects of public housing construction on median rent and income in the nearby neighborhoods. I find statistically significant declines in median income in nearby neighborhoods, with estimated effects of -2.5% immediately and -3.5% one decade after construction. Effects in later decades become smaller and statistically insignificant, though the point estimates remain negative. This may indicate some degree of economic decline or sorting in these nearby neighborhoods. However, I find no effect on median rent in these nearby neighborhoods, suggesting that the housing market effects of public housing construction were not widespread.

This evidence suggests that public housing construction did affect nearby neighborhoods by precipitating white flight. However, the modest effects on incomes and absence of rent effects suggest that the "urban decline" narrative surrounding public housing (Jackson 1985) overstates the program's economic impact on surrounding areas. What occurred in nearby neighborhoods was primarily racial transition, not broad economic deterioration.

6.4 Robustness Checks

One might worry that the results are sensitive to the specific matching specification. In particular, one might be concerned that the exact matching criteria are too restrictive, resulting in imperfect nearest-neighbor matches. Indeed, the balance achieved in the main specifications above is imperfect, with some standardized mean differences between 0.1 and 0.2. I run several alternative matching specifications to test the robustness of these results. First, I run a specification that drops poor matches: in particular, I apply a caliper that excludes matches with a propensity score difference of more than 0.25 standard deviations. Second, I loosen the exact-match restriction on county and instead require a match from a neighborhood within any other metropolitan area. This expands the donor pool, particularly for treated neighborhoods in smaller cities. A comparison of the results from these alternative specifications is shown in Figure 13 for treated neighborhoods and Figure 14 for nearby neighborhoods. Overall, the estimates are largely similar across these alternative specifications, both in sign and magnitude.

Another potential concern is that the use of the inverse hyperbolic sine transformation for population and monetary outcomes could affect the interpretation of results. Recent work by Chen and Roth (2024) highlights that logarithmic transformations in difference-in-differences designs can produce estimates that are sensitive to the choice of units and may conflate extensive and intensive margin effects. McConnell (2024) further shows that when treated and control groups have different baseline means, log specifications can yield misleading estimates and even produce opposite signs relative to levels specifications. To address this concern, I re-estimate all specifications using outcome variables in levels rather than asinh-transformed values. The results show that both specifications yield qualitatively identical conclusions: all outcomes show the same sign of treatment effect. Furthermore, when the level coefficients are expressed as a percentage of the baseline control mean—an approach suggested by Chen and Roth (2024) to ensure unit-invariance—the magnitudes are broadly consistent with the asinh estimates. For example, the estimated effect on median income was -14.8% in the levels specification compared to -15.1% in the asinh specification, while Black population increased by 28.5% (levels) versus 27.8% (asinh). These results confirm that my findings are not sensitive to the choice of functional form.

One more fundamental concern with this design is that, if individuals are re-sorted across neighborhoods as a result of the construction of public housing, then the within-metro control neighborhoods may in fact be affected by the treatment as well. For example, one might be concerned that, in response to the construction of public housing, individuals who initially lived in a treated neighborhood might relocate to the initially similar control neighborhood. This would be a violation of the stable unit treatment value assumption (SUTVA).

I show two pieces of evidence that these potential SUTVA violations are not driving my results. First, I run a specification in which control neighborhoods are only selected from metropolitan areas with low public housing intensity. In these cities, control areas are less likely to be affected by any potential spillovers from public housing. I calculate each CBSA's intensity as total public housing units divided by the total population of the neighborhoods in that metropolitan area in my balanced sample in 1950. Across the 47 metropolitan areas in my sample, the intensity ranges from 0 to 20 units per 1,000 people, with a median of 7.0. I restrict the donor pool to metros in the bottom quintile of intensity, which includes large metros such as Los Angeles, Detroit, St. Louis, and Indianapolis. I then match treated and nearby neighborhoods from all metros to controls in these low-intensity metros, requiring that matches come from different metropolitan areas.

If SUTVA violations were inflating my baseline estimates—for example, because Black residents leaving control tracts made those controls appear less Black, thereby exaggerating the treatment-control differences—then I would expect smaller effects using these less-contaminated controls. Instead, I find that the effects on Black population shares are larger than my baseline. These results are presented in Appendix Table 19.

Second, if results are driven by SUTVA violations, we might expect offsetting movements between the control tracts and treated tracts. For example, we might see the Black population share in matched controls decline as residents move to the treated neighborhoods, which could drive the event study results. Appendix Figure 27 plots raw Black shares for treated neighborhoods and their matched controls over event time. Black shares increase in both treated and control neighborhoods, but more so in the treated neighborhoods. The absence of declining Black shares in controls suggests that within-metro resorting is not the primary driver of the results.

A separate concern is that unobserved shocks occurring between census dates could be correlated with both project site selection and neighborhood outcomes. For example, if a neighborhood experienced an influx of Black residents after the baseline census, this demographic shift could have influenced where public housing was sited. Because the matching is based on baseline census characteristics, it would fail to account for these changes, and the estimates could reflect pre-existing demographic trends rather than the causal effect of public housing. To assess this, I exploit the fact that projects open at different points within the decade. A project that opens shortly after a census is matched on covariates that are nearly contemporaneous, while one that opens just before the next census is matched on covariates that are up to nine years stale. I classify treated tracts as early openers (5 to 9 years before the next census) or late openers (0 to 4 years before the next census), pooling across treatment decades, and estimate separate event studies for each group. If within-decade confounders were driving the results, late openers, whose matching covariates are most stale, should show different pre-trends or systematically different treatment effects. Across all outcomes, I find no pre-trend violations in either subgroup and no systematic differences in post-treatment effects between early and late openers. In fact, late openers show slightly larger initial effects on Black population share than early openers, the opposite of what a stale-matching bias story would predict. Results are similar for nearby neighborhoods. These results, shown in Appendix Figure 29, suggest that within-decade timing of project openings is not a meaningful source of bias.

7 Mechanisms and Heterogeneity

In this section, I explore the mechanisms outlined in Section 3 by examining heterogeneity in the neighborhood effects of public housing. I present results at $t = 1$ and $t = 3$ to highlight both short/medium-run and long-run effects.

7.1 Mechanical vs Behavioral Effects

I first explore whether effects were purely mechanical. To do so, I examine whether treatment effects vary with the share of tract residents living in public housing in the post-treatment period ($t = 0$). If effects were purely mechanical, we would expect them to scale proportionally with this share. Figure 15 presents treatment effects separately for treated neighborhoods in the bottom, middle, and top terciles in the public housing population share at $t = 0$. For median income and rents, all three terciles show declines of similar magnitudes. This pattern provides suggestive evidence against purely mechanical effects: if income declines simply reflected the mechanical addition of low-income public housing residents, we would expect larger effects in neighborhoods where public housing residents constitute a larger share of the population. The similar magnitudes across terciles are consistent with public housing triggering broader sorting responses in the private housing market, though I cannot definitively rule out mechanical explanations.

For population outcomes, the pattern is less clear. Black and total population increases are larger in neighborhoods with higher public housing shares, consistent with mechanical effects. However, behavioral responses could amplify mechanical population changes, for example, if the arrival of Black public housing residents attracted additional Black in-migration or accelerated white out-migration. The income and rent results suggest that such amplification is plausible.

7.2 Neighborhood Tipping

I next examine whether these behavioral responses operated through racial tipping dynamics. Models of neighborhood tipping suggest that the initial racial composition of a neighborhood may influence whether demographic shocks precipitate broader racial transition (Schelling 1971; Card, Mas, and J. Rothstein 2008). Some historical accounts of public housing describe dynamics consistent with tipping, where public housing construction coincided with a rapid neighborhood racial transition (R. Rothstein 2017; Jackson 1985). To test this tipping hypothesis, I divide treated neighborhoods into three groups based on their baseline Black population share: almost entirely non-Black (less than 1%; $N = 257$), those in a potential "tipping range" (1–12%; $N = 186$), and those with higher initial Black shares (above 12%; $N = 356$). I choose 12% as the upper bound of the tipping range because it corresponds to the top of the threshold estimated by Card, Mas, and J. Rothstein (2008) in 1970. I also analyze nearby neighborhoods using the same cutoffs. I use population by race in levels rather than inverse hyperbolic sine here, since percentage changes in Black/white population will mechanically be larger/smaller in neighborhoods with low/high initial Black population shares.

Figure 16 presents the results, which show that the effects of public housing on racial composition depend strikingly on the initial neighborhood context. In treated neighborhoods in the tipping range, I find large increases in Black population and large declines in white population in both treated and nearby neighborhoods, consistent with racial tipping triggering white flight. In contrast, treated neighborhoods that were almost entirely non-Black at baseline saw moderate increases in Black population but no significant change in white population, while treated neighborhoods with high initial Black shares saw no change in Black population and potentially small increases in white population.

Appendix Figure 28 confirms this pattern using Black population share directly as the outcome: treated neighborhoods in the tipping range experienced a 20 percentage point increase in Black share by $t = 30$, while neighborhoods with high initial Black shares saw no significant change.

Nearby neighborhoods experienced similar patterns: nearby neighborhoods in the tipping range also saw sizable declines in white population, providing evidence that in this range, public housing construction triggered broader white flight dynamics beyond the treated neighborhoods themselves. These nearby neighborhoods also saw increases in Black population, though notably smaller than in treated neighborhoods. Overall, these results suggest that racial tipping dynamics played a significant role in shaping the neighborhood effects of public housing construction. These results complement findings on modern affordable housing programs that the neighborhood effects of new affordable housing construction depend on initial neighborhood characteristics (Diamond and McQuade 2019).

7.3 Additional Heterogeneity Analyses

Finally, I explore several additional dimensions of heterogeneity. I first examine whether effects differed in New York City and other cities in my sample. The New York City Housing Authority (NYCHA) was, and remains, the largest housing authority in the country, and it has often been thought of as better managed and more successful than other American public housing programs (Bloom 2008; Hunt 2018). Figure 17 shows results separately for New York City and for other cities. The results are somewhat nuanced. Comparing the long-run effects, treated neighborhoods in New York City experienced smaller and less statistically significant declines in median income and rents compared to those in other cities. However, white flight patterns are more pronounced in New York City, with nearby neighborhoods around New York City projects experiencing much larger and more significant white population declines than in other cities. Overall, these results paint a mixed picture, and suggest that the neighborhood effects of public housing may not neatly correspond with the perceived success of local housing authorities.

I also test whether the effects of the projects varied depending on when they were built. These results are shown in Figure 18. I find that the effects of public housing differed substantially based on when projects were built, particularly for population and racial composition. Early-period projects led to larger and more persistent Black population increases in treated neighborhoods, while the effects for later-period projects were smaller and faded over time. More strikingly, nearby neighborhoods around projects built in the early period saw notable long-run white outflows, while nearby neighborhoods around late-period projects saw no significant white population changes. On the other hand, the effects on median incomes and rents were similar across the two periods. One possible explanation for these differences is that urban neighborhoods had lower Black population shares in the 1940s and 1950s than later, and so public housing construction in that period may have been more likely to trigger the tipping dynamics shown in Section 7.2.

Finally, I test whether these effects differed based on whether the neighborhoods were also affected by the urban renewal program. This analysis both highlights potential interactions between the two programs and addresses concerns that my baseline estimates may be conflated with the effects of urban renewal. Figure 19 presents the results, separately estimating Equation 3 for neighborhoods that were urban renewal tracts and those that were not. The main finding is that the neighborhood effects of public housing construction seem largely to be concentrated in non-urban renewal areas. Non-urban renewal tracts show the large Black population increases and income declines seen in the main results, while urban renewal tracts show no significant demographic changes. This pattern is consistent with LaVoice (2024), who finds that urban renewal led to declines in Black population shares. In neighborhoods where both programs operated, the displacement effects of urban renewal may have counteracted the demographic changes associated with public housing construction.

8 Public Housing and Long-Run Economic Opportunity

I have shown that public housing had significant and persistent effects on neighborhood population, racial and economic composition, and housing markets. I next examine whether neighborhoods that received public housing also became lower-opportunity environments for children growing up in later decades.

I explore this question by merging tract-level data from the Opportunity Atlas (Chetty et al. 2018). These data include measures of long-run upward mobility and incarceration for children born from 1978 to 1983 whose parents were at the 25th percentile of the national income distribution. I will refer to these children as "low-income children". In particular, I use tract-level measures of the mean household income rank in adulthood in 2014-2015 and the share of these children who were incarcerated as of April 1st, 2010, at the 2010 Census tract level. I concord these data to 1950 tract boundaries using my tract crosswalks, along with population counts of relevant children in each tract from the Opportunity Atlas data. I then match these data to the matched pairs of public housing tracts, nearby tracts, and control tracts used in Section 6.

I then estimate the following regression model separately for public housing tracts and nearby tracts:

$$Y_{im} = \beta \text{Treated}_i + X'_{i,1980}\gamma + \mu_m + \epsilon_{im} \tag{4}$$

where Y_i denotes the Opportunity Atlas outcome of interest, either the mean income rank or the incarceration rate for low-income children, in tract i, μ_m denotes a matched-pair fixed effect, and $X_{i,1980}$ is a vector of tract characteristics in 1980 (Black share, median income, total population, unemployment rate, and median rent). The variable Treated_i is an indicator for whether tract i is a public housing tract (or a nearby tract). Without 1980 controls, β captures the average difference in outcomes between treated tracts and their matched controls. With controls, it captures the remaining difference conditional on 1980 neighborhood characteristics. Since most public housing projects were built before 1980, the 1980 characteristics are themselves affected by public housing and should therefore be considered "bad controls" in the usual causal inference sense. However, including these controls helps assess how much of the raw difference is accounted for by observable neighborhood composition versus the presence of public housing itself.

Tables 6 and 7 present results for public housing tracts and nearby tracts, respectively. I estimate specifications both without controls (Columns 1 and 3) and with controls for 1980 neighborhood characteristics (Columns 2 and 4).

Children who grew up in tracts containing public housing experienced significantly worse outcomes: a 1.8 percentage point lower income rank and a 0.6 percentage point higher incarceration rate compared to matched controls (Table 6, Columns 1 and 3). About a quarter of this difference remains after controlling for 1980 neighborhood characteristics, such as Black share and median income (Columns 2 and 4), suggesting that public housing had effects on neighborhood upward mobility and incarceration rates even beyond simply changing observable demographics.

By contrast, nearby tracts show worse outcomes only before adding 1980 neighborhood controls. While nearby tracts initially have lower mobility and higher incarceration than their matched controls (Table 7, Columns 1 and 3), these differences disappear when controlling for 1980 neighborhood characteristics (Columns 2 and 4). This pattern suggests that the nearby-tract differences are largely explained by later neighborhood composition, rather than by an independent association with proximity to public housing.

Taken together, these results suggest that the neighborhood changes documented above are reflected in later measures of economic opportunity. The strongest associations are concentrated in the tracts that contained public housing projects, while the nearby-tract differences appear to operate through broader neighborhood composition. Because families who remained in or moved to these neighborhoods by the late 1970s may differ systematically from families in control neighborhoods, and because children experienced different exposure lengths depending on when projects were built, I interpret these estimates as suggestive evidence on long-run neighborhood opportunity rather than as individual-level causal effects.

9 Conclusion

This paper has examined the long-run neighborhood effects of the mid-20th-century U.S. public housing program. Using a newly constructed national dataset of more than 8,000 projects built between 1935 and 1973, linked to a consistent panel of tract-level census data spanning 1930–2010, I documented two central findings. First, public housing projects were systematically targeted toward neighborhoods that were initially poorer, more populated, disproportionately Black, and slated for urban renewal. These patterns confirm the program's slum-clearance origins and the racialized politics of site selection. Second, public housing construction had significant and persistent effects on neighborhood trajectories: treated tracts experienced long-run increases in Black population shares and sustained declines in incomes and rents, with modest spillover effects on nearby neighborhoods including white population decline and income reductions. Importantly, I find evidence consistent with racial tipping dynamics: neighborhoods with moderate baseline Black shares experienced substantial white population outflows following public housing construction, while neighborhoods with higher initial Black shares showed more muted responses.

These results suggest that public housing not only reflected but reinforced existing patterns of segregation and disinvestment, adding quantitative evidence to longstanding debates about the role of federal policy in creating residential segregation (R. Rothstein 2017; Trounstine 2018; T. D. Logan and Parman 2025). Rather than improving neighborhoods as the Housing Act of 1949 envisioned, the program contributed to concentrated poverty and persistent racial segregation in American cities. These patterns have implications for contemporary affordable housing policy. Modern reforms to public housing have emphasized smaller-scale interventions, mixed-income developments, and designs that blend into existing neighborhoods rather than large, segregated projects. My findings help explain the rationale behind these reforms: large concentrations of subsidized housing in neighborhoods undergoing racial transition accelerated segregation and demographic change.

One crucial caveat is that my results do not necessarily imply that public housing failed to benefit individual residents. For many low-income residents, public housing may have provided better housing quality, stability, and affordability than available private-market alternatives. In ongoing work, I am linking individuals to public housing projects using full-count Census data to study the effects on project residents and their neighbors. This individual-level analysis will complement the neighborhood-level findings presented here by directly measuring the benefits and costs experienced by public housing residents themselves, providing a more complete assessment of the program's legacy.

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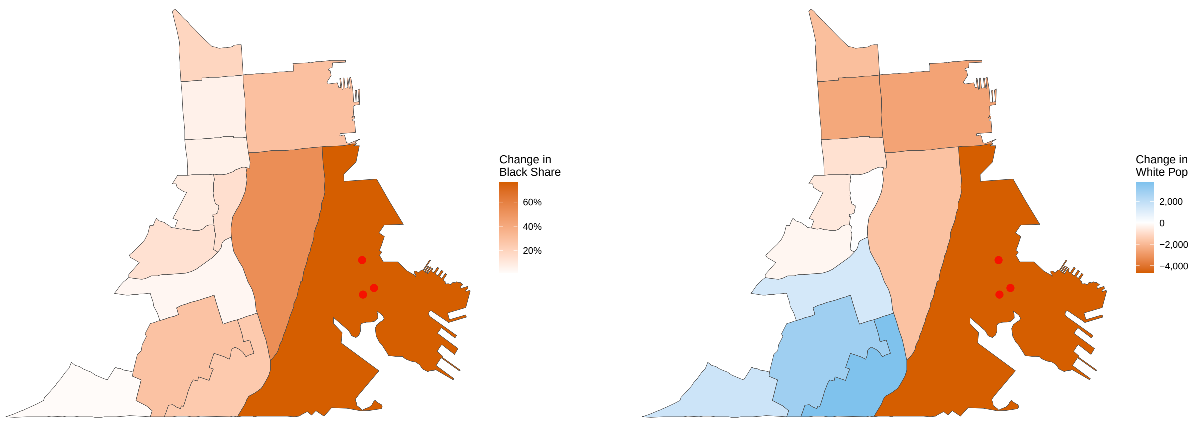
Figures and Tables

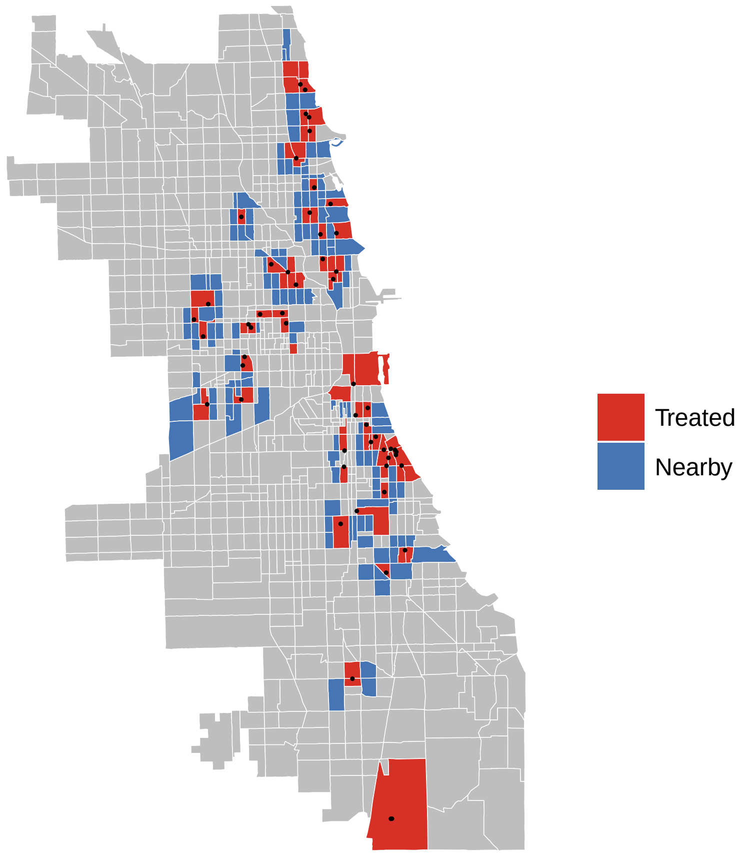
Figure 1: Motivating Example: Hunters Point, San Francisco

(a) Change in Black Population Share (b) Change in White Population



Note: These maps show demographic changes from 1940 to 1970 in the census tract containing Hunters Point public housing (built 1942–1944) and surrounding tracts within 2 kilometers. Panel (a) shows the change in Black population share; panel (b) shows the change in white population. Red dots indicate public housing project locations. See Section 2 for historical context.

Figure 2: Public Housing Projects and Spillover Areas: Chicago



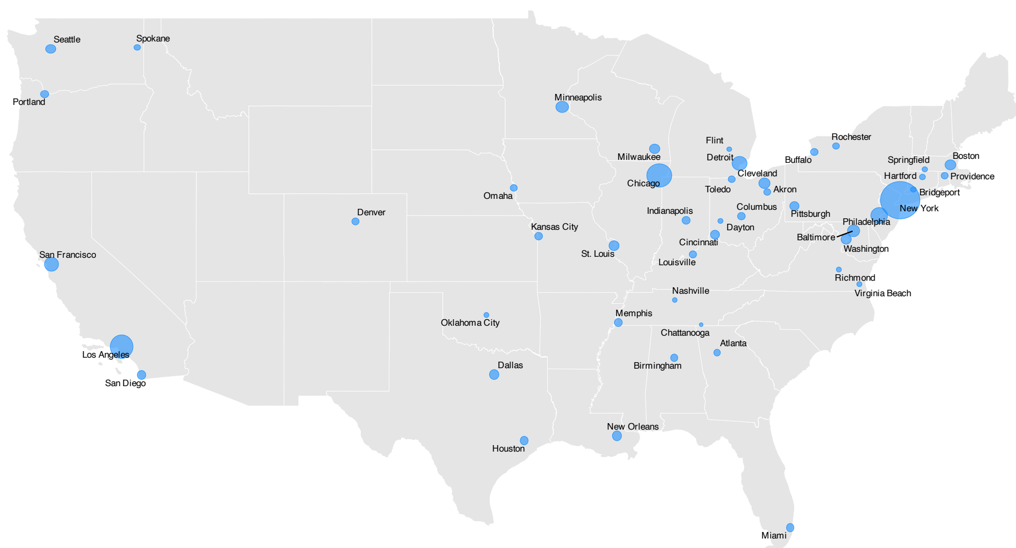
Note: This map shows census tracts in Chicago containing public housing projects built between 1941 and 1973 (red, N=72 tracts) and nearby spillover areas (blue). Spillover areas are defined as census tracts that share a border with a treated tract and are within 1 kilometer of the nearest public housing project. See Section 4 for sample construction details.

Table 1: Sample Attrition

Step	Tracts	Pop. 1940 (M)	CBSAs	Treated Tracts	Projects	Units
Original sample	12,062	38.1	60	1,396	1,540	516,084
Exclude unmatched PSH projects	11,309	35.2	60	1,396	1,503	513,086
Balanced on years (1930-2010)	8,510	31.4	50	1,145	1,182	464,150
Complete variables	8,040	30.3	50	1,119	1,153	449,658
Exclude treatments ≤1940	7,932	29.8	50	1,011	1,042	392,797
Drop CBSAs <30 tracts	7,881	29.5	47	996	1,015	388,699
Drop population outliers	6,080	25.3	47	816	781	295,458

Note: This table shows the impact of sequential sample restrictions on the analysis sample. Columns report the number of census tracts, 1940 population (in millions), Core-Based Statistical Areas (CBSAs), treated tracts (those receiving public housing), public housing projects, and total housing units at each stage. The final balanced sample represents approximately 19% of the 1940 U.S. population. See Section 4 for detailed discussion of each restriction.

Figure 3: Geographic Coverage of CBSAs in Analysis Sample



Note: This map shows the 47 Core-Based Statistical Areas (CBSAs) included in the balanced panel analysis, containing 6,080 census tracts and 816 treated tracts. Shaded areas represent metropolitan areas meeting the sample selection criteria: CBSAs with at least one public housing project built between 1941 and 1973, at least 30 total tracts, and complete census tract data for all study years (1930–2010). See Section 4 for sample construction details.

Table 2: Comparison of Public Housing and Non-Public Housing Neighborhoods (1940 Baseline)

Variable	Untreated	Treated	Std. Diff.
N (Tracts)	5252	814	
Total Population (asinh)	8.81 (0.65)	9.09 (0.60)	0.450***
Black Share	0.06 (0.17)	0.16 (0.28)	0.456***
Black Population (asinh)	3.17 (2.54)	4.88 (2.85)	0.635***
Median Income (asinh)	3.62 (0.30)	3.42 (0.31)	0.625***
Median Rent (asinh)	6.59 (0.42)	6.39 (0.43)	0.468***
Unemployment Rate	0.11 (0.05)	0.14 (0.06)	0.544***
Labor Force Participation Rate	0.56 (0.05)	0.56 (0.05)	0.144***
Distance from CBD (asinh)	8.08 (2.83)	7.00 (3.41)	0.346***
Share Needing Major Repairs	0.07 (0.09)	0.11 (0.11)	0.402***

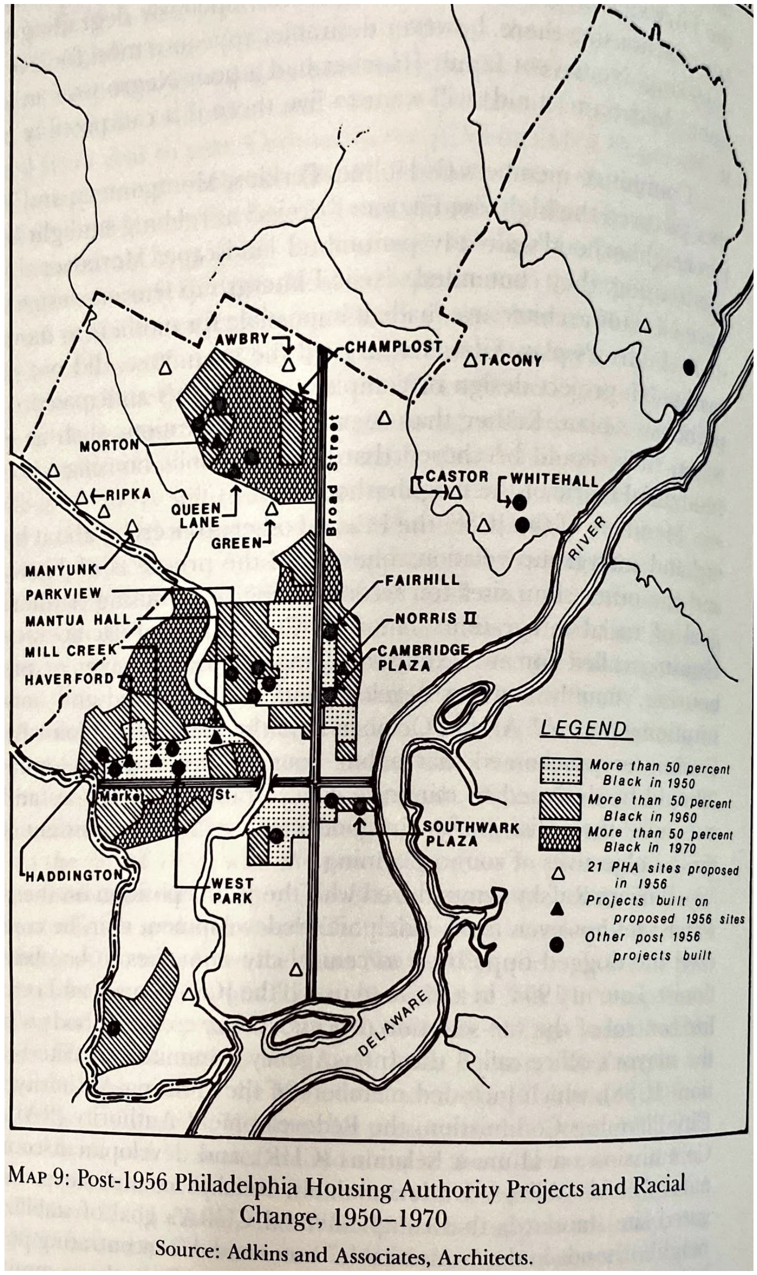
Note: This table compares 1940 baseline characteristics between census tracts that eventually received public housing projects between 1941 and 1973 and tracts that did not. Sample sizes are reported in the first row. Asinh denotes the inverse hyperbolic sine transformation. Standard deviations are shown in parentheses. The Std. Diff. column reports standardized mean differences (difference in means divided by pooled standard deviation). Statistical significance from two-sample t-tests: *p<0.1, **p<0.05, ***p<0.01. See Section 5 for analysis of site selection patterns.

Table 3: Site Selection: Predicting Public Housing Placement from 1940 Neighborhood Characteristics

	(1)	(2)	(3)	(4)
Black Share	0.239***	0.202***	0.197***	0.185***
	(0.040)	(0.041)	(0.043)	(0.039)
Asinh Total Population	0.048***	0.035***	0.038***	0.034***
	(0.012)	(0.011)	(0.012)	(0.011)
Asinh Median Income	-0.115***	-0.067***	-0.065***	-0.056***
	(0.021)	(0.020)	(0.020)	(0.020)
Asinh Median Rent	-0.083***	-0.013	0.000	-0.009
	(0.018)	(0.030)	(0.033)	(0.030)
Pct Graduated HS		0.123*	0.122*	0.099
		(0.064)	(0.069)	(0.065)
Unemployment Rate		0.696***	0.716***	0.664***
		(0.185)	(0.189)	(0.180)
LFP Rate		-0.481***	-0.421***	-0.443***
		(0.150)	(0.149)	(0.143)
Redlined (HOLC)		0.025	0.028	0.023
		(0.020)	(0.020)	(0.020)
Asinh Dist. from CBD		0.003	0.003	0.002
		(0.003)	(0.003)	(0.003)
CBD Indicator		-0.008	-0.014	-0.018
		(0.032)	(0.034)	(0.034)
Share Needing Major Repairs			0.072	
			(0.080)	
Asinh Dist. to Highway				0.014
				(0.010)
Urban Renewal Area				0.104***
				(0.026)
Homeownership Rate		-0.252***	-0.243***	-0.231***
		(0.044)	(0.045)	(0.044)
Share Under 18		0.424***	0.429***	0.463***
		(0.147)	(0.148)	(0.144)
Share 65+		-0.415	-0.364	-0.337
		(0.484)	(0.557)	(0.480)
Num.Obs.	6080	6080	5767	6080
R2	0.129	0.145	0.143	0.152
R2 Adj.	0.120	0.135	0.134	0.142
County fixed effects	Yes	Yes	Yes	Yes

Note: This table reports linear probability model estimates of the relationship between 1940 neighborhood characteristics and public housing placement. The dependent variable equals 1 if a census tract received a public housing project between 1941 and 1973 (N=816) and 0 otherwise (N=5,264). The baseline probability of treatment is 13.4%. All specifications include county fixed effects. Standard errors in parentheses are adjusted for spatial correlation following Conley (1999) with a 2-kilometer cutoff. Columns progressively add controls: (1) core demographics; (2) additional neighborhood characteristics; (3) housing conditions; and (4) urban renewal and highway exposure. Statistical significance: *p<0.1, **p<0.05, ***p<0.01. See Section 5 for interpretation.

Figure 4: Historical Map of Proposed and Actual Philadelphia Public Housing Sites, 1956



Note: This map shows the distribution of proposed and actual public housing sites in Philadelphia. Following the 1954 Housing Act, the Philadelphia Housing Authority proposed 21 projects across the city, but many faced significant backlash from white communities and were eventually cancelled. The proposed-but-not-built sites had systematically different baseline characteristics from actual sites (particularly lower Black population shares), illustrating both the political dynamics of site selection and why such sites would not serve as valid counterfactuals in a research design. Source: Bauman (1987), georeferenced by author.

Table 4: 1940 Neighborhood Characteristics: Proposed vs Actual Public Housing Locations in Philadelphia

Variable	Actual Public Housing	Proposed Only	Std. Diff.
Black Share	0.28 (0.29)	0.01 (0.02)	1.310***
Asinh Total Population	9.16 (1.27)	7.98 (1.78)	0.762**
Asinh Median Income	3.47 (0.30)	3.52 (0.28)	0.148
Unemployment Rate	0.21 (0.10)	0.14 (0.05)	0.897**
Asinh Median Rent	6.41 (0.32)	6.45 (0.40)	0.102
Asinh Distance from CBD	8.31 (2.04)	9.75 (0.26)	0.992**
Redlined (HOLC)	0.40 (0.50)	0.08 (0.29)	0.791**
Urban Renewal Area	0.34 (0.48)	0.17 (0.39)	0.398

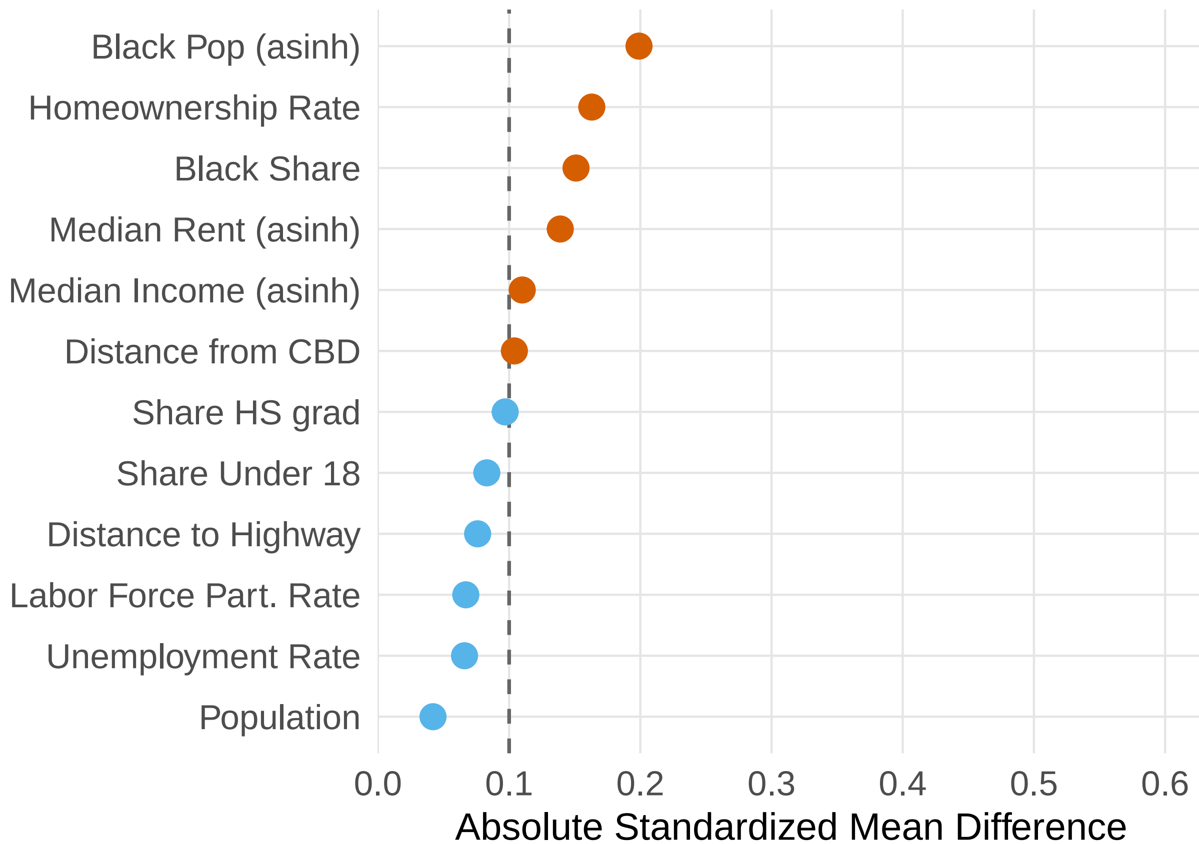
Note: This table compares 1940 neighborhood characteristics between proposed-but-not-built public housing sites (N=12) identified from a 1956 Philadelphia Housing Authority map (Bauman (1987)) and actually-built sites (N=47) in Philadelphia. The first two columns show means with standard deviations in parentheses. The Std. Diff. column reports standardized mean differences. The large differences, particularly in Black population share, illustrate how political opposition in white neighborhoods shaped site selection, and why proposed-but-rejected sites would not serve as valid counterfactuals. Statistical significance: *p<0.1, **p<0.05, ***p<0.01. See Section 5.2 for discussion.

Table 5: Project Demographics: Predicting Racial Composition of Public Housing from Baseline Neighborhood Characteristics

	(1)	(2)	(3)
(Intercept)	1.560***	1.719***	
	(0.169)	(0.335)	
Black Share	0.435***	0.442***	0.231***
	(0.062)	(0.070)	(0.059)
Asinh Median Income	-0.293***	-0.254***	-0.228***
	(0.041)	(0.056)	(0.058)
Asinh Median Rent		-0.145**	-0.097
		(0.072)	(0.061)
Asinh Population		0.050**	0.057***
		(0.024)	(0.019)
Unemployment Rate		-0.174	0.308**
		(0.177)	(0.125)
LFP Rate		0.237	0.396
		(0.236)	(0.236)
Asinh Dist. to CBD		0.011*	0.018***
		(0.006)	(0.005)
CBD		-0.078	-0.116*
		(0.072)	(0.066)
Asinh Dist. to Highway		-0.012	-0.049
		(0.033)	(0.034)
Redlined (HOLC)		-0.033	0.020
		(0.053)	(0.050)
Urban Renewal Area		-0.010	0.015
		(0.045)	(0.031)
N	755	755	755
R^2	0.358	0.386	0.583
Adj. R^2	0.356	0.377	0.546
County FE	No	No	Yes

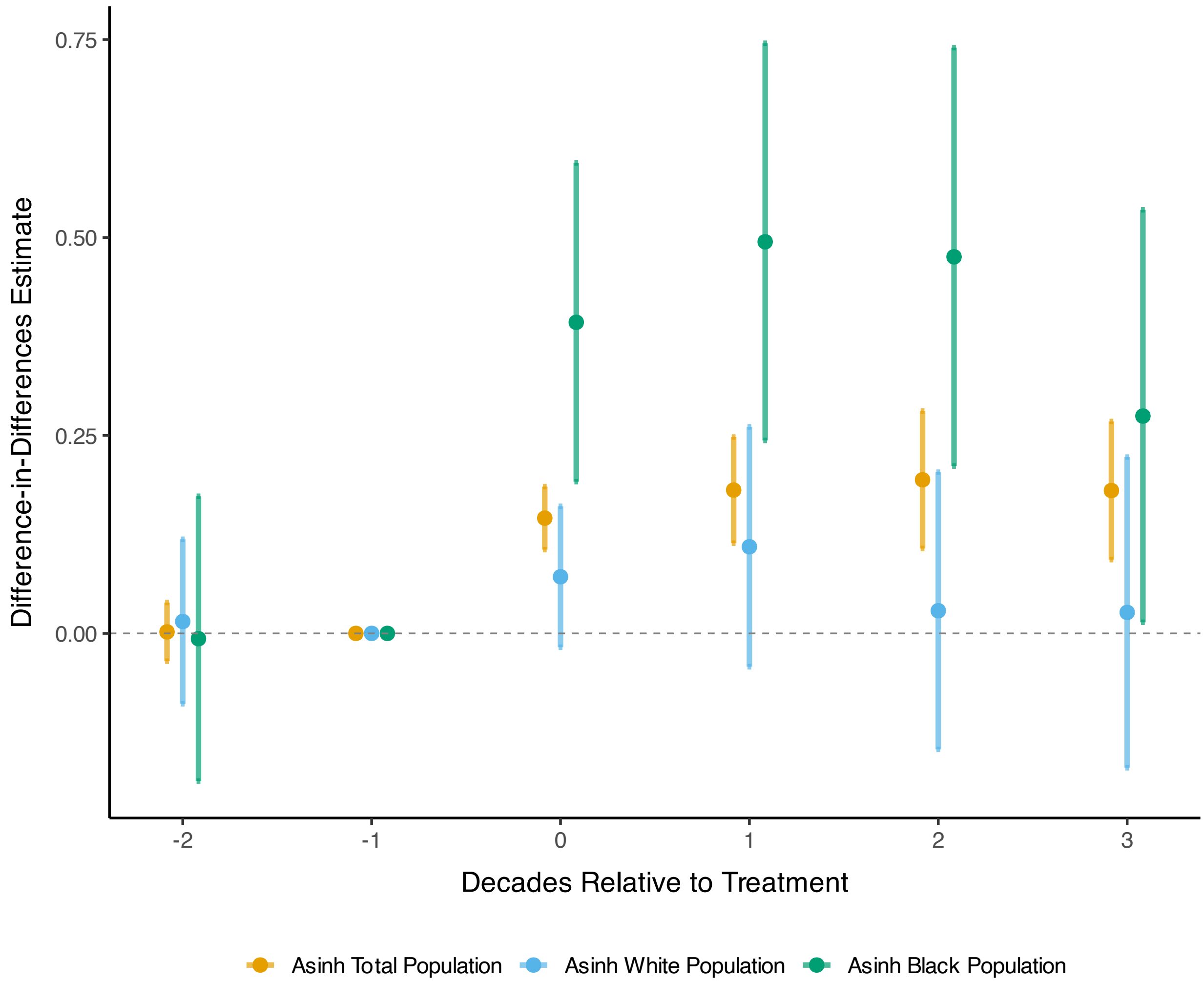
Note: This table reports OLS estimates of the relationship between baseline neighborhood characteristics (measured at $t-1$, the decade before construction) and project racial composition. The sample includes treated tracts with available project demographic data. The dependent variable is the Black population share within the public housing project, measured from 1970s administrative data (1977 Picture of Subsidized Households for most projects, 1971 data for NYC, 1973 annual reports for Chicago). Column (1) includes baseline Black share and income only; Column (2) adds neighborhood controls; Column (3) adds county fixed effects. Standard errors clustered at the county level in parentheses. Statistical significance: *p<0.1, **p<0.05, ***p<0.01. See Section 5.3 for interpretation.

Figure 5: Pre-period Balance: Treated Neighborhoods vs Matched Controls



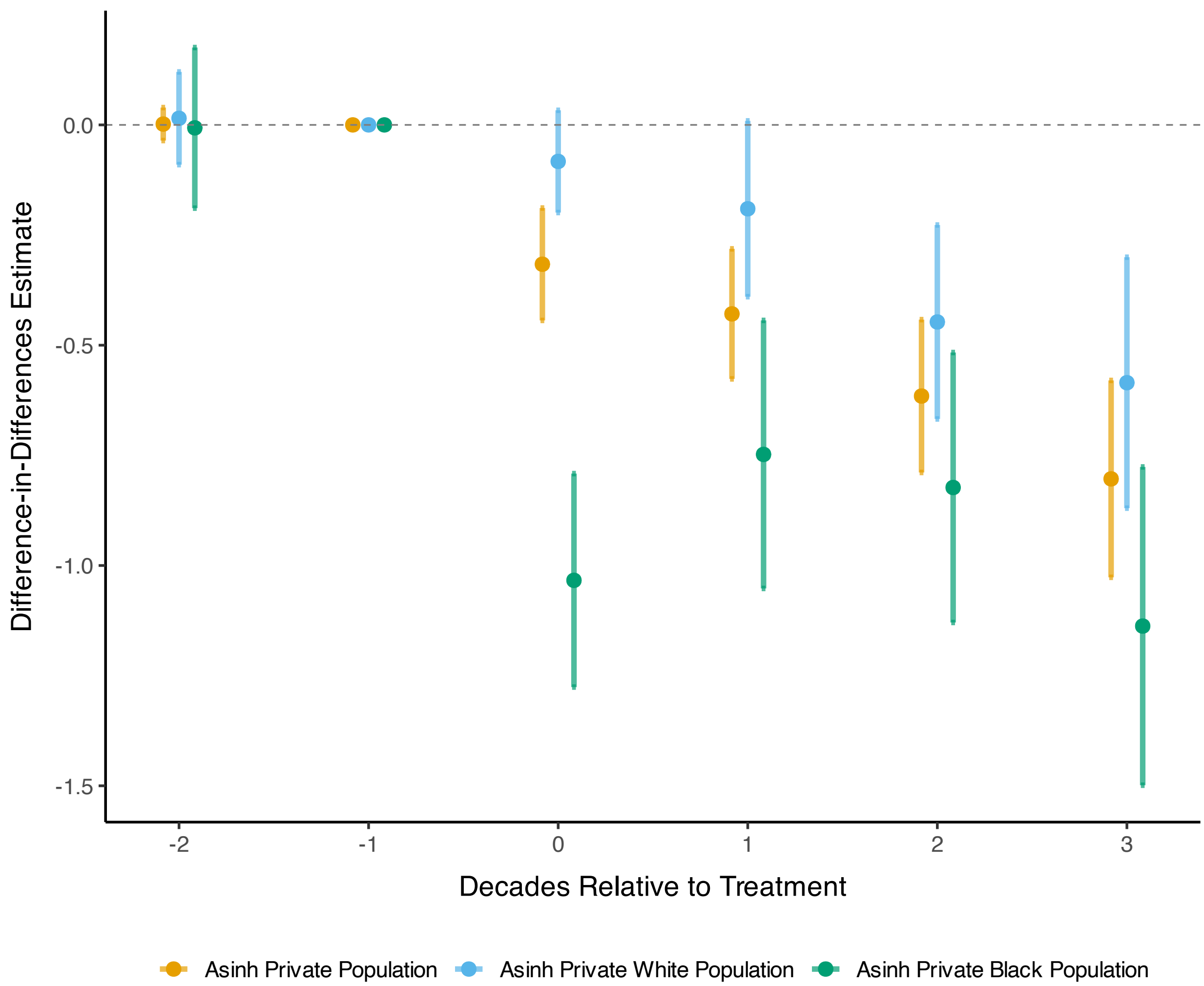
Note: This figure displays standardized mean differences (SMD) between treated neighborhoods (N=799) and their matched controls across baseline covariates. Each point represents the SMD for a covariate measured in the two pre-treatment decades. SMDs near zero indicate successful balance; values above 0.1 (vertical dashed lines) suggest meaningful imbalance. Matching uses propensity scores with exact matching on county and urban renewal status. See Section 6.1 for matching procedure details.

Figure 6: Population and Racial Composition Effects in Treated Neighborhoods



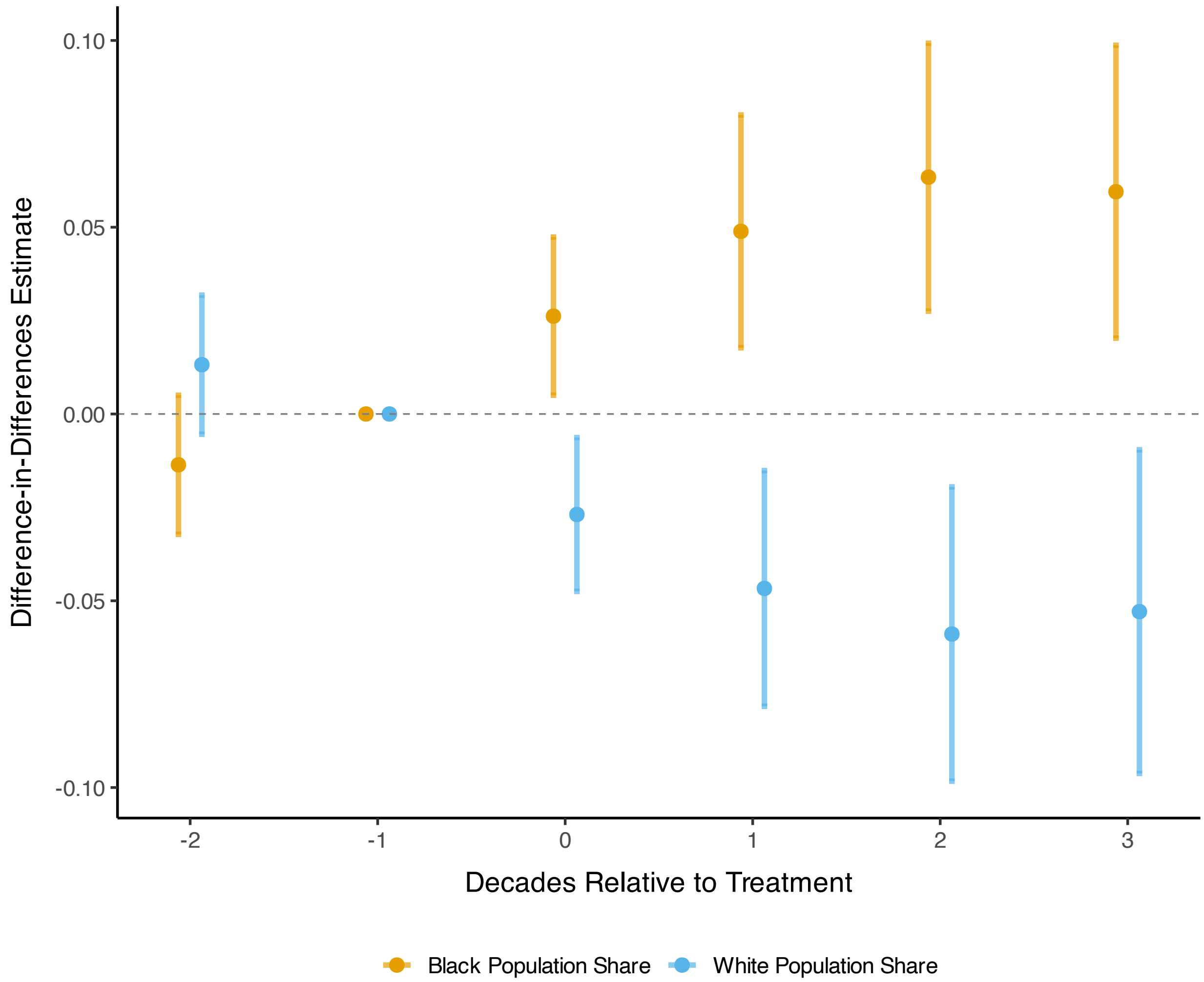
Note: This figure displays event study estimates of public housing effects on population by race in treated neighborhoods (N=799) compared to matched controls, estimated using the stacked difference-in-differences specification in Equation 3. Outcomes are the inverse hyperbolic sine (asinh) of total population, Black population, and white population. Shaded areas represent 95% confidence intervals based on standard errors adjusted for spatial correlation following Conley (1999) with a 2-kilometer cutoff. The reference period is $t = -1$ (one decade before construction). See Section 6.2 for interpretation.

Figure 7: Effects on Estimated Private Population, Treated Neighborhoods



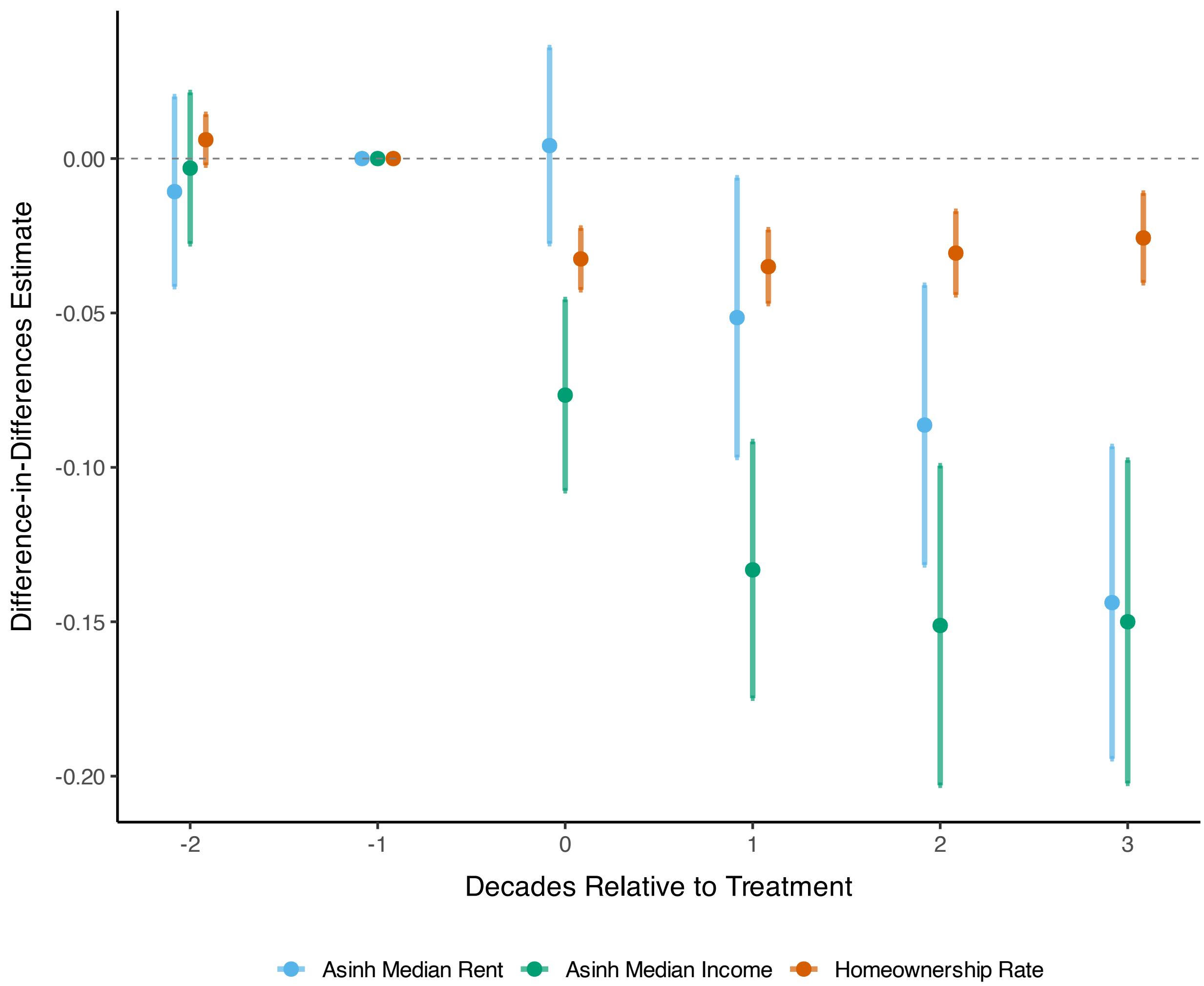
Note: This figure shows event study estimates of public housing effects on the asinh of private (non-public housing) population by race in treated neighborhoods (N=799) compared to matched controls. Private population is estimated by subtracting public housing residents from total tract population; public housing population from 1970s administrative data is set to zero before construction and held constant thereafter. Estimates follow Equation 3. Shaded areas represent 95% confidence intervals with Conley standard errors (2km cutoff). Reference period is $t = -1$. See Section 6.2 for interpretation.

Figure 8: Effects on Racial Composition Shares in Treated Neighborhoods



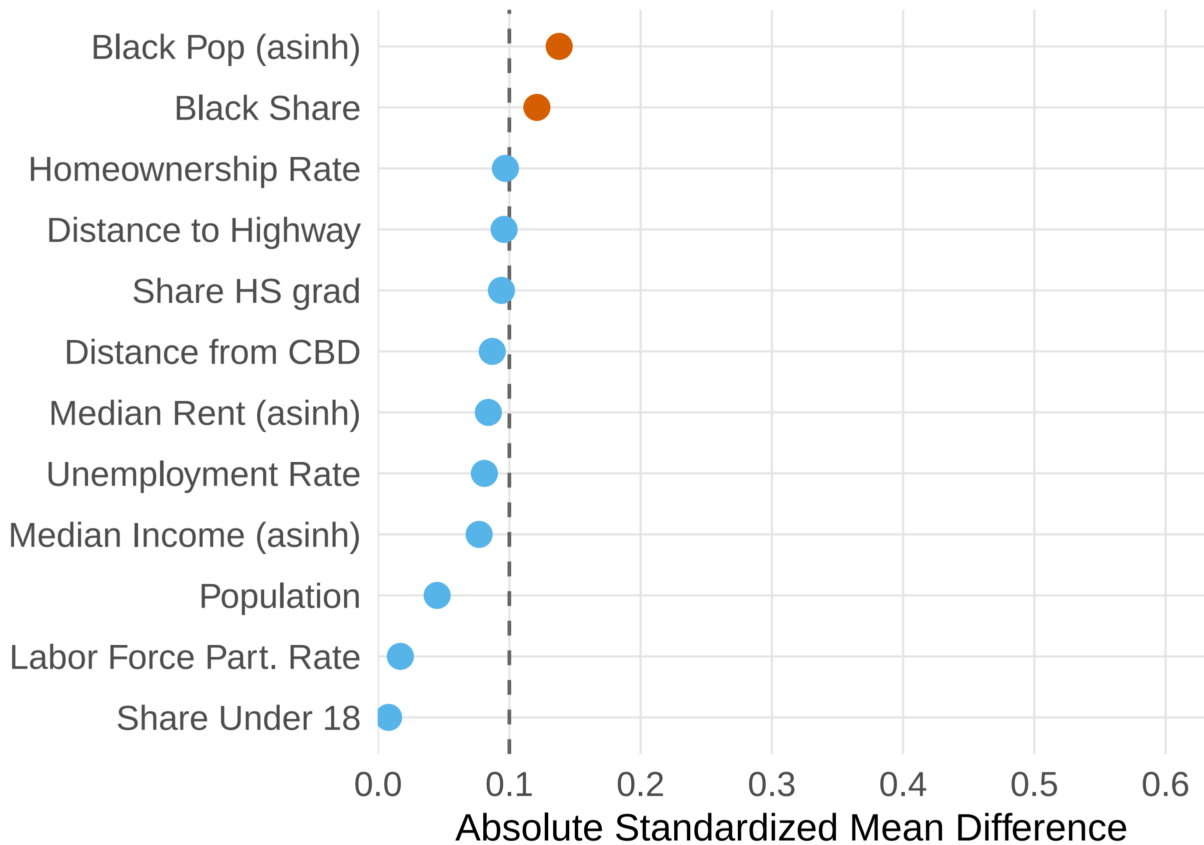
Note: This figure shows event study estimates of public housing effects on racial composition shares in treated neighborhoods (N=799) compared to matched controls, estimated using Equation 3. Coefficients represent percentage point changes in Black and white population shares. Shaded areas represent 95% confidence intervals with Conley standard errors (2km cutoff). Reference period is $t = -1$. See Section 6.2 for interpretation.

Figure 9: Economic and Housing Effects in Treated Neighborhoods



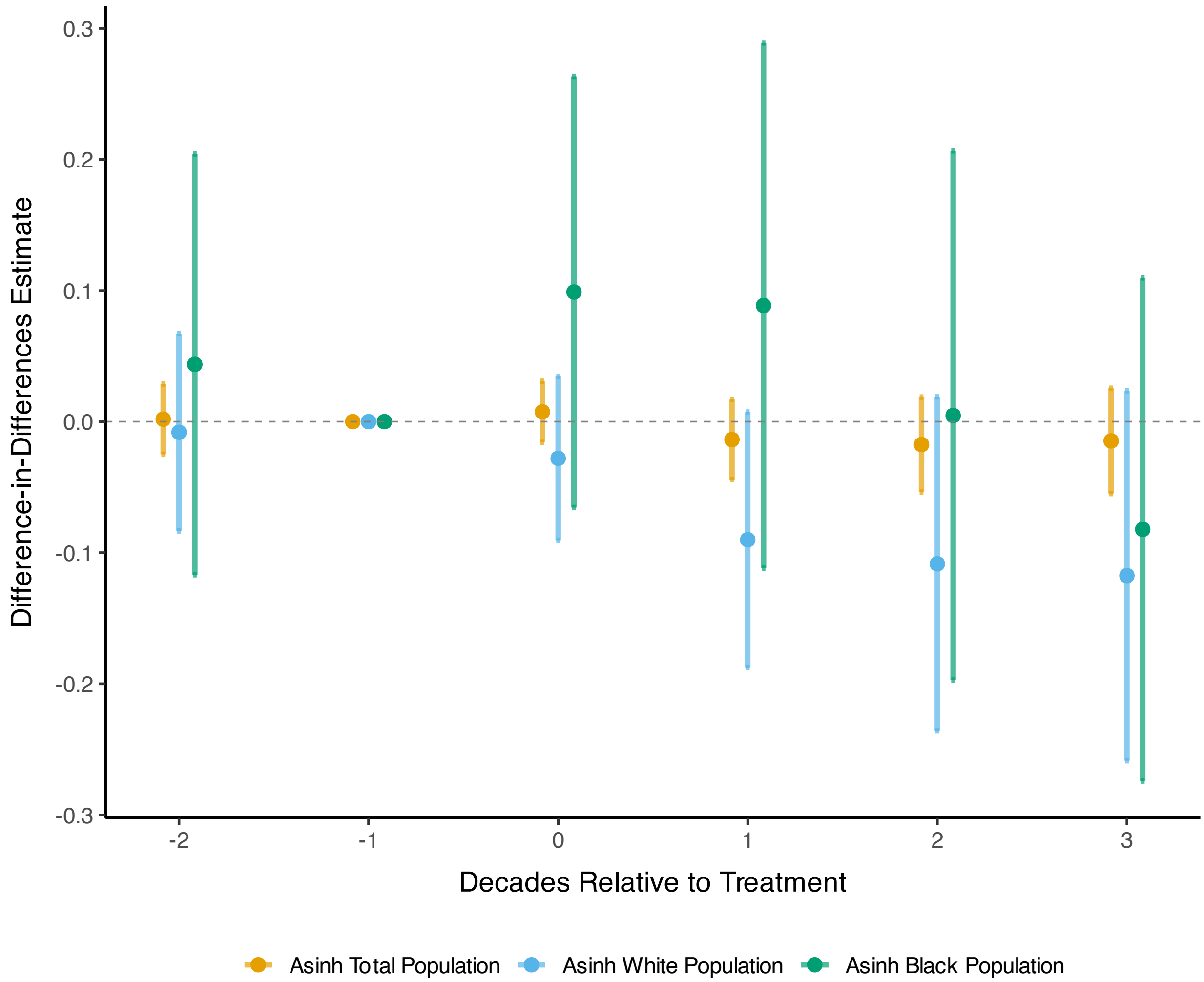
Note: This figure displays event study estimates of public housing effects on the asinh of median rent, the asinh of median household income, and the homeownership rate in treated neighborhoods (N=799) compared to matched controls, estimated using Equation 3. Monetary values are in 2000 dollars. Shaded areas represent 95% confidence intervals with Conley standard errors (2km cutoff). Reference period is $t = -1$. See Section 6.2 for interpretation.

Figure 10: Pre-period Balance: Nearby Neighborhoods vs Matched Controls



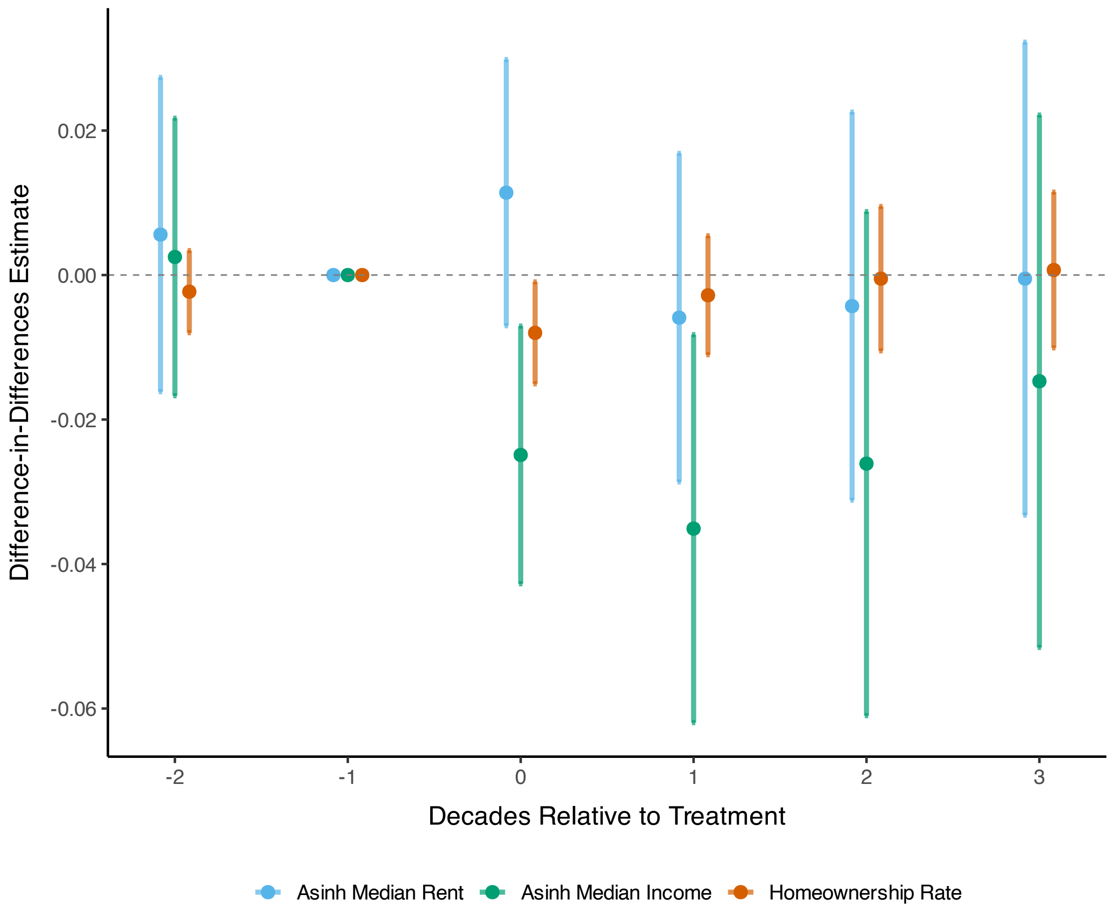
Note: This figure displays standardized mean differences (SMD) between nearby neighborhoods (N=1,218; those sharing a border with treated tracts and within 1km of the nearest project) and their matched controls across baseline covariates. Each point represents the SMD for a covariate measured in the two pre-treatment decades. SMDs near zero indicate successful balance; values above 0.1 (vertical dashed lines) suggest meaningful imbalance. See Section 6.1 for matching procedure.

Figure 11: Spillover Effects: Population and Racial Composition in Nearby Neighborhoods



Note: This figure displays event study estimates of public housing spillover effects on population by race in nearby neighborhoods (N=1,218; those sharing a border with treated tracts and within 1km of projects) compared to matched controls, estimated using Equation 3. Outcomes are the asinh of total, Black, and white population. Shaded areas represent 95% confidence intervals with Conley standard errors (2km cutoff). Reference period is $t = -1$. See Section 6.3 for interpretation.

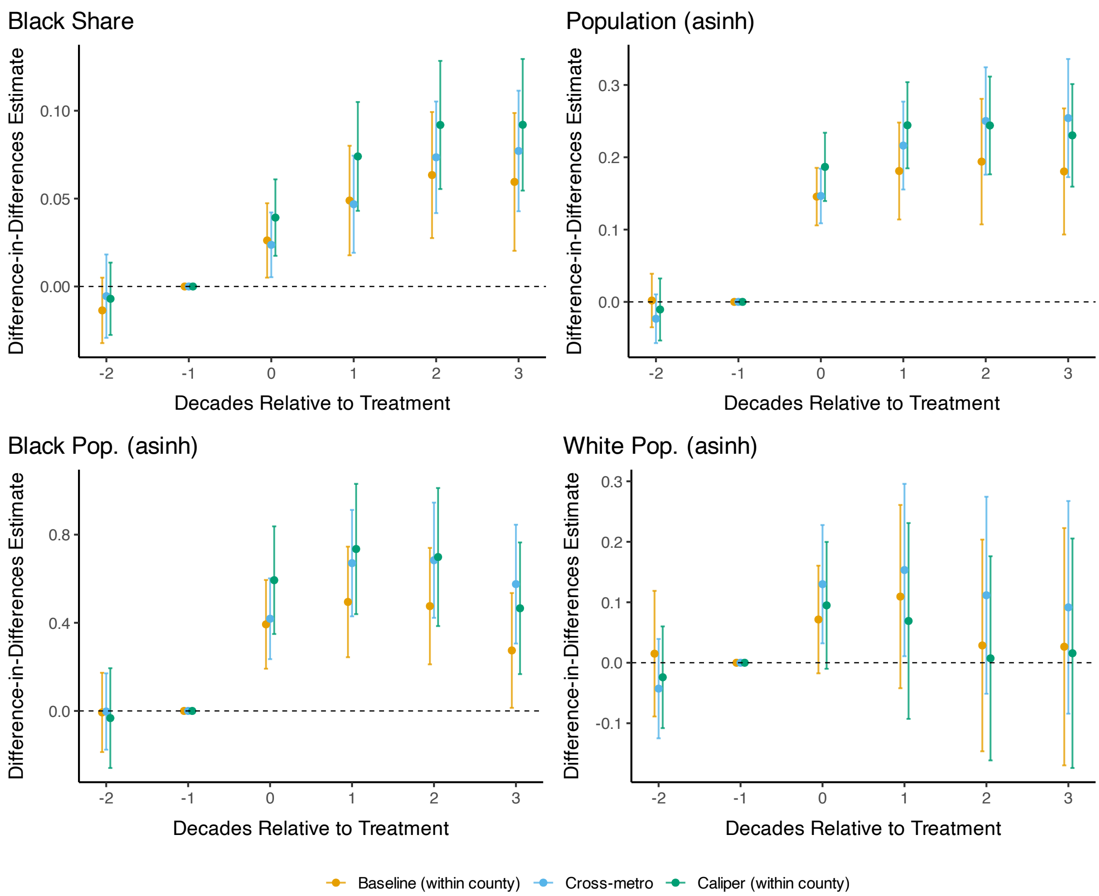
Figure 12: Spillover Effects: Economic and Housing Outcomes in Nearby Neighborhoods



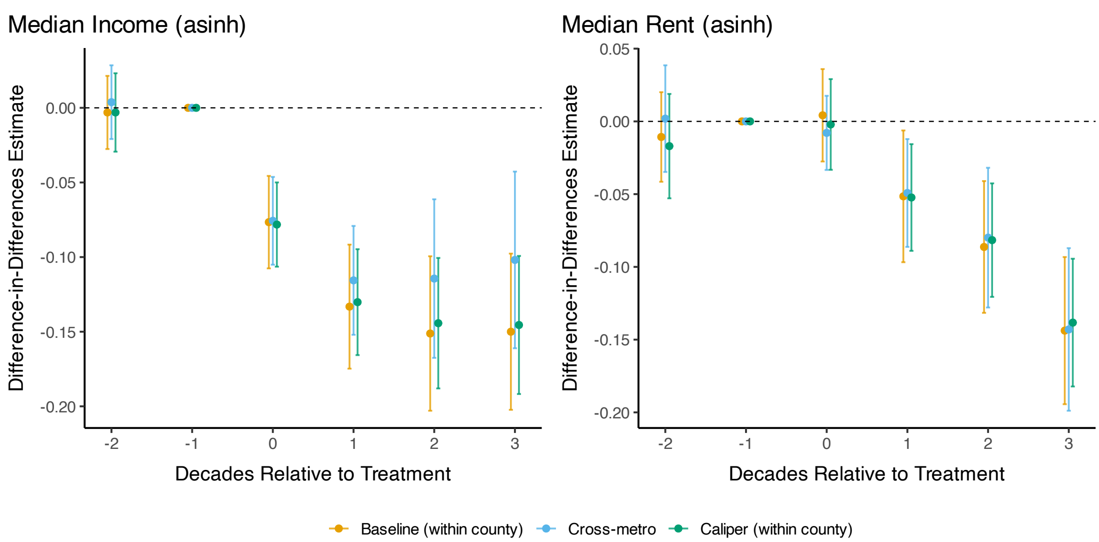
Note: This figure displays event study estimates of public housing spillover effects on the asinh of median rent, the asinh of median household income, and the homeownership rate in nearby neighborhoods (N=1,218) compared to matched controls, estimated using Equation 3. Monetary values are in 2000 dollars. Shaded areas represent 95% confidence intervals with Conley standard errors (2km cutoff). Reference period is $t = -1$. See Section 6.3 for interpretation.

Figure 13: Alternative Matching Specifications: Treated neighborhoods

(a) Demographics



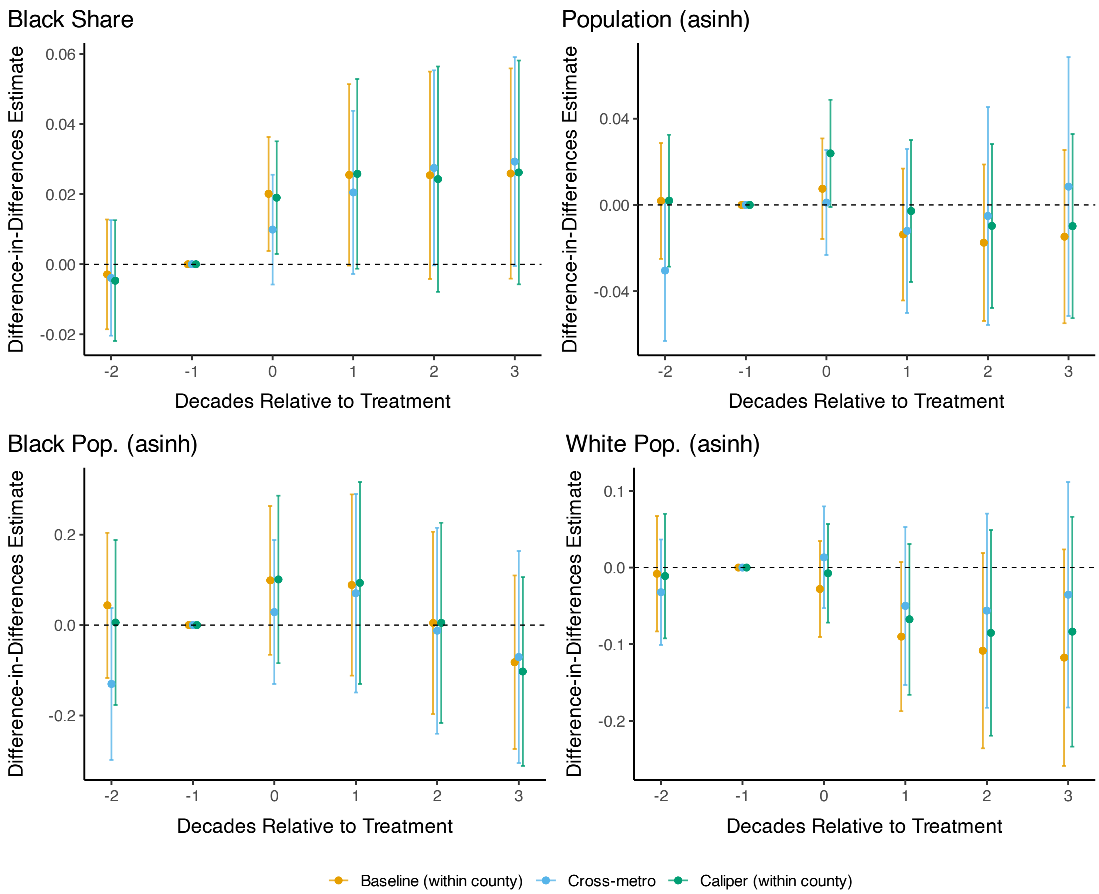
(b) Income & Rent



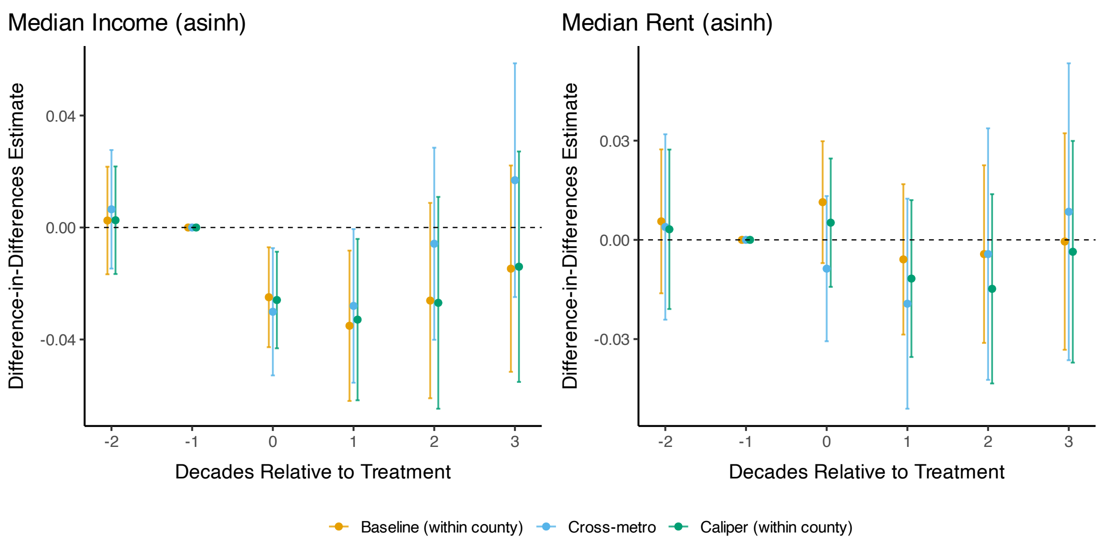
Note: Event study estimates for treated neighborhoods across three matching specifications: (1) Baseline: exact match on county and urban renewal status, N=799; (2) Cross-metro: controls drawn from other CBSAs, exact on urban renewal, N=815; (3) Caliper: baseline plus 0.25 SD propensity score caliper, N=521. All specifications use 1:1 nearest-neighbor matching with replacement. Reference period: $t = -1$. Shaded areas show 95% confidence intervals with Conley standard errors (2km cutoff). See Section 6.4 for discussion.

Figure 14: Alternative Matching Specifications: Nearby neighborhoods

(a) Demographics



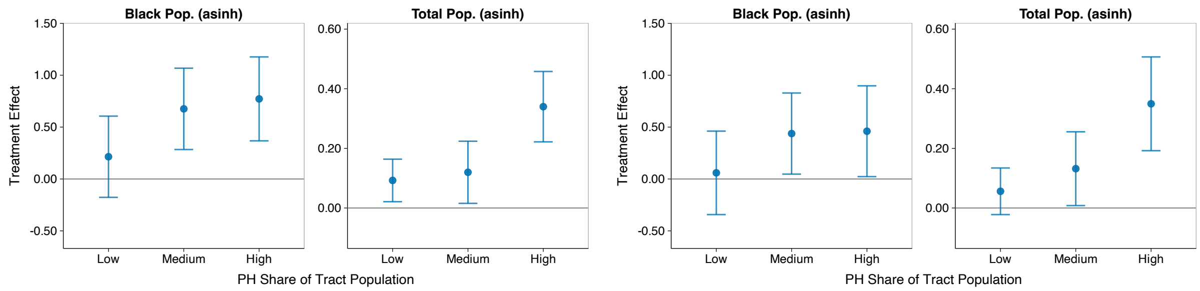
(b) Income & Rent



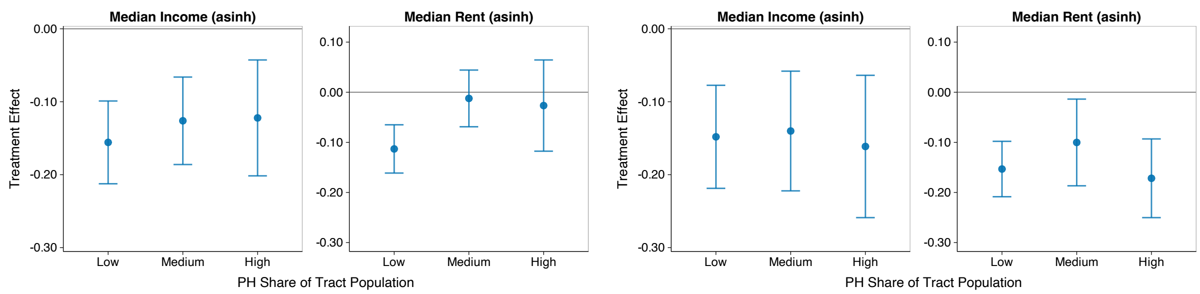
Note: Event study estimates for nearby neighborhoods (N=1,218; share border with treated tracts and within 1km of projects) across three matching specifications: (1) Baseline: exact match on county and urban renewal; (2) Cross-metro: controls from other CBSAs; (3) Caliper: baseline plus 0.25 SD propensity score caliper. All specifications use 1:1 nearest-neighbor matching with replacement. Reference period: $t = -1$. Shaded areas show 95% confidence intervals with Conley standard errors (2km cutoff). See Section 6.4 for discussion.

Figure 15: Heterogeneity by Public Housing Population Share

(a) Population



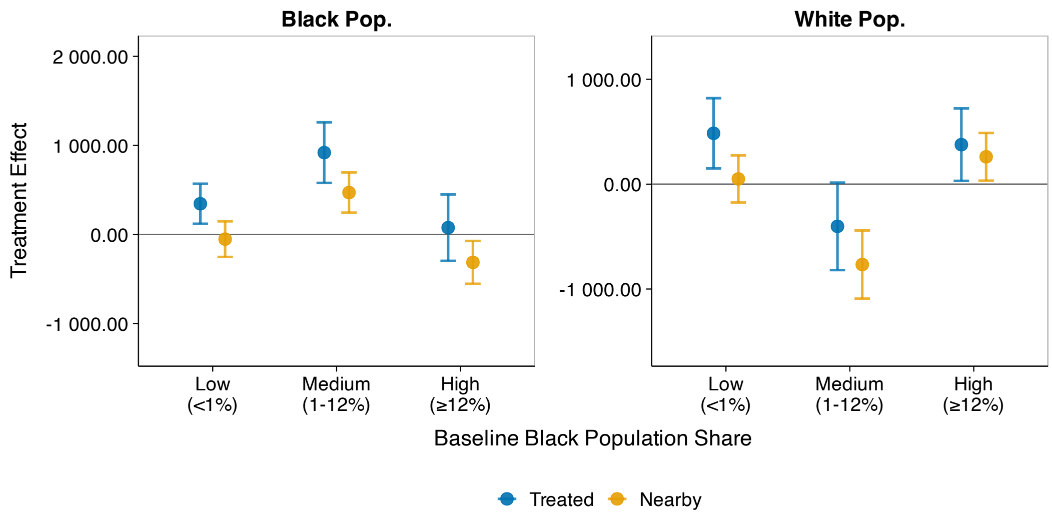
(b) Income and Rent



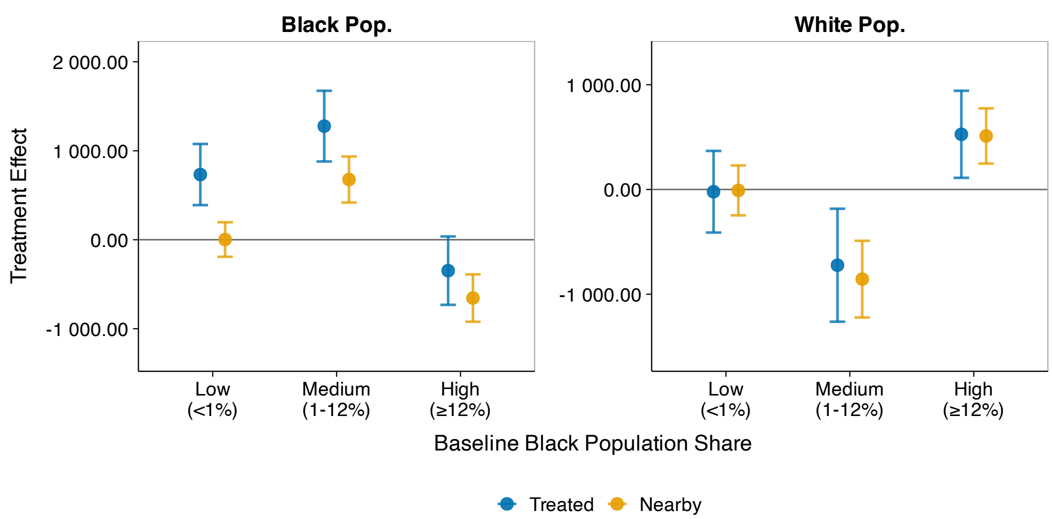
Note: This figure displays heterogeneous treatment effects by the share of tract population living in public housing at $t = 0$, testing whether effects are purely mechanical. Treated neighborhoods (N=799) are divided into terciles. Estimates follow Equation 3. Left panels show effects at $t = 1$ (one decade after construction); right panels at $t = 3$ (three decades after). For income and rent, similar magnitudes across terciles provide suggestive evidence against purely mechanical effects. Bars show 95% confidence intervals with Conley standard errors (2km cutoff). See Section 7 for interpretation.

Figure 16: Heterogeneity by Baseline Black Population Share

(a) Medium-run Effects ($t = 1$)



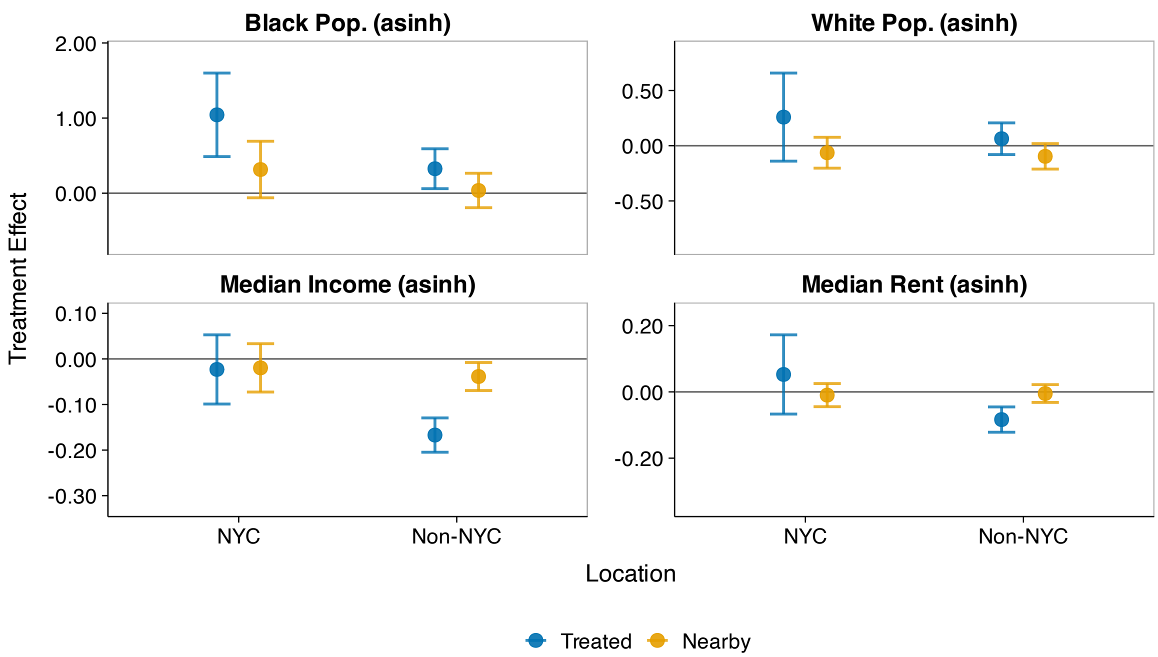
(b) Long-run Effects ($t = 3$)



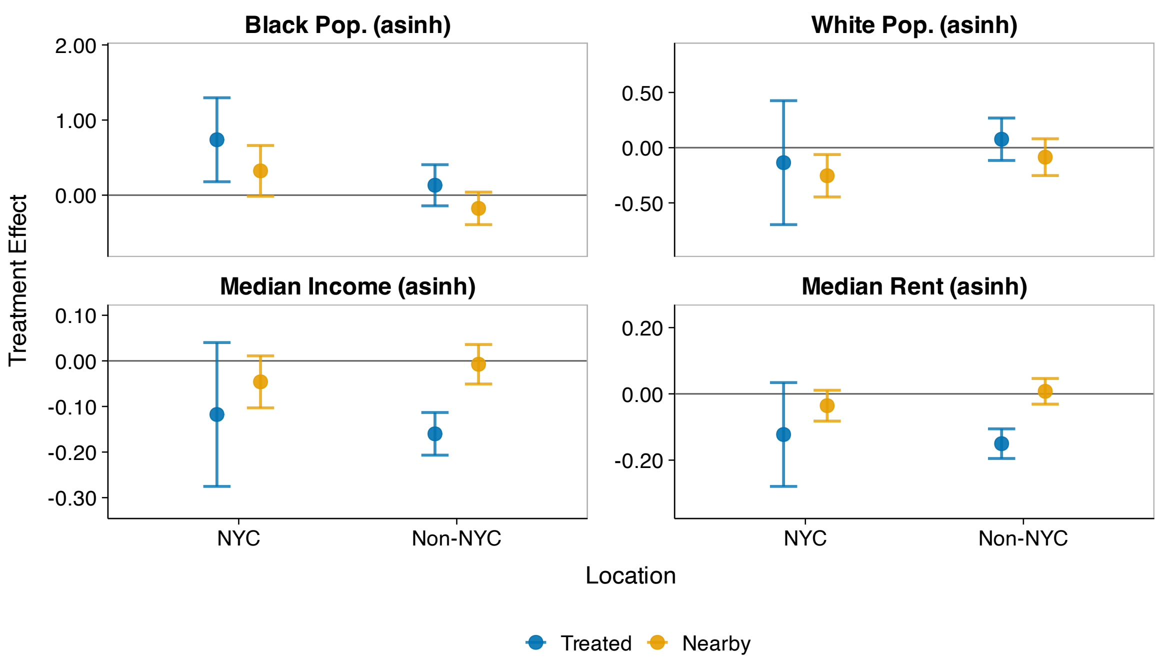
Note: This figure displays heterogeneous treatment effects by baseline Black population share, testing for neighborhood tipping dynamics. Treated neighborhoods are divided into three groups: very low (<1%, N=257), medium baseline Black share (1–12%, N=186), and high (≥12%, N=356). These bins distinguish nearly all-white neighborhoods, neighborhoods with low but nontrivial Black populations, and neighborhoods with higher baseline Black shares. Estimates follow Equation 3. Panel (a) shows effects at $t = 1$; panel (b) at $t = 3$. Nearby neighborhoods classified by their own baseline Black share. Population outcomes use raw counts (not asinh). Bars show 95% confidence intervals with Conley standard errors (2km cutoff). See Section 7.2 for interpretation.

Figure 17: New York City vs Other Cities

(a) Medium-Run ($t = 1$)



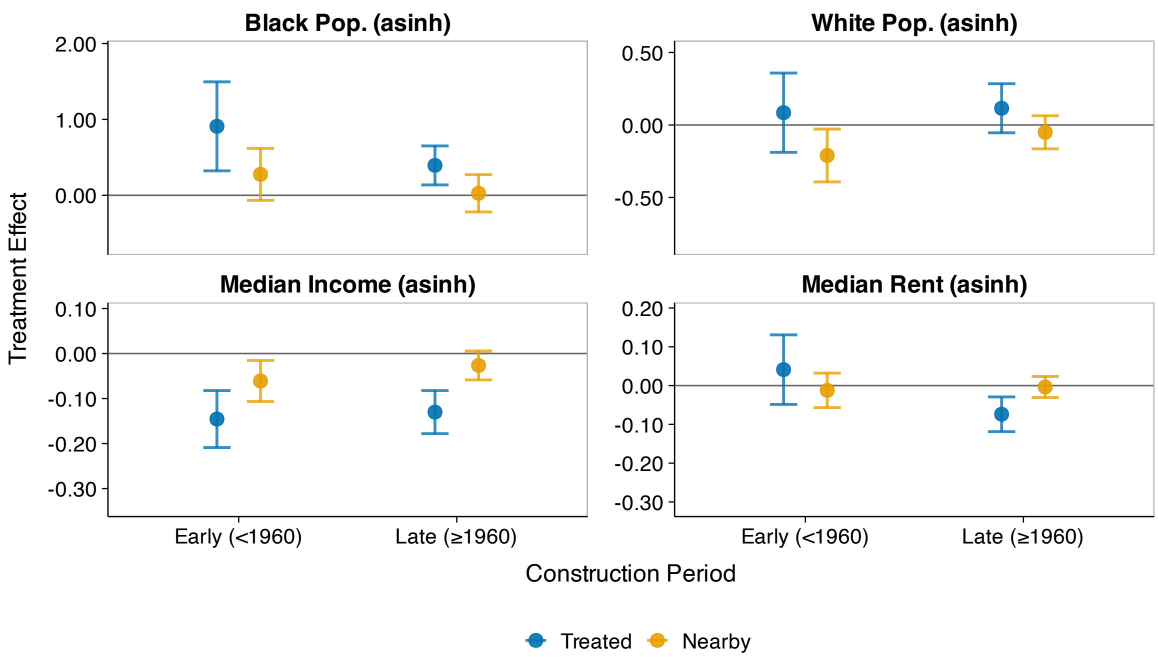
(b) Long-Run ($t = 3$)



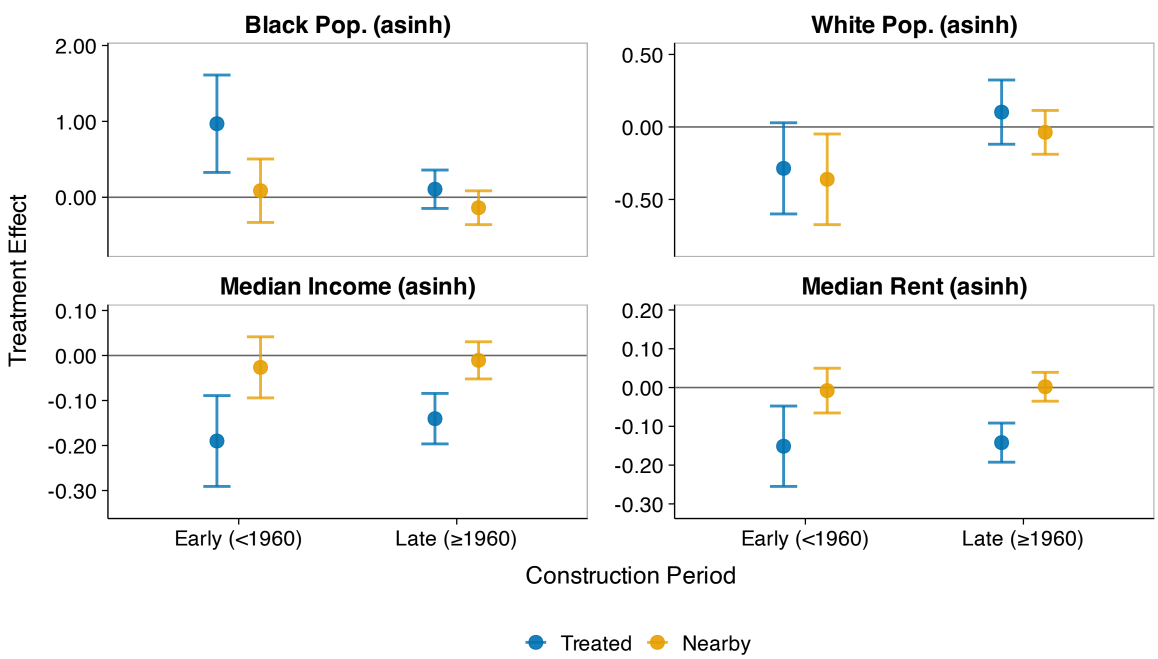
Note: This figure displays heterogeneous treatment effects comparing New York City to other cities in the sample. NYCHA was the largest U.S. housing authority and is often considered better-managed than other PHAs (Bloom 2008). Estimates follow Equation 3. Panel (a) shows effects at $t = 1$; panel (b) at $t = 3$. Population, income, and rent outcomes use asinh transformation. Bars show 95% confidence intervals with Conley standard errors (2km cutoff). See Section 7.3 for interpretation.

Figure 18: Heterogeneity by Construction Decade

(a) Medium-Run ($t = 1$)



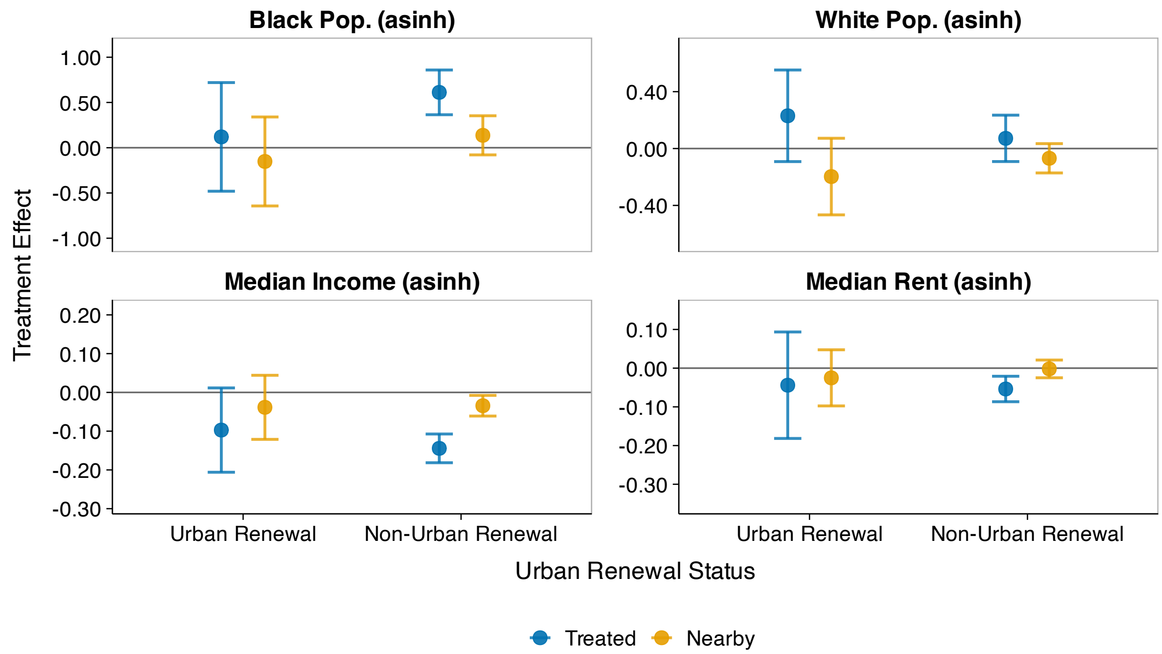
(b) Long-Run ($t = 3$)



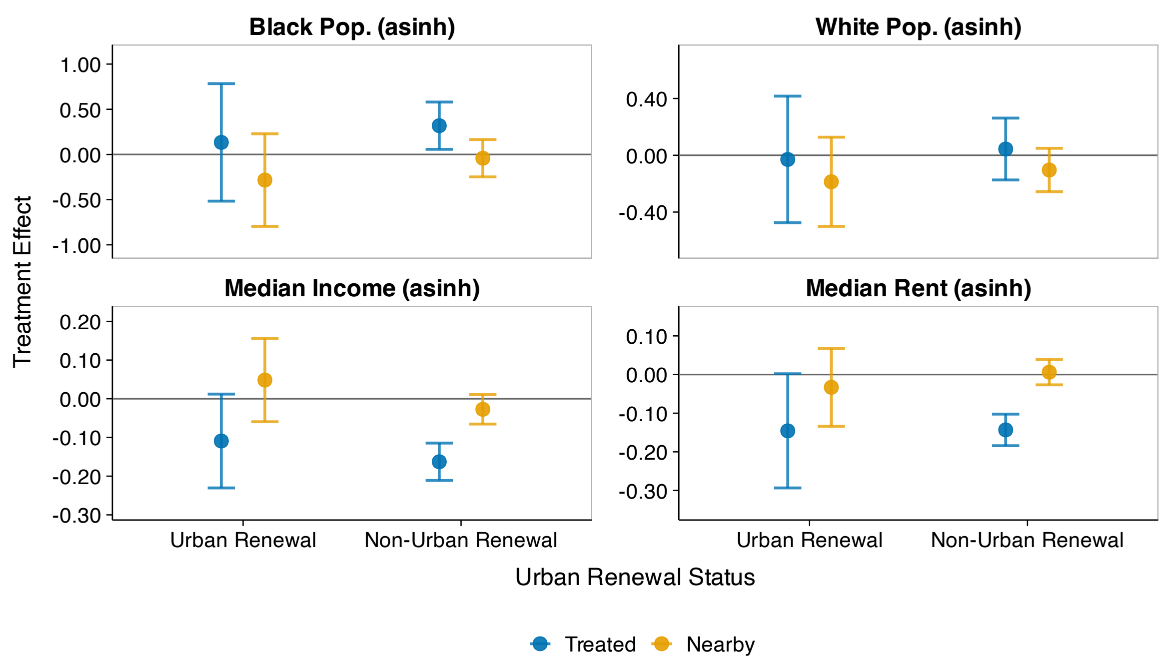
Note: This figure displays heterogeneous treatment effects by construction timing, comparing projects built before 1960 to those built 1960 or later. Early-period projects may have had different effects due to lower baseline Black population shares in urban neighborhoods during the 1940s–1950s. Estimates follow Equation 3. Panel (a) shows effects at $t = 1$; panel (b) at $t = 3$. Population, income, and rent outcomes use asinh transformation. Bars show 95% confidence intervals with Conley standard errors (2km cutoff). See Section 7.3 for interpretation.

Figure 19: Heterogeneity by Urban Renewal Status

(a) Medium-Run ($t = 1$)



(b) Long-Run ($t = 3$)



Note: This figure displays heterogeneous treatment effects by urban renewal status, comparing tracts with >5% overlap with urban renewal project boundaries to non-urban renewal tracts. This analysis addresses potential confounding between the two programs and tests whether effects differ where both operated. Estimates follow Equation 3. Panel (a) shows effects at $t = 1$; panel (b) at $t = 3$. Population, income, and rent outcomes use asinh transformation. Bars show 95% confidence intervals with Conley standard errors (2km cutoff). See Section 7.3 for interpretation.

Table 6: Opportunity Atlas Outcomes: Public Housing Tracts vs Matched Controls

	Mobility		Incarceration	
	(1)	(2)	(3)	(4)
Treated	-0.018***	-0.004**	0.006***	0.002**
	(0.002)	(0.002)	(0.001)	(0.001)
Num.Obs.	1540	1540	1540	1540
R2	0.758	0.877	0.660	0.736

Note: This table reports matched-pair comparisons of Opportunity Atlas outcomes for children who grew up in tracts containing public housing and children in matched control tracts. OLS estimates compare Opportunity Atlas outcomes in public housing tracts (N=799) to matched controls. Columns (1)–(2): mean income rank in adulthood (2014–2015) for children from families at the 25th percentile of the national income distribution, born 1978–1983. Columns (3)–(4): incarceration rate as of April 1, 2010. Odd columns show unconditional differences; even columns add 1980 neighborhood controls (Black share, median income, population, unemployment, rent). All specifications include matched-pair fixed effects. Standard errors clustered at the matched-pair level in parentheses. Statistical significance: *p<0.1, **p<0.05, ***p<0.01.

Table 7: Opportunity Atlas Outcomes: Nearby Tracts vs Matched Controls

	Mobility		Incarceration	
	(1)	(2)	(3)	(4)
Nearby	-0.006***	-0.001	0.002**	0.001
	(0.002)	(0.001)	(0.001)	(0.001)
Num.Obs.	2756	2756	2756	2756
R2	0.756	0.871	0.680	0.755

Note: This table reports matched-pair comparisons of Opportunity Atlas outcomes for children who grew up in tracts within 1km of public housing and children in matched control tracts. OLS estimates compare Opportunity Atlas outcomes in nearby tracts (N=1,218; within 1km of public housing) to matched controls. Columns (1)–(2): mean income rank in adulthood (2014–2015) for children from families at the 25th percentile of the national income distribution, born 1978–1983. Columns (3)–(4): incarceration rate as of April 1, 2010. Odd columns show unconditional differences; even columns add 1980 neighborhood controls (Black share, median income, population, unemployment, rent). All specifications include matched-pair fixed effects. Standard errors clustered at the matched-pair level in parentheses. Statistical significance: *p<0.1, **p<0.05, ***p<0.01.

A Public Housing Data Construction

A.1 Estimating Public Housing Project Populations

This section describes the methodology for estimating the population of each public housing project using the 1977 Picture of Subsidized Households (PIC77) dataset. The PIC77 dataset contains household-level demographic information for public housing residents, including the number of households in the project, racial composition, household size, income, and age structure. However, it does not provide direct population counts.

To estimate total, white, and Black populations for each project p, I multiply the number of households for group r by the average household size:

$$\text{Population}_{pr} = \text{Households}_{pr} \times \text{Average Household Size}_{p}.$$

One challenge is that approximately 42% of projects lack household size data. To address this, I impute missing values using two categorical variables that explain substantial variation in household composition: (i) *elderly designation* (all elderly, some elderly, or none) and (ii) *racial composition* (simplified into six categories: all white, all Black, all other race, mixed, no white, and other). For each elderly designation × racial composition cell, I calculate the mean household size across all projects with non-missing data. Missing household size values are then imputed using the corresponding cell mean. This imputation strategy leverages the fact that elderly projects have systematically smaller household sizes, and household size varies by the racial composition of the project.

This approach assumes that average household size is constant across racial groups within each project. While household size may vary by race in the general population, this assumption is reasonable within public housing projects where unit size allocations are determined by family size rather than race, and eligibility criteria are applied uniformly across racial groups.

To validate the imputation strategy, I conducted a hold-out test where 20% of projects with observed household size were randomly set to missing and then imputed using the cell means calculated from the remaining 80%. The imputation has an R^2 of 0.46 and is essentially unbiased, with a mean error of -0.01 persons.

B Census Tract Data Construction and Harmonization

B.1 Data Sources and Variable Availability

I construct tract-level neighborhood characteristics from two primary sources: published Census tabulations from NHGIS (Manson et al. 2022) and full-count Census microdata from IPUMS (Ruggles et al. 2022). The two sources play complementary roles.

For 1950 onward, NHGIS provides comprehensive tract-level tabulations for all tracted areas, including population by race, housing characteristics, and socioeconomic measures. I use these tabulations as the primary data source for these decades.

For 1930 and 1940, NHGIS tract coverage is limited to select cities. I therefore supplement NHGIS data with tract-level aggregates constructed from full-count Census microdata, using enumeration district to tract crosswalks from the Urban Transition Historical GIS Project (J. R. Logan et al. 2024). For tracts where both sources are available, I use the NHGIS tabulations; for tracts covered only in the full-count data, I use the microdata-based estimates. This approach maximizes geographic coverage while maintaining consistency with published Census figures where available.

Table 8 summarizes the data format for key variables by decade. The format determines how I calculate tract-level values when harmonizing to 1950 boundaries (see Section B.2).

Table 8: Census Tract Variable Data Formats by Decade

Variable	1930	1940	1950	1960	1970	1980	1990	2000	2010
Population by race	Count[a]	Count[a]	Count	Count	Count	Count	Count	Count	Count
Median income	FC	FC	Dist.	Dist.	Dist.	Rep.	Rep.	Rep.	Rep.
Median rent	Dist.[a]	Rep.	Rep.	Dist.	Dist.	Rep.	Rep.	Rep.	Rep.
Median home value	Dist.[a]	Rep.	Rep.	Dist.	Dist.	Rep.	Rep.	Rep.	Rep.
Age structure	FC	Count[a]	Count	Count	Count	Count	Count	Count	Count
Homeownership	FC	Count[a]	Count	Count	Count	Count	Count	Count	Count
Employment/LFP	FC	FC	Count	Count	Count	Count	Count	Count	Count
Education	—	FC	Count	Count	Count	Count	Count	Count	Count

Note: "Count" = raw counts from NHGIS (Manson et al. 2022). "Rep." = reported median from NHGIS. "Dist." = median calculated from frequency distributions. "FC" = constructed from full-count Census microdata (Ruggles et al. 2022). [a]NHGIS coverage limited to select cities; full-count microdata used for remaining cities.

The Census Bureau's reporting practices for median variables changed substantially over time. Median family income was first reported at the tract level in 1980; earlier decades provide only frequency distributions (1950–1970) or require construction from microdata (1930–1940).

For housing variables, the Census reported medians in 1940, 1950, and 1980 onward, but provided only distributions in 1960 and 1970.

For variables reported as frequency distributions, I calculate medians using midpoint interpolation (described below). For all economic variables, I construct a composite measure that uses reported medians when available and calculates medians from distributions otherwise, ensuring consistent coverage across decades.

B.1.1 Calculating Medians from Categorical Distributions

For census years that report only frequency distributions across income, rent, or home value ranges, I calculate medians using a midpoint interpolation method. For each tract, I first calculate cumulative household counts across the income, rent, or value ranges reported by the Census Bureau. I then identify the first range where the cumulative frequency exceeds 50% of total households. Finally, I assign the midpoint of that range as the median value.

Formally, let n_i denote the number of households in range i, with bounds $[L_i, U_i]$. The median is calculated as:

$$\text{Median} = \frac{L_m + U_m}{2} \tag{5}$$

where m is the first range satisfying $\sum_{i=1}^{m} n_i \geq \frac{1}{2}\sum_{i=1}^{N} n_i$. For top-coded categories (e.g., "$25,000 or more"), I use the lower bound since the upper bound is undefined.

For example, the 1960 Census reports median income as 13 frequency bins: <$1,000, $1,000-$2,000, $2,000-$3,000, ..., $25,000+. If a tract has 1,000 families and the cumulative count first exceeds 500 at the $5,000-$7,500 range, I assign median income as $6,250. Similarly, for 1970 median rent, the Census provides 15 bins ranging from <$20 to $500+, and the same midpoint interpolation method is applied.

When crosswalking census tracts across decades to consistent 1950 boundaries (see Section B.2 for details), the methodology differs depending on whether medians are reported or calculated. For years with reported medians (income: 1980+; rent/value: 1940, 1950, 1980+), I use population-weighted averaging of median values when aggregating across source tracts. For years with distributions (income: 1950-1970; rent/value: 1930, 1960, 1970), I reallocate the count of households in each income or rent range using area weights, sum across source tracts, and then calculate the median from the resulting distribution. This distinction is critical because medians are intensive variables that should be averaged rather than summed, as explained in Section B.2.

All monetary values are converted to real 2000 dollars using the Consumer Price Index (CPI-U) from Officer and Williamson (2025). The adjustment is applied after calculating medians (for distribution years) or using reported medians (for other years), but before crosswalking to 1950 tract boundaries.

B.2 Tract Boundary Harmonization

This section describes the geographic harmonization of Census tract-level data to consistent boundaries. My baseline analysis harmonizes all tract-level data from 1930 to 2010 to 1950 Census tract boundaries using an area reweighting approach following Eckert et al. (2020).

I proceed as follows. First, for each source year, I spatially intersect census tract shapefiles with 1950 tract boundaries to identify overlapping areas. Next, for each source tract i, I calculate weights as the proportion of the tract's area that falls within each 1950 tract j:

$$w_{ij} = \frac{\text{Area}(\text{Tract}_i \cap \text{Tract}_{1950,j})}{\text{Area}(\text{Tract}_i)}$$

Weights are normalized to sum to one for each source tract to account for minor geometric inconsistencies.

Finally, I use these weights to estimate count and median variables in the 1950 tract geography. This proceeds slightly differently for count-based variables and median-based variables.

For each 1950 tract j, count variables such as population, housing units, and employment are harmonized by simply summing the weighted contributions from all overlapping source tracts:

$$X_j = \sum_{i \in I_j} w_{ij} \cdot X_i$$

where X_i represents the count variable in source tract i and I_j is the set of all source tracts overlapping with 1950 target tract j. This method assumes that outcome variables are uniformly distributed within each source tract, which may introduce measurement error in areas where populations are spatially concentrated within tract boundaries. For instance, if a source tract contains a spatially concentrated Black population but overlaps evenly with multiple 1950 tracts, the method would allocate residents uniformly across those tracts, potentially attenuating spatial estimates of segregation.

For median variables such as median income, median rent, and median home value, tract-level estimates are calculated as weighted averages across source tracts:

$$\tilde{X}_j = \frac{\sum_{i \in I_j} w_{ij} \cdot \tilde{X}_i}{\sum_{i \in I_j} w_{ij}}$$

The weights w_{ij} are based solely on area overlap. This follows from the uniform-within-tract assumption described above: under the assumption that population (and the relevant subpopulations for each median) are uniformly distributed within each source tract, the share of a tract's population falling into a target tract is proportional to the area share w_{ij}.[12] This weighted-average approach is mathematically equivalent to using target-tract area shares, as recommended by Eckert et al. (2020) for intensive variables, because the denominator renormalizes contributions to sum to one for each target tract. This approach is applied to median family income (1980, 1990, 2000), median contract rent (1940, 1980, 1990, 2000), and median home value (1940, 1980, 1990, 2000).

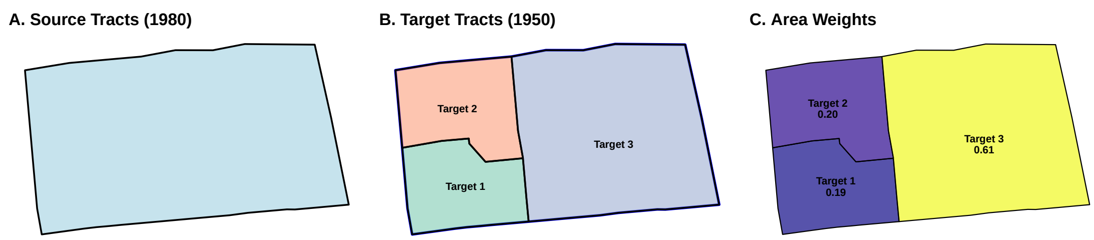
C Alternative Geographic Counterfactuals

This appendix evaluates alternative geographic counterfactuals for the matched difference-in-differences design. These specifications answer a natural question: what happens if I define the comparison group using proximity to public housing projects rather than matching on baseline neighborhood characteristics? I estimate two versions. The first uses tract contiguity to define treated, adjacent, and outer-ring neighborhoods. The second uses fixed distance bands around each project. Both designs produce post-treatment patterns that are broadly similar to the matched estimates, but both exhibit meaningful pre-treatment differences. I therefore treat them as diagnostic stress tests rather than as preferred estimates.

The designs rest on a different identifying assumption than the matched difference-in-differences. The main analysis assumes that matched neighborhoods provide valid counterfactuals conditional on observed pre-treatment characteristics. The geographic designs instead assume that nearby neighborhoods would have evolved similarly absent public housing construction. This assumption is attractive because the comparisons are local, but it is demanding in this setting because public housing sites were selected within cities in ways that are correlated with neighborhood trajectories.

[12] Population weighting would require sub-tract spatial information on these subpopulations (renter households for median rent, families for median income), which is unavailable.

(a) Split Pattern: One 1980 tract → Multiple 1950 tracts



(b) Merge Pattern: Multiple 1980 tracts → One 1950 tract

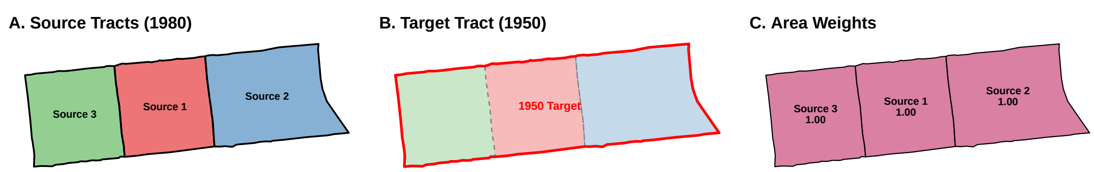


Figure 20: Census Tract Harmonization Examples

Note: These figures demonstrate the area-reweighting methodology for harmonizing 1980 census tracts to 1950 boundaries in Chicago. Panel (a) shows a split pattern where one 1980 source tract is divided across multiple 1950 target tracts, with weights representing the proportion of area allocated to each target. Panel (b) shows a merge pattern where multiple 1980 source tracts contribute to a single 1950 target tract.

C.1 Research Design

Contiguity-ring design. The first specification uses census tract adjacency to construct three spatial units for each project: treated tracts that contain public housing, inner-ring tracts that are first-order adjacent neighbors, and outer-ring tracts that are second-order adjacent neighbors. I construct these rings using tract contiguity rather than simple distance bands, as administrative boundaries better capture neighborhood economic geography. To address irregularly shaped tracts, I impose distance filters: inner-ring tracts must be within 1km of the project, and outer-ring tracts must be within 2km. The median distances in the final sample are 407 meters for inner-ring tracts and 1,076 meters for outer-ring tracts.

Distance-band design. The second specification uses fixed distance bands around each project. To align the estimand with the main analysis, tracts within 0–100 meters of a project form the treated ring, matching the 100-meter buffer used in the baseline treatment definition. Tracts 100–800 meters away form the nearby ring, and tracts 800–1500 meters away form the control ring. This design is closer to the simple concentric-ring approach often used in place-based-policy settings. It is useful because it asks whether the results survive a more mechanical geographic definition of the counterfactual group.

Contamination cleaning. In both spatial designs, I remove contaminated controls. A tract that ever receives public housing is excluded from serving as a control for another project, and a tract that appears in a nearer ring is excluded from farther control rings. This keeps the comparison groups geographically nested around each project.

Econometric specification. For tract i associated with project p in year t, I estimate stacked event studies of the form

$$y_{ipt} = \alpha + \sum_{r\in R}\sum_{\tau\neq -1} \beta_{r\tau} \mathbf{1}[\text{Ring}_{ip} = r] \times \mathbf{1}[\text{EventTime}_p = \tau] + \text{FEs} + \varepsilon_{ipt}, \tag{6}$$

where event time is measured in decades and the omitted category is the control ring at $t = -1$. In the contiguity-ring design, R includes treated and inner-ring tracts, with outer-ring tracts as the control group. In the distance-band design, R includes the 0–100m and 100–800m rings, with the 800–1500m ring as the control group.

The contiguity-ring specification uses the conservative fixed effects structure from Wing, Freedman, and Hollingsworth (2024): project-by-year fixed effects, project-by-ring fixed effects, baseline-outcome-decile-by-project-by-year fixed effects, and project-by-policy-by-year interactions for urban renewal, highway proximity, and redlining. The distance-band specification uses the same stacked event-study logic with project-by-year and project-by-ring fixed effects, along with project-by-policy-by-year controls. Standard errors are clustered at the project level.

Sample. The contiguity-ring sample includes 816 public housing projects across 45 metropolitan areas, with treatment cohorts in 1950, 1960, 1970, and 1980, observed from 1930 to 1990. Each project contributes a separate geographic comparison.

C.2 Balance and Pre-trends

A key limitation of geographic counterfactuals is that proximity does not guarantee comparability. Table 9 compares standardized mean differences between treated tracts and controls for the contiguity-ring spatial design and the matched difference-in-differences design at baseline.

Table 9: Treated-Tract Balance Comparison: Spatial DiD vs Matched DiD

Variable	Spatial DiD	Matched DiD
Total Population (asinh)	0.298	0.042
Black Share	0.277	0.151
Black Population (asinh)	0.411	0.199
Median Income (asinh)	0.445	0.110
Median Rent (asinh)	0.449	0.139
Unemployment Rate	0.350	0.066
Labor Force Participation	0.146	0.067
Distance from CBD (asinh)	0.221	0.104
Distance to Highway (km)	0.046	0.076
HS Graduation Share	0.352	0.097
Mean SMD	**0.300**	**0.105**
Max SMD	**0.449**	**0.199**

Note: The table compares absolute standardized mean differences in baseline covariates between treated tracts and controls under the contiguity-ring spatial difference-in-differences design and the preferred matched difference-in-differences design. The spatial DiD column compares public-housing tracts to second-order adjacent outer-ring tracts within 2 kilometers of the project. The matched DiD column compares public-housing tracts to propensity-score matched controls. Covariates are measured in the pre-treatment baseline period used for each project cohort. Standardized mean differences are differences in means divided by the pooled standard deviation; values closer to zero indicate better balance.

The contiguity-ring design shows substantially worse balance than the matched difference-in-differences. Treated tracts have higher Black population share (SMD = 0.28 vs 0.15), lower income (SMD = 0.44 vs 0.11), and lower rent (SMD = 0.45 vs 0.14) compared to their respective controls. The mean SMD is also larger in the spatial design (0.30 vs 0.11).

Table 10 summarizes the key pre- and post-treatment estimates for Black population share across the preferred matched design and the two geographic designs. The post-treatment estimates are broadly similar in sign and magnitude: all three designs show increases in Black population share after public housing construction. The pre-treatment estimates, however, are not reassuring for the geographic designs. The contiguity-ring design shows a statistically significant positive pre-trend in Black share, while the distance-band design shows a statistically significant negative pre-trend. The matched design has a much smaller and statistically insignificant pre-period coefficient.

Table 10: Alternative Geographic Counterfactuals: Black Population Share

Design	$t = -2$	$t = +2$	$t = +3$
Matched DiD	-1.4	6.3***	5.9***
	(0.9)	(1.8)	(2.0)
Contiguity-ring spatial DiD	2.9***	3.1**	3.9***
	(1.0)	(1.3)	(1.4)
Distance-band spatial DiD	-2.9***	3.4*	2.1
	(0.9)	(2.0)	(2.6)

Note: The table reports event-study coefficients for Black population share, in percentage points. Standard errors are in parentheses. The matched DiD row is the preferred matched specification. The contiguity-ring spatial DiD compares treated tracts to second-order adjacent outer-ring tracts using the conservative Spec D fixed-effects structure. The distance-band spatial DiD uses the same 100-meter treatment buffer as the main analysis and compares treated tracts to control tracts 800–1500 meters away. All designs use $t = -1$ as the omitted event time. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The same pattern appears in the broader diagnostics. In the contiguity-ring specification, the conservative fixed effects structure substantially improves pre-trends for economic outcomes, but some differences remain. At $t = -2$, treated tracts show no statistically significant pre-trend for income: the asinh median income coefficient is -0.019 ($p = 0.068$). Rent pre-trends remain marginally significant (asinh coefficient of -0.043, $p = 0.047$). For racial composition, Black share shows a $+2.9$ percentage point pre-trend ($p = 0.005$) and white share shows a -2.6 percentage point pre-trend ($p = 0.019$). By contrast, the matched difference-in-differences has small and statistically insignificant $t = -2$ coefficients for Black share (-0.014, $p = 0.154$), white share ($+0.013$, $p = 0.165$), and asinh median income (-0.003, $p = 0.802$).

C.3 Results

Figures 21–24 present event study estimates from the contiguity-ring spatial difference-in-differences specification. Figure 22 presents the corresponding Black-share event study from the distance-band specification. These figures reinforce the diagnostic pattern in Table 10: post-treatment estimates move in the same direction as the matched estimates, but pre-treatment differences are more pronounced than in the matched design.

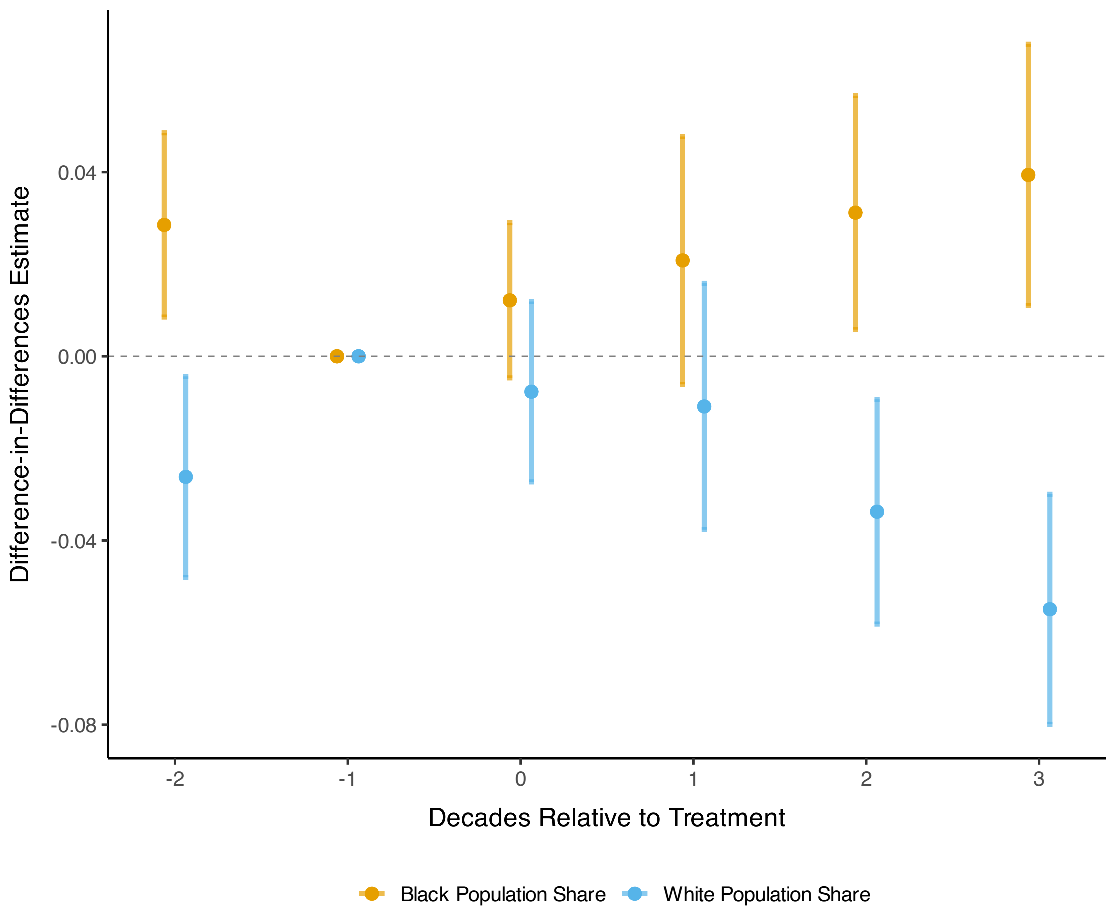


Figure 21: Contiguity-Ring Spatial DiD: Effects on Racial Composition in Treated Tracts

Note: Event study estimates from the contiguity-ring spatial difference-in-differences specification. Coefficients represent the differential change in treated tracts relative to outer-ring controls, with $t = -1$ as the reference period. Outer-ring controls are second-order adjacent tracts within 2km of the public housing project. Standard errors are clustered at the project level.

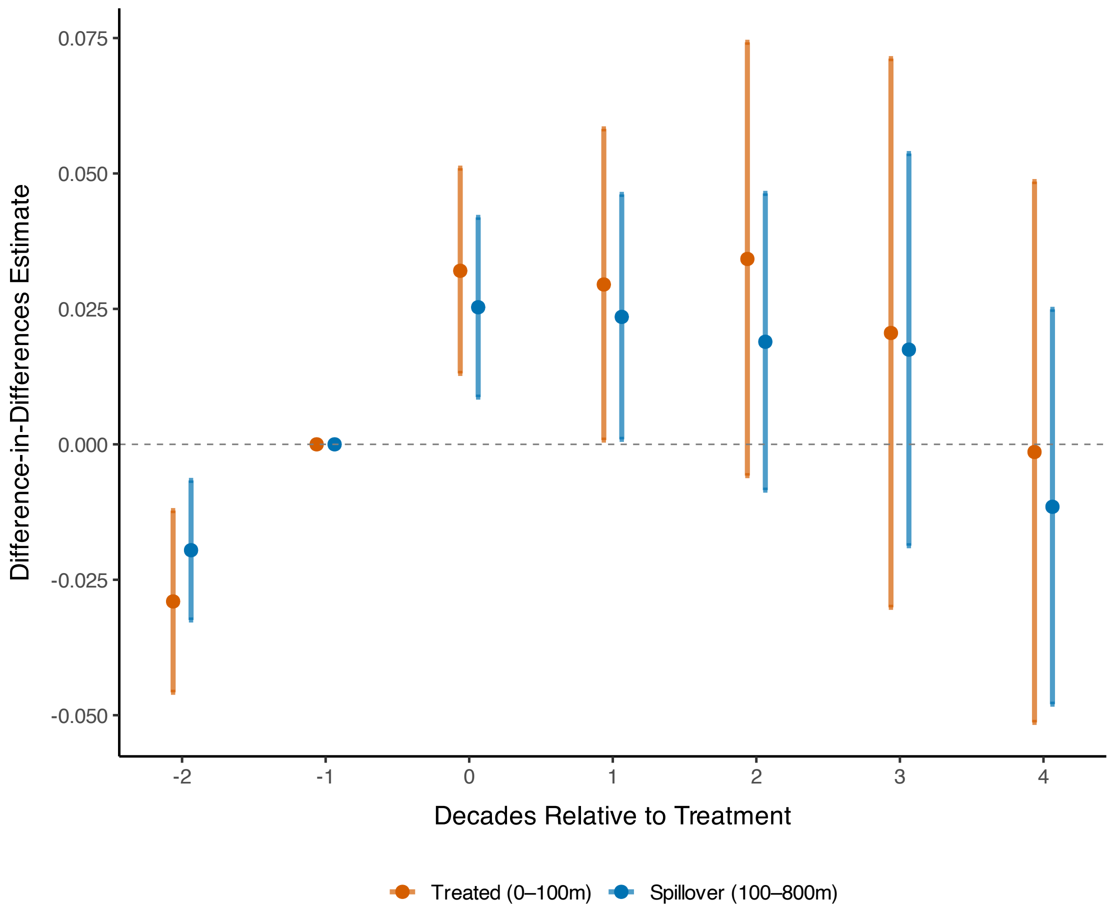


Figure 22: Distance-Band Spatial DiD: Effects on Black Population Share

Note: Event study estimates from the distance-band spatial difference-in-differences specification. The treated ring includes tracts within 0–100 meters of a project, matching the baseline treatment buffer. The nearby ring includes tracts 100–800 meters away, and the control ring includes tracts 800–1500 meters away. Coefficients are measured relative to the control ring at $t = -1$. Standard errors are clustered at the project level.

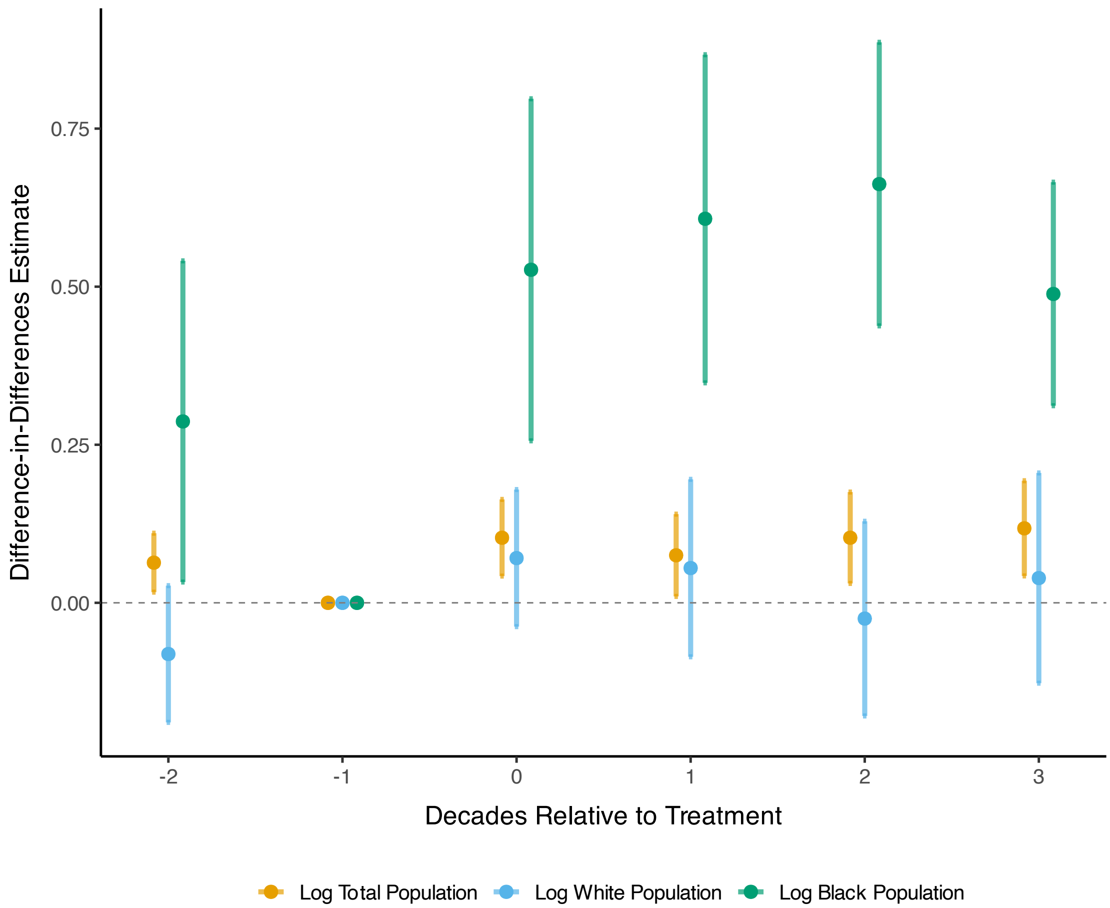


Figure 23: Contiguity-Ring Spatial DiD: Effects on Population by Race in Treated Tracts

Note: Event study estimates from the contiguity-ring spatial difference-in-differences specification. Coefficients represent the differential change in treated tracts relative to outer-ring controls, with $t = -1$ as the reference period. Outcome variables are inverse hyperbolic sine transformations. Standard errors are clustered at the project level.

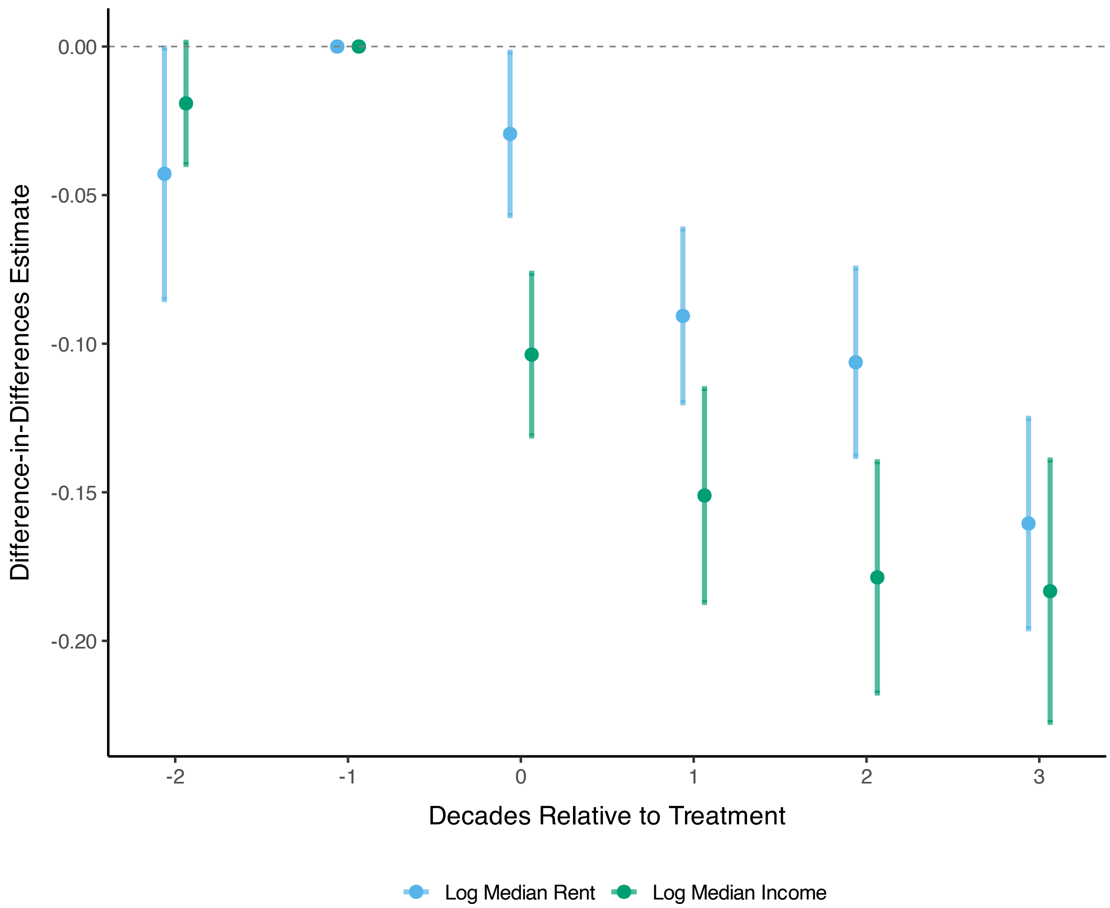


Figure 24: Contiguity-Ring Spatial DiD: Effects on Income and Rents in Treated Tracts

Note: Event study estimates from the contiguity-ring spatial difference-in-differences specification. Coefficients represent the differential change in treated tracts relative to outer-ring controls, with $t = -1$ as the reference period. Standard errors are clustered at the project level.

Racial Composition. The contiguity-ring design estimates a 3.1 percentage point increase in Black population share at $t = +2$ and a 3.9 percentage point increase at $t = +3$. The matched difference-in-differences estimates are larger but similar in direction: 6.3 percentage points at $t = +2$ and 5.9 percentage points at $t = +3$. The distance-band estimates in Table 10 are also positive in the post period. Thus, alternative geographic counterfactuals do not overturn the post-treatment racial-composition pattern, but their pre-trends make them less credible as standalone causal estimates.

Population, Income, and Rents. The contiguity-ring design also produces post-treatment changes in the same direction as the matched estimates for key demographic and economic outcomes. It estimates an increase in total tract population of 10.3% at $t = +2$ and 11.8%

at $t = +3$. It estimates declines in median income of 18% at $t = +2$ and 18.4% at $t = +3$, and declines in median rent of 10.6% and 16.1% in the same periods. These magnitudes are somewhat smaller than the matched estimates, but the qualitative pattern is similar.

Spillovers to Nearby Rings. The geographic designs also allow examination of nearby rings. In the contiguity-ring design, inner-ring tracts show a statistically insignificant 1.2 percentage point increase in Black share at $t = +2$. In the distance-band design, the 100–800m nearby ring similarly shows smaller and noisier estimates than the treated ring. This is consistent with the main matched spillover analysis, where nearby-neighborhood effects are present but smaller than the direct treated-tract effects.

Summary. These geographic designs are useful stress tests. They show that the post-treatment increase in Black population share and decline in economic status are not unique to the matched design. However, both spatial approaches exhibit meaningful pre-treatment differences, especially for racial composition. I therefore do not interpret the spatial estimates as preferred causal estimates. Instead, they show that plausible geographic counterfactuals point in the same post-treatment direction while also illustrating why the main analysis relies on matching to construct more comparable controls.

D Callaway-Sant'Anna Staggered Difference-in-Differences

As an alternative to the matched difference-in-differences approach, I implement the Callaway and Sant'Anna (2021) estimator for staggered treatment adoption. This estimator aggregates group-time average treatment effects across treatment cohorts using never-treated and not-yet-treated units as the comparison group. I use doubly-robust estimation, which combines inverse probability weighting with outcome regression to provide consistent estimates if either the propensity score or outcome model is correctly specified. Controls include CBSA fixed effects, distance to CBD, HOLC redlining status, urban renewal exposure, and 1940 baseline neighborhood characteristics.

Figures 25 and 26 present event study estimates. Treatment effects are qualitatively consistent with the main results: public housing increased Black population and reduced median income and rents. However, significant pre-trends appear two decades before treatment, particularly for log Black population (-0.245, $p < 0.001$) and log median income ($+0.041$, $p < 0.001$). These pre-trends indicate that treated tracts had lower Black populations and higher incomes than control tracts even before public housing was built—a selection pattern consistent with the site selection analysis in Section 5. This motivates the matched difference-in-differences approach, which restricts comparisons to observationally similar neighborhoods and produces flat pre-trends.

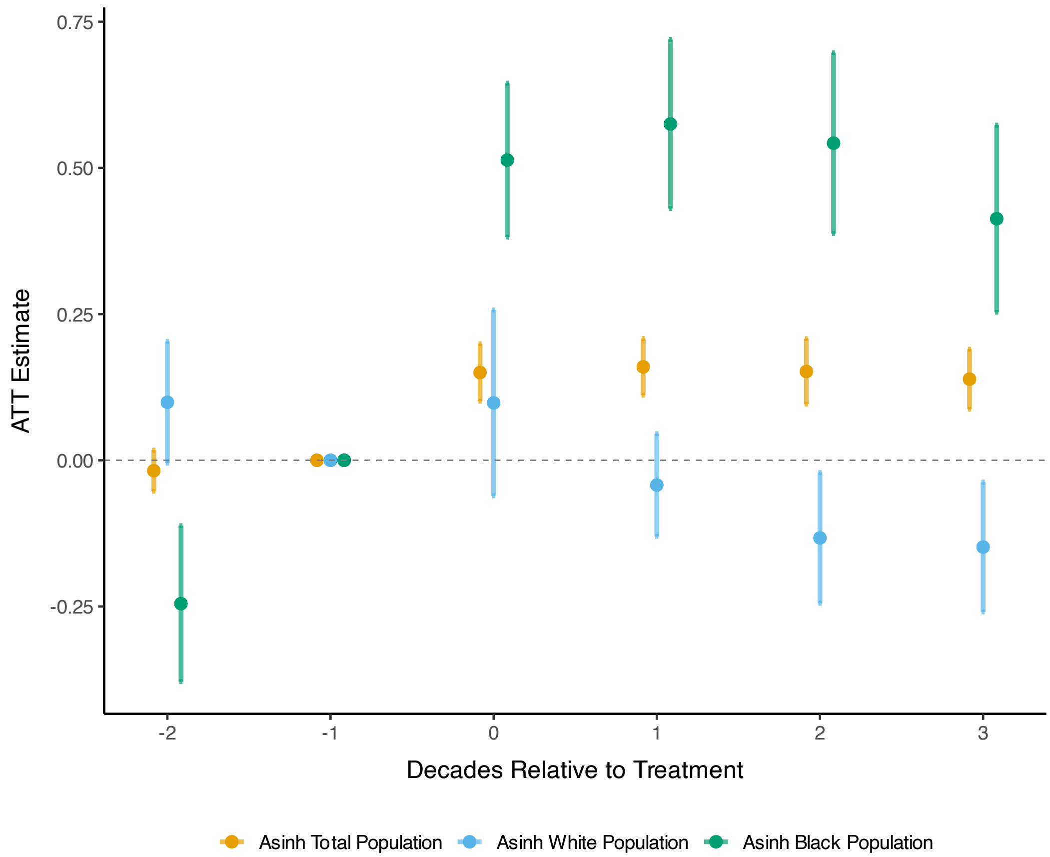


Figure 25: Callaway-Sant'Anna Event Study: Population by Race

Note: Event study estimates from the Callaway and Sant'Anna (2021) staggered-adoption estimator with doubly robust estimation. Outcomes are inverse hyperbolic sine transformations of total, White, and Black population. Comparison units are never-treated and not-yet-treated tracts. Controls include CBSA indicators, distance to the central business district, HOLC redlining status, urban renewal exposure, and 1940 baseline neighborhood characteristics. Vertical bars show 95% confidence intervals based on tract-clustered standard errors. The reference period is $t = -1$, one decade before treatment.

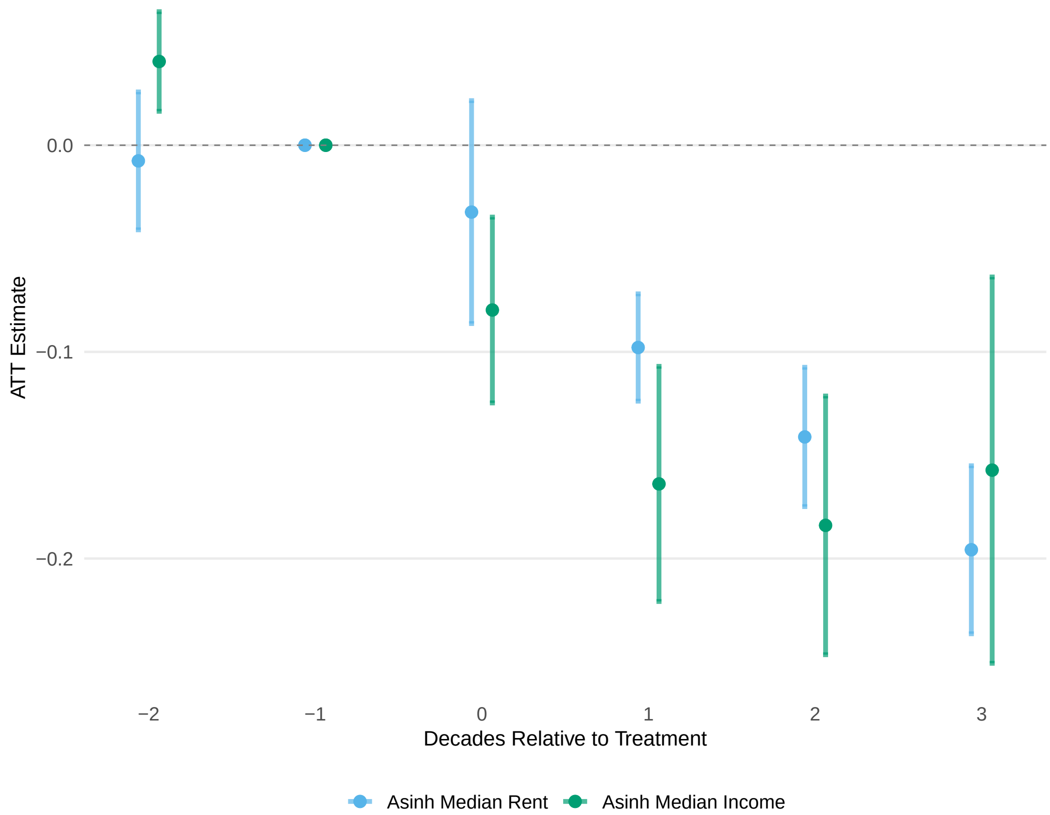


Figure 26: Callaway-Sant'Anna Event Study: Rent and Income

Note: Event study estimates from the Callaway and Sant'Anna (2021) staggered-adoption estimator with doubly robust estimation. Outcomes are inverse hyperbolic sine transformations of median rent and median income. Comparison units are never-treated and not-yet-treated tracts. Controls include CBSA indicators, distance to the central business district, HOLC redlining status, urban renewal exposure, and 1940 baseline neighborhood characteristics. Vertical bars show 95% confidence intervals based on tract-clustered standard errors. The reference period is $t = -1$, one decade before treatment.

E Robustness Checks: Alternative Matching Specifications and Standard Errors

This appendix presents full event study coefficient tables for alternative matching specifications discussed in the main text.

E.1 Alternative Matching Specifications

Here, I present the full set of coefficients from two alternative specifications that address potential limitations of the baseline matching approach. First, I present results using the baseline matching specification but adding a caliper restriction of 0.25 standard deviations of the propensity score to drop poor matches to improve covariate balance. Second, I run a specification in which I match public housing and nearby tracts to control tracts from different metropolitan areas, rather than within the same county. This specification expands the donor pool for each tract, potentially allowing for closer matches on pre-treatment characteristics, particularly for treated tracts in counties with limited control tracts.

Table 11: Caliper Matching: Treated Neighborhoods, Population and Demographics (Conley SEs)

Event Time	Asinh Black Pop	Asinh Total Pop	Asinh White Pop	Black Share
t = -2	-0.033 (0.116)	-0.011 (0.022)	-0.024 (0.043)	-0.007 (0.011)
t = 0	0.593*** (0.125)	0.187*** (0.024)	0.095* (0.053)	0.039*** (0.011)
t = +1	0.735*** (0.151)	0.244*** (0.030)	0.069 (0.083)	0.074*** (0.016)
t = +2	0.698*** (0.160)	0.244*** (0.035)	0.007 (0.086)	0.092*** (0.019)
t = +3	0.466*** (0.152)	0.230*** (0.036)	0.016 (0.097)	0.092*** (0.019)

Note: This table tests sensitivity of main results to stricter matching by dropping poor matches. Event study coefficients from Equation 3 for treated neighborhoods (N=521) using caliper matching (baseline specification plus 0.25 SD propensity score caliper). Sample: 47 CBSAs, 1930–2010. Outcomes: asinh(Black Pop), asinh(Total Pop), asinh(White Pop), and Black Share. Conley standard errors (2km cutoff) in parentheses. Reference period: $t = -1$. Statistical significance: *p<0.1, **p<0.05, ***p<0.01.

Table 12: Caliper Matching: Treated Neighborhoods, Economic and Housing Outcomes (Conley SEs)

Event Time	Asinh Median Income	Asinh Median Rent	HS Grad Rate	Unemp Rate
t = -2	-0.003 (0.013)	-0.017 (0.018)	0.006 (0.005)	0.005 (0.004)
t = 0	-0.078*** (0.014)	-0.002 (0.016)	-0.001 (0.005)	0.009** (0.004)
t = +1	-0.130*** (0.018)	-0.052*** (0.019)	-0.014** (0.006)	0.010** (0.004)
t = +2	-0.144*** (0.022)	-0.082*** (0.020)	-0.022*** (0.008)	0.016*** (0.005)
t = +3	-0.145*** (0.024)	-0.138*** (0.022)	-0.015* (0.009)	0.021*** (0.007)

Note: This table tests sensitivity of main results to stricter matching by dropping poor matches. Event study coefficients from Equation 3 for treated neighborhoods (N=521) using caliper matching (baseline specification plus 0.25 SD propensity score caliper). Sample: 47 CBSAs, 1930–2010. Outcomes: asinh(Median Income), asinh(Median Rent), HS Grad Rate, and Unemp Rate. Conley standard errors (2km cutoff) in parentheses. Reference period: $t = -1$. Statistical significance: *p<0.1, **p<0.05, ***p<0.01.

Table 13: Caliper Matching: Spillover Neighborhoods, Population and Demographics (Conley SEs)

Event Time	Asinh Black Pop	Asinh Total Pop	Asinh White Pop	Black Share
t = -2	0.006 (0.093)	0.002 (0.016)	-0.011 (0.042)	-0.005 (0.009)
t = 0	0.101 (0.095)	0.024* (0.013)	-0.007 (0.033)	0.019** (0.008)
t = +1	0.093 (0.114)	-0.003 (0.017)	-0.068 (0.050)	0.026* (0.014)
t = +2	0.005 (0.113)	-0.010 (0.019)	-0.085 (0.068)	0.024 (0.016)
t = +3	-0.103 (0.106)	-0.010 (0.022)	-0.084 (0.076)	0.026 (0.016)

Note: This table tests sensitivity of spillover results to stricter matching by dropping poor matches. Event study coefficients from Equation 3 for spillover neighborhoods (N=971; adjacent to treated tracts, within 1km of projects) using caliper matching (baseline specification plus 0.25 SD propensity score caliper). Sample: 47 CBSAs, 1930–2010. Outcomes: asinh(Black Pop), asinh(Total Pop), asinh(White Pop), and Black Share. Conley standard errors (2km cutoff) in parentheses. Reference period: $t = -1$. Statistical significance: *p<0.1, **p<0.05, ***p<0.01.

Table 14: Caliper Matching: Spillover Neighborhoods, Economic and Housing Outcomes (Conley SEs)

Event Time	Asinh Median Income	Asinh Median Rent	HS Grad Rate	Unemp Rate
t = -2	0.003 (0.010)	0.003 (0.012)	-0.003 (0.004)	0.001 (0.003)
t = 0	-0.026*** (0.009)	0.005 (0.010)	-0.004 (0.004)	-0.002 (0.003)
t = +1	-0.033** (0.015)	-0.012 (0.012)	-0.003 (0.006)	-0.001 (0.003)
t = +2	-0.027 (0.019)	-0.015 (0.015)	-0.003 (0.007)	0.000 (0.004)
t = +3	-0.014 (0.021)	-0.004 (0.017)	0.001 (0.008)	0.002 (0.004)

Note: This table tests sensitivity of spillover results to stricter matching by dropping poor matches. Event study coefficients from Equation 3 for spillover neighborhoods (N=971; adjacent to treated tracts, within 1km of projects) using caliper matching (baseline specification plus 0.25 SD propensity score caliper). Sample: 47 CBSAs, 1930–2010. Outcomes: asinh(Median Income), asinh(Median Rent), HS Grad Rate, and Unemp Rate. Conley standard errors (2km cutoff) in parentheses. Reference period: $t = -1$. Statistical significance: *p<0.1, **p<0.05, ***p<0.01.

Table 15: No CBSA Restriction: Treated Neighborhoods, Population and Demographics (Conley SEs)

Event Time	Asinh Black Pop	Asinh Total Pop	Asinh White Pop	Black Share
t = -2	-0.003 (0.088)	-0.023 (0.017)	-0.043 (0.042)	-0.005 (0.012)
t = 0	0.418*** (0.093)	0.146*** (0.019)	0.130*** (0.050)	0.024** (0.009)
t = +1	0.671*** (0.123)	0.216*** (0.031)	0.153** (0.073)	0.047*** (0.014)
t = +2	0.684*** (0.133)	0.250*** (0.038)	0.112 (0.083)	0.073*** (0.016)
t = +3	0.575*** (0.138)	0.254*** (0.042)	0.092 (0.090)	0.077*** (0.018)

Note: This table tests whether results hold when controls are drawn from different metropolitan areas, expanding the donor pool. Event study coefficients from Equation 3 for treated neighborhoods (N=815) using cross-metro matching (controls from other CBSAs, exact match on urban renewal). Sample: 47 CBSAs, 1930–2010. Outcomes: asinh(Black Pop), asinh(Total Pop), asinh(White Pop), and Black Share. Conley standard errors (2km cutoff) in parentheses. Reference period: $t = -1$. Statistical significance: *p<0.1, **p<0.05, ***p<0.01.

Table 16: No CBSA Restriction: Treated Neighborhoods, Economic and Housing Outcomes (Conley SEs)

Event Time	Asinh Median Income	Asinh Median Rent	HS Grad Rate	Unemp Rate
t = -2	0.004 (0.013)	0.002 (0.019)	0.001 (0.005)	0.002 (0.005)
t = 0	-0.076*** (0.015)	-0.008 (0.013)	0.001 (0.004)	-0.005 (0.006)
t = +1	-0.116*** (0.019)	-0.049*** (0.019)	-0.004 (0.006)	-0.009 (0.007)
t = +2	-0.114*** (0.027)	-0.080*** (0.025)	-0.007 (0.008)	0.000 (0.007)
t = +3	-0.102*** (0.030)	-0.143*** (0.029)	-0.021** (0.010)	0.004 (0.007)

Note: This table tests whether results hold when controls are drawn from different metropolitan areas, expanding the donor pool. Event study coefficients from Equation 3 for treated neighborhoods (N=815) using cross-metro matching (controls from other CBSAs, exact match on urban renewal). Sample: 47 CBSAs, 1930–2010. Outcomes: asinh(Median Income), asinh(Median Rent), HS Grad Rate, and Unemp Rate. Conley standard errors (2km cutoff) in parentheses. Reference period: $t = -1$. Statistical significance: *p<0.1, **p<0.05, ***p<0.01.

Table 17: No CBSA Restriction: Spillover Neighborhoods, Population and Demographics (Conley SEs)

Event Time	Asinh Black Pop	Asinh Total Pop	Asinh White Pop	Black Share
t = -2	-0.130 (0.086)	-0.030* (0.017)	-0.032 (0.035)	-0.004 (0.008)
t = 0	0.029 (0.081)	0.001 (0.012)	0.013 (0.034)	0.010 (0.008)
t = +1	0.070 (0.112)	-0.012 (0.019)	-0.050 (0.053)	0.021* (0.012)
t = +2	-0.012 (0.116)	-0.005 (0.026)	-0.056 (0.065)	0.028* (0.014)
t = +3	-0.071 (0.120)	0.009 (0.031)	-0.035 (0.075)	0.029* (0.015)

Note: This table tests whether spillover results hold when controls are drawn from different metropolitan areas, addressing concerns about CBSA-specific confounders. Event study coefficients from Equation 3 for spillover neighborhoods (N=1,228; adjacent to treated tracts, within 1km of projects) using cross-metro matching. Sample: 47 CBSAs, 1930–2010. Outcomes: asinh(Black Pop), asinh(Total Pop), asinh(White Pop), and Black Share. Conley standard errors (2km cutoff) in parentheses. Reference period: $t = -1$. Statistical significance: *p<0.1, **p<0.05, ***p<0.01.

Table 18: No CBSA Restriction: Spillover Neighborhoods, Economic and Housing Outcomes (Conley SEs)

Event Time	Asinh Median Income	Asinh Median Rent	HS Grad Rate	Unemp Rate
t = -2	0.006 (0.011)	0.004 (0.014)	-0.007 (0.004)	-0.002 (0.004)
t = 0	-0.030*** (0.012)	-0.009 (0.011)	0.007* (0.004)	-0.007 (0.005)
t = +1	-0.028** (0.014)	-0.019 (0.016)	0.011* (0.006)	-0.006 (0.004)
t = +2	-0.006 (0.018)	-0.004 (0.019)	0.015** (0.008)	-0.006 (0.004)
t = +3	0.017 (0.021)	0.009 (0.023)	0.015* (0.009)	-0.003 (0.005)

Note: This table tests whether spillover results hold when controls are drawn from different metropolitan areas, addressing concerns about CBSA-specific confounders. Event study coefficients from Equation 3 for spillover neighborhoods (N=1,228; adjacent to treated tracts, within 1km of projects) using cross-metro matching. Sample: 47 CBSAs, 1930–2010. Outcomes: asinh(Median Income), asinh(Median Rent), HS Grad Rate, and Unemp Rate. Conley standard errors (2km cutoff) in parentheses. Reference period: $t = -1$. Statistical significance: *p<0.1, **p<0.05, ***p<0.01.

E.2 Low-Intensity Metro Controls: Addressing SUTVA Concerns

This section presents results using control neighborhoods drawn only from metropolitan areas with low public housing intensity, as described in Section 6.4. I calculate each metro's intensity as total public housing units divided by the 1950 population of sampled neighborhoods, expressed per 1,000 residents. I restrict the control pool to metros in the bottom quintile of intensity (fewer than 3.9 units per 1,000). Treated neighborhoods from all metros are matched to controls in these low-intensity metros, with matches required to come from different metropolitan areas.

Table 19: Low-Intensity Controls: Treated Neighborhoods, Population and Demographics (Conley SEs)

Event Time	Asinh Black Pop	Asinh Total Pop	Asinh White Pop	Black Share
t = -2	0.105 (0.126)	-0.050* (0.026)	-0.053 (0.042)	0.001 (0.014)
t = 0	0.344*** (0.100)	0.136*** (0.028)	0.060 (0.053)	0.007 (0.013)
t = +1	0.473*** (0.150)	0.099* (0.054)	-0.043 (0.095)	0.041** (0.020)
t = +2	0.599*** (0.186)	0.105 (0.071)	-0.159 (0.128)	0.077*** (0.027)
t = +3	0.566*** (0.207)	0.122 (0.086)	-0.121 (0.152)	0.083*** (0.031)

Note: This table addresses SUTVA concerns by restricting controls to metros with minimal public housing exposure. Event study coefficients from Equation 3 for treated neighborhoods (N=820) using low-intensity matching (controls from metros in bottom quintile of PH intensity, $<$ 3.9 units per 1,000 residents). Sample: 47 CBSAs, 1930–2010. Outcomes: asinh(Black Pop), asinh(Total Pop), asinh(White Pop), and Black Share. Conley standard errors (2km cutoff) in parentheses. Reference period: $t = -1$. Statistical significance: *p<0.1, **p<0.05, ***p<0.01.

Table 20: Low-Intensity Controls: Treated Neighborhoods, Economic and Housing Outcomes (Conley SEs)

Event Time	Asinh Median Income	Asinh Median Rent	HS Grad Rate	Unemp Rate
t = -2	0.017 (0.016)	0.184 (0.136)	-0.004 (0.007)	-0.002 (0.007)
t = 0	-0.038 (0.029)	-0.085* (0.045)	0.009* (0.005)	-0.012 (0.008)
t = +1	-0.060*** (0.023)	-0.097** (0.043)	0.026** (0.012)	-0.019** (0.009)
t = +2	-0.043 (0.032)	-0.120** (0.051)	0.041** (0.017)	-0.016* (0.009)
t = +3	0.054 (0.038)	-0.121*** (0.045)	0.057*** (0.020)	-0.009 (0.011)

Note: This table addresses SUTVA concerns by restricting controls to metros with minimal public housing exposure. Event study coefficients from Equation 3 for treated neighborhoods (N=820) using low-intensity matching (controls from metros in bottom quintile of PH intensity). Sample: 47 CBSAs, 1930–2010. Outcomes: asinh(Median Income), asinh(Median Rent), HS Grad Rate, and Unemp Rate. Conley standard errors (2km cutoff) in parentheses. Reference period: $t = -1$. Statistical significance: *p<0.1, **p<0.05, ***p<0.01.

Table 21: Low-Intensity Controls: Spillover Neighborhoods, Population and Demographics (Conley SEs)

Event Time	Asinh Black Pop	Asinh Total Pop	Asinh White Pop	Black Share
t = -2	-0.052 (0.095)	-0.036** (0.016)	-0.052 (0.047)	-0.010 (0.010)
t = 0	-0.024 (0.086)	0.003 (0.019)	0.036 (0.058)	0.005 (0.010)
t = +1	-0.017 (0.132)	0.006 (0.032)	-0.066 (0.076)	0.017 (0.016)
t = +2	-0.142 (0.158)	-0.018 (0.042)	-0.195* (0.111)	0.037* (0.022)
t = +3	-0.141 (0.180)	0.007 (0.057)	-0.164 (0.137)	0.042 (0.027)

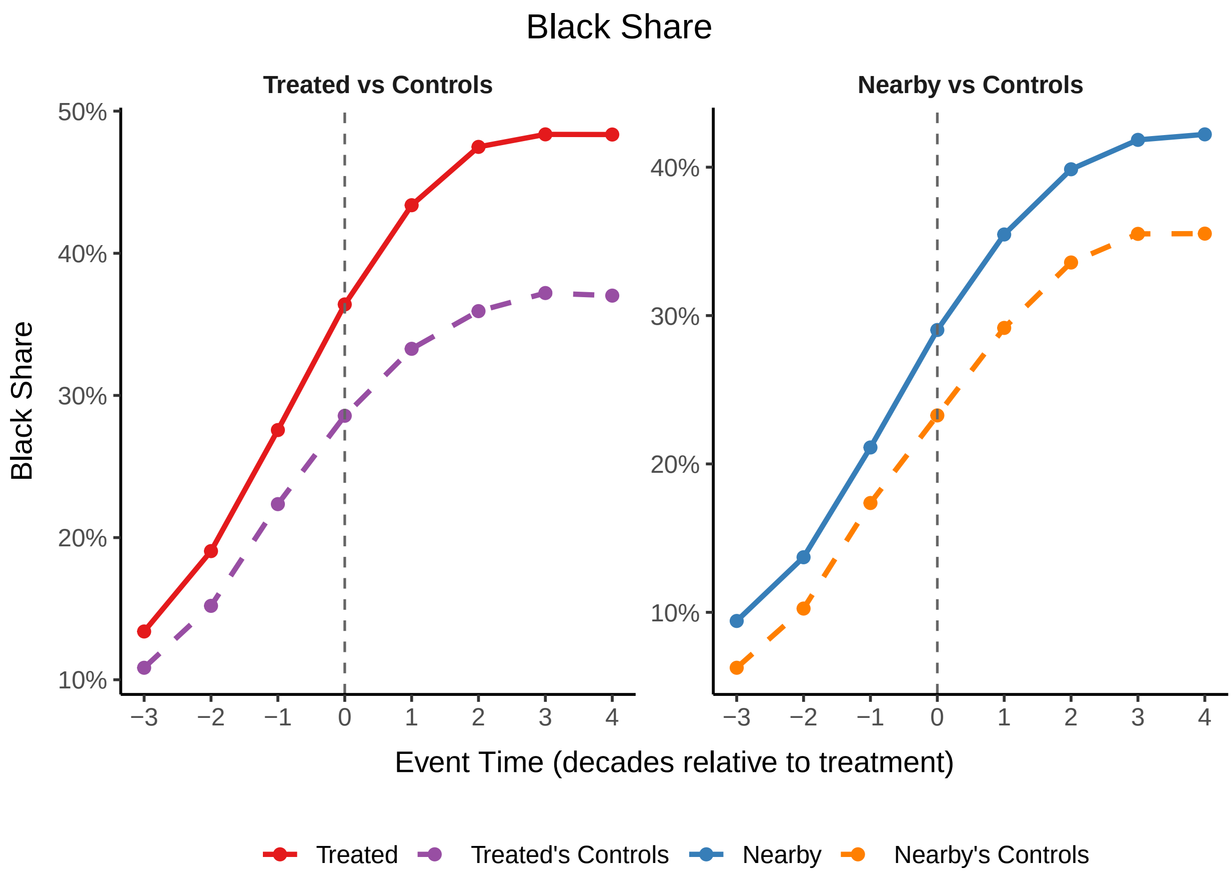
Note: This table addresses SUTVA concerns for spillover neighborhoods by restricting controls to metros with minimal public housing exposure. Event study coefficients from Equation 3 for spillover neighborhoods (N=1,427; adjacent to treated tracts, within 1km of projects) using low-intensity matching (controls from metros in bottom quintile of PH intensity). Sample: 47 CBSAs, 1930–2010. Outcomes: asinh(Black Pop), asinh(Total Pop), asinh(White Pop), and Black Share. Conley standard errors (2km cutoff) in parentheses. Reference period: $t = -1$. Statistical significance: *p<0.1, **p<0.05, ***p<0.01.

Table 22: Low-Intensity Controls: Spillover Neighborhoods, Economic and Housing Outcomes (Conley SEs)

Event Time	Asinh Median Income	Asinh Median Rent	HS Grad Rate	Unemp Rate
t = -2	0.002 (0.011)	-0.065 (0.041)	-0.013** (0.005)	-0.005 (0.006)
t = 0	-0.016 (0.010)	-0.046 (0.043)	0.013*** (0.005)	-0.020*** (0.006)
t = +1	0.025 (0.015)	0.002 (0.049)	0.033*** (0.008)	-0.032*** (0.007)
t = +2	0.060*** (0.022)	0.021 (0.053)	0.052*** (0.012)	-0.032*** (0.006)
t = +3	0.158*** (0.026)	0.054 (0.053)	0.073*** (0.015)	-0.027*** (0.007)

Note: This table addresses SUTVA concerns for spillover neighborhoods by restricting controls to metros with minimal public housing exposure. Event study coefficients from Equation 3 for spillover neighborhoods (N=1,427; adjacent to treated tracts, within 1km of projects) using low-intensity matching (controls from metros in bottom quintile of PH intensity). Sample: 47 CBSAs, 1930–2010. Outcomes: asinh(Median Income), asinh(Median Rent), HS Grad Rate, and Unemp Rate. Conley standard errors (2km cutoff) in parentheses. Reference period: $t = -1$. Statistical significance: *p<0.1, **p<0.05, ***p<0.01.

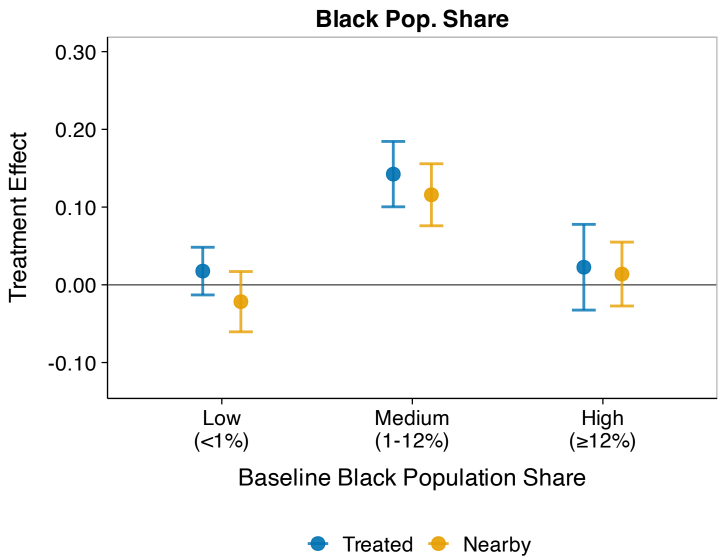
Figure 27: Black Population Share: Treated and Nearby Neighborhoods vs Matched Controls Over Event Time



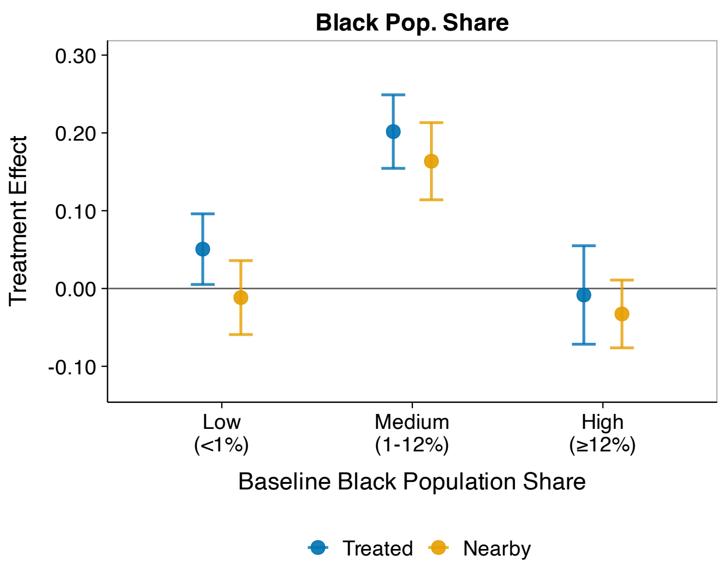
Note: Mean Black population share for treated neighborhoods and their matched controls (left panel) and nearby neighborhoods and their matched controls (right panel), aligned by event time (decades relative to public housing construction). Shaded bands show 95% confidence intervals. Both treated and control neighborhoods show increasing Black shares over time, consistent with the Great Migration. The treatment effect is identified by the differential increase in treated relative to controls.

Figure 28: Tipping Heterogeneity: Black Population Share as Outcome

(a) Medium-run Effects ($t = 1$)

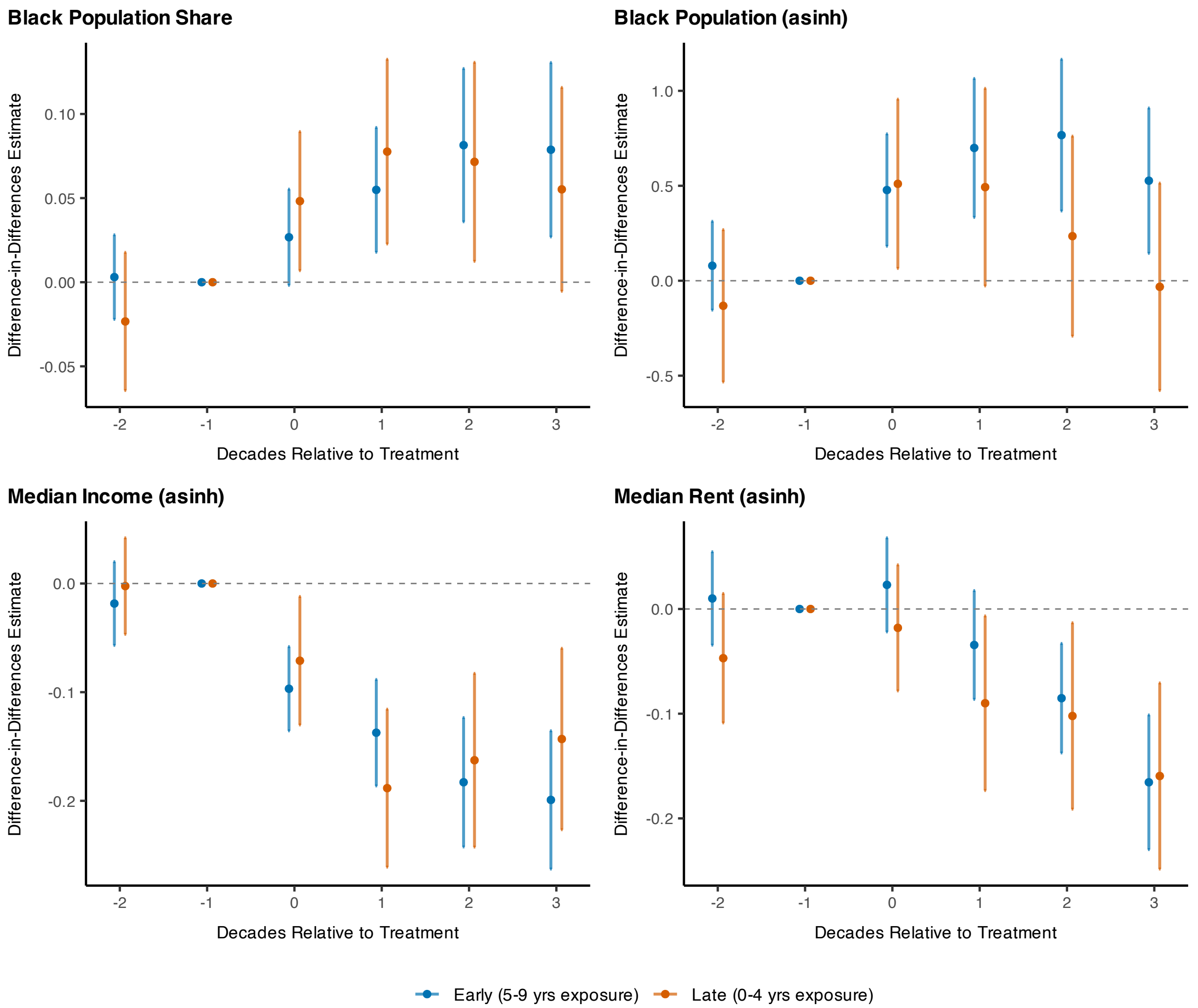


(b) Long-run Effects ($t = 3$)



Note: Treatment effects from Equation 3 with Black population share as the outcome, estimated separately by baseline Black share: very low (<1%; $N = 257$), medium (1–12%; $N = 186$), and high (≥12%; $N = 356$). These bins distinguish nearly all-white neighborhoods, neighborhoods with low but nontrivial Black populations, and neighborhoods with higher baseline Black shares. Conley standard errors (2km cutoff). Reference period: $t = -1$.

Figure 29: Within-Decade Timing: Early vs Late Openers (Pooled)



Note: Event study coefficients from Equation 3, estimated separately for early openers (projects opening 5 to 9 years before the next census, 288 treated tracts) and late openers (0 to 4 years, 182 treated tracts), pooling across treatment decades. Panels show Black population share (top left), Black population in asinh (top right), median income in asinh (bottom left), and median rent in asinh (bottom right). 95% confidence intervals based on Conley standard errors (2km cutoff). Reference period: $t = -1$.

E.3 Baseline Specification: Tract-Level Clustered Standard Errors

This section presents event study coefficients for the baseline matching specification using standard errors clustered at the census tract level, rather than using Conley (1999) spatially-correlated standard errors, as used in the main text.

Table 23: Baseline Matching: Treated Neighborhoods, Population and Demographics (Tract-Clustered SEs)

Event Time	Asinh Black Pop	Asinh Total Pop	Asinh White Pop	Black Share
t = -2	-0.007 (0.083)	0.002 (0.018)	0.015 (0.043)	-0.014* (0.008)
t = 0	0.393*** (0.083)	0.146*** (0.018)	0.071* (0.039)	0.026*** (0.008)
t = +1	0.495*** (0.099)	0.181*** (0.025)	0.109* (0.060)	0.049*** (0.011)
t = +2	0.476*** (0.104)	0.194*** (0.030)	0.029 (0.068)	0.063*** (0.013)
t = +3	0.275*** (0.102)	0.180*** (0.031)	0.026 (0.072)	0.059*** (0.014)

Note: Event study coefficients from Equation 3 for treated neighborhoods (N=799) using baseline matching (exact on county and urban renewal). Outcomes: asinh(Black Pop), asinh(Total Pop), asinh(White Pop), and Black Share. Standard errors clustered at the census tract level in parentheses (cf. Conley SEs in main text). Reference period: $t = -1$. Statistical significance: *p<0.1, **p<0.05, ***p<0.01.

Table 24: Baseline Matching: Treated Neighborhoods, Economic and Housing Outcomes (Tract-Clustered SEs)

Event Time	Asinh Median Income	Asinh Median Rent	HS Grad Rate	Unemp Rate
t = -2	-0.003 (0.011)	-0.011 (0.013)	0.003 (0.005)	0.005* (0.003)
t = 0	-0.077*** (0.012)	0.004 (0.012)	-0.006 (0.004)	0.004 (0.003)
t = +1	-0.133*** (0.015)	-0.051*** (0.015)	-0.016*** (0.005)	0.005* (0.003)
t = +2	-0.151*** (0.019)	-0.086*** (0.016)	-0.025*** (0.006)	0.011*** (0.004)
t = +3	-0.150*** (0.020)	-0.144*** (0.018)	-0.022*** (0.007)	0.015*** (0.005)

Note: Event study coefficients from Equation 3 for treated neighborhoods (N=799) using baseline matching (exact on county and urban renewal). Sample: 47 CBSAs, 1930–2010. Outcomes: asinh(Median Income), asinh(Median Rent), HS Grad Rate, and Unemp Rate. Standard errors clustered at the census tract level in parentheses (cf. Conley SEs in main text). Reference period: $t = -1$. Statistical significance: *p<0.1, **p<0.05, ***p<0.01.

Table 25: Baseline Matching: Spillover Neighborhoods, Population and Demographics (Tract-Clustered SEs)

Event Time	Asinh Black Pop	Asinh Total Pop	Asinh White Pop	Black Share
t = -2	0.044 (0.072)	0.002 (0.012)	-0.008 (0.029)	-0.003 (0.006)
t = 0	0.099 (0.071)	0.007 (0.011)	-0.028 (0.026)	0.020*** (0.007)
t = +1	0.089 (0.084)	-0.014 (0.015)	-0.090** (0.041)	0.025*** (0.010)
t = +2	0.005 (0.090)	-0.018 (0.017)	-0.108** (0.051)	0.025** (0.011)
t = +3	-0.082 (0.088)	-0.015 (0.018)	-0.117** (0.056)	0.026** (0.012)

Note: Event study coefficients from Equation 3 for spillover neighborhoods (N=1,218; adjacent to treated tracts, within 1km of projects) using baseline matching (exact on county and urban renewal). Sample: 47 CBSAs, 1930–2010. Outcomes: asinh(Black Pop), asinh(Total Pop), asinh(White Pop), and Black Share. Standard errors clustered at the census tract level in parentheses (cf. Conley SEs in main text). Reference period: $t = -1$. Statistical significance: *p<0.1, **p<0.05, ***p<0.01.

Table 26: Baseline Matching: Spillover Neighborhoods, Economic and Housing Outcomes (Tract-Clustered SEs)

Event Time	Asinh Median Income	Asinh Median Rent	HS Grad Rate	Unemp Rate
t = -2	0.003 (0.009)	0.006 (0.011)	-0.001 (0.003)	0.000 (0.002)
t = 0	-0.025*** (0.008)	0.011 (0.008)	-0.004 (0.003)	-0.003 (0.002)
t = +1	-0.035*** (0.011)	-0.006 (0.010)	-0.002 (0.004)	-0.002 (0.002)
t = +2	-0.026* (0.014)	-0.004 (0.011)	-0.003 (0.005)	-0.002 (0.003)
t = +3	-0.015 (0.015)	-0.001 (0.013)	-0.003 (0.006)	0.002 (0.003)

Note: Event study coefficients from Equation 3 for spillover neighborhoods (N=1,218; adjacent to treated tracts, within 1km of projects) using baseline matching (exact on county and urban renewal). Sample: 47 CBSAs, 1930–2010. Outcomes: asinh(Median Income), asinh(Median Rent), HS Grad Rate, and Unemp Rate. Standard errors clustered at the census tract level in parentheses (cf. Conley SEs in main text). Reference period: $t = -1$. Statistical significance: *p<0.1, **p<0.05, ***p<0.01.